DLS Buddy — Theory and Equations Guide

Applies to v1.21.0 and later

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Reading This Guide

This guide explains the physics and statistics behind every analysis the program offers, and it is the program's equation of record: every equation is numbered, labeled and cited, and each rule the program applies — the flag, the refusal or the default — is stated once, on a marked **In the program:** line in the section it governs. A reader who wants the *why* reads the prose; a reader who wants the *rule* reads the equation block and its marked line. The chapters stand in the order their material is needed; where a section must use a quantity a later chapter defines, it names that chapter's section at the first use:

- **Chapter 1, Inputs and Calibration** — the solvent's refractive index and viscosity (§1.1), the scattering vector (§1.2), the Stokes–Einstein relation (§1.3), the refractive-index correction (§1.4), the optical constant and what a blank $\mathrm{d}n/\mathrm{d}c$ does (§1.5), the polarization convention and the toluene Rayleigh ratio (§1.6), and the calibration of raw intensities to an excess Rayleigh ratio (§1.7), which is also the one home of what an uncalibrated measurement still yields.
- **Chapter 2, The Measurement and Its Quality** — the count-rate trace and its diagnostics (§2.1), the correlogram, the Siegert relation and the multi-tau lag axis (§2.2), and the noise a correlator puts on each channel (§2.3), which the DLS fits weight by.
- **Chapter 3, Dynamic Light Scattering** — what every DLS fit shares (§3.1: the delay window, the determined baseline, the decay-completeness check and the size range), the cumulant methods (§3.2), the exponential fits (§3.3), the distribution methods and the choice of regularization (§3.4), combining angles and concentrations (§3.5), depolarized DLS and its shape models (§3.6), and the intensity-to-number reweighting of a distribution (§3.7).
- **Chapter 4, Static Light Scattering** — intensities derived from DLS count rates (§4.1) and partitioned by mode (§4.2), the single-concentration Debye plot (§4.3), the Zimm and Berry double extrapolation (§4.4), the calibration-free $2A_2M_\mathrm{w}$ product (§4.5), the Guinier plot (§4.6), the depolarization correction (§4.7), and the form-factor fits — sphere and coil (§4.8), ellipsoid (§4.9), Mie (§4.10) and core–shell (§4.11).
- **Chapter 5, Derived and Cross-Sample Quantities** — $\rho = R_\mathrm{g}/R_\mathrm{h}$ (§5.1), the scaling laws (§5.2), the branching contraction factor g (§5.3) and its hydrodynamic sibling h (§5.4), all of which consume results of chapters 3 and 4.
- **Chapter 6, Uncertainty** — the consolidation: which ± the program reports and which it withholds, the HC3 estimator (§6.1) and the classical opt-in (§6.1.1), the delta method (§6.2), the Zimm two-route check (§6.3), the replicate ± (§6.4), the roster of where a ± is and is not reported (§6.5) and the ill-conditioning flag (§6.6).
- **Chapter 7, Synthetic Data Generation** — what the built-in generator draws (§7.1), its per-channel noise models (§7.2) and its correlated-noise control (§7.3), so every analysis above can be checked against a known answer.

The **Notation** table below lists the symbols the guide uses throughout; the Equations Index at the end lists every equation with its name and section. The **User-Manual** is the companion for the controls: each section here names the program's rule, and the Manual says where the control is and what the screen shows.

> **Note.** Equations are numbered (1), (2), … and rendered in LaTeX; an in-text reference like "Eq. 2" resolves to the numbered display equation, and citations resolve to the numbered reference list at the end.

Notation

One symbol per quantity, throughout the guide. Where a cited source writes a quantity with a different letter, the source's symbol is named in words beside the citation and the guide keeps the symbol below. A symbol local to one section is defined where it is used and is not listed here.

Symbol	Quantity	Units	Defined in
λ	vacuum wavelength of the laser	nm	§1.2
n	refractive index of the solvent (medium) at λ and T	—	§1.1
η	dynamic viscosity of the solvent at T	Pa·s	§1.1
T	absolute temperature	K	§1.3
θ	scattering angle	°	§1.2
q	scattering vector magnitude, $(4\pi n/\lambda)\sin(\theta/2)$	nm^{-1}	§1.2
k_B	Boltzmann constant	$\mathrm{J\ K}^{-1}$	§1.3
D	translational diffusion coefficient	$\mathrm{m}^2\ \mathrm{s}^{-1}$	§1.3
R_h	hydrodynamic radius	nm	§1.3
R_g	radius of gyration (Burchard's $\langle S^2\rangle_z^{1/2}$, Sedlák's R_G)	nm	§4.3
M_w	weight-average molar mass; $M_\mathrm{w,app}$ its apparent value	$\mathrm{g\ mol}^{-1}$	§4.3
A_2	second virial coefficient	$\mathrm{mol\ mL\ g}^{-2}$	§4.3
c	solute mass concentration	$\mathrm{g\ mL}^{-1}$	§1.5
$\mathrm{d}n/\mathrm{d}c$	refractive-index increment of the solute	$\mathrm{mL\ g}^{-1}$	§1.5
K	optical constant	$\mathrm{mol\ mL\ g}^{-2}\ \mathrm{cm}^{-1}$	§1.5
f_n	refractive-index correction $(n_\mathrm{solvent}/n_\mathrm{standard})^2$	—	§1.4
I_VV, I_VH	polarized and depolarized scattered intensity; subscripts (incident, analyzer)	file units	§1.6
ρ_v, ρ_u	depolarization ratio for vertically polarized and for unpolarized incident light	—	§1.6
R_θ, ΔR_θ	Rayleigh ratio; excess Rayleigh ratio of the solution over the solvent	cm^{-1}	§1.7
k_c	calibration constant from the calibrant measurement	cm^{-1} per file unit	§1.7
$\bar{r}$	mean count rate of a trace	counts s^{-1}	§2.1
k_σ	the outlier band's multiplier (Iglewicz and Hoaglin's D)	—	§2.1.1
τ	delay (lag) time of a correlogram channel	µs	§2.2

Symbol	Quantity	Units	Defined in
Δt_k	sample time channel k was accumulated over (in the noise formulas k counts channels within their block)	µs	§2.2
L	channels per octave of a multi-tau correlator	—	§2.2
T_{run}	run duration of an acquisition	s	§2.3
$G(\tau)$, G_∞	raw intensity autocorrelation and its long-delay level (Koppel's B, ISO's A)	—	§2.2
$g_2(\tau) - 1$, $g_1(\tau)$	normalized intensity autocorrelation minus one; field autocorrelation	—	§2.2
β	coherence (intercept) factor of the Siegert relation (ISO's B)	—	§2.2
Γ, $\bar{\Gamma}$	decay rate of g_1; the mean decay rate of a cumulant fit	s^{-1}	§3.2
Θ_k	decay per sample time, $\bar{\Gamma}\,\Delta t_k$ (the dimensionless Γ of Schätzel and of Biganzoli and Ferri)	—	§2.3
$\bar{N}_k$	mean photon count per sample time of channel k (Biganzoli and Ferri's $\langle n \rangle$)	—	§2.3
μ_2, μ_3, μ_4	second, third and fourth central moments of the decay-rate distribution	s^{-2}, s^{-3}, s^{-4}	§3.2
PDI	polydispersity index, $\mu_2/\bar{\Gamma}^2$	—	§3.2
$\varkappa_4$	the fourth cumulant, $\mu_4 - 3{\mu_2}^2$	s^{-4}	§3.2.6
B	the additive baseline of $g_2 - 1$ the Nonlinear fit floats; ε the same offset pinned for the exponential fits	—	§3.2.5, §3.1.3
$G(\Gamma)$	scattered-intensity distribution of decay rates (ISO's $C(\Gamma)$)	—	§3.2.2, §3.4.1
α	Tikhonov regularization strength of a CONTIN fit (Lukas's λ is α^2)	—	§3.4
γ_{r}	robustness parameter of robust GCV (Lukas's γ)	—	§3.4.7
t_2	normalized trace of the squared influence matrix (Lukas's μ_2)	—	§3.4.7
D_{t}, D_{r}	translational and rotational diffusion coefficients (DDLS; Koenig's $D_\perp$)	$\mathrm{m}^2\ \mathrm{s}^{-1}$, s^{-1}	§3.6
p	axial ratio of a rod or an ellipsoid of revolution, polar over equatorial; $\phi = 1/p$ is Perrin's and Koenig's ρ	—	§3.6.5, §3.6.6, §4.9
x	size parameter $2\pi n R/\lambda$ (in-medium)	—	§3.7, §4.10
m	relative refractive index n_{solute}/n	—	§3.7, §4.10
ρ	shape ratio $R_{\mathrm{g}}/R_{\mathrm{h}}$	—	§5.1
g, h	branching contraction factors on R_{g} and on R_{h}	—	§5.3, §5.4
f_{a}	arm number of a regular star (Zimm and Stockmayer's functionality f)	—	§5.3
m_{b}	branching index, the average number of branch units per molecule (Zimm and Stockmayer's m; the column **n (branch)**)	—	§5.3.1

Symbol	Quantity	Units	Defined in
N	the number of independent points a statistic rests on: samples in a trace, points in a regression, replicates in a set	—	§2.1.4, §6.1, §6.4
SE, SEM	standard error of a fitted coefficient; standard error of a replicate mean	as the quantity	§6.1, §6.4
h_{ii}	leverage of point i in a regression (a different symbol from h)	—	§6.1
$\mathrm{cond}(\widehat{\mathbf{X}})$	condition number of the column-normalized design	—	§6.6

Symbols local to one section

Symbols local to one section are defined there and are not listed: the Cauchy coefficients A_{n}, B_{n}, C_{n} (§1.1.1) beside the Aden–Kerker ratio functions A_n, B_n of §4.11 and the Mie series order n of §4.10–§4.11 (a subscript, never the refractive index); the trace samples I_k and their median $\tilde{I}$ (§2.1.1) and the block count m of §2.1.4; the dark count d (§1.7.2) beside the particle diameter d (§3.2.2) and the rod diameter d (§3.6.5); the rod length L (§3.6.5) beside the channels per octave; the Zimm spacing constant k (§4.4) and the wavenumber $k = 2\pi n/\lambda$ (§4.10–§4.11) beside the channel index; the parameter count p (§6.1) beside the axial ratio; the number distribution $N(R)$ (§3.7) and the leading-channel skip N (§3.1.1) beside the point count; the lognormal weight w_i (§3.4.10) beside the residual weight w_k; the weight anchor a_{w} (§3.4.3); the singular values s_j of §3.4.5 and §6.1 and the matrix $\mathbf{S}$ that holds them; the regression coefficients a, b, d and their primed Berry forms (§4.4), the sphere's $u = qR$ and the coil's $v = (qR_{\mathrm{g}})^2$ (§4.8), Welch's segment quantities L_{s}, N_{s}, Δ_{s} and $C_{\mathrm{ov}}(j)$ (§2.1.6), the Hampel half-width H (§2.1.2), the stretch exponent s of the KWW fit (§3.3.3) beside the sample SD s of §6.4, the Riccati–Bessel functions and Bohren & Huffman's angle functions π_n, τ_n (§4.10), and the lag index k and the matrix indices i, j wherever they appear inside one equation.

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1. Inputs and Calibration

Every quantity the program reports rests on a handful of inputs that no fit can recover from the data: the solvent's refractive index and viscosity, the laser wavelength, the scattering angle, the temperature, the solute's refractive-index increment, and — for absolute static results — a calibration of the detector's intensity scale. This chapter defines them in the order they are consumed. §1.1 states where the two solvent properties come from and how their tolerance is shown; §1.2 and §1.3 give the two relations everything else builds on, the scattering vector and the Stokes–Einstein radius; §1.4 and §1.5 introduce the refractive-index correction and the optical constant, and say what the program does when the increment is absent; §1.6 fixes the polarization convention and the toluene Rayleigh ratio every calibration uses; §1.7 puts a raw intensity on the absolute scale, and is the one place the guide says what an **uncalibrated** measurement still yields. The chapter closes with what chapters 3 and 4 take from it (§1.8).

1.1 The Solvent Inputs: Refractive Index and Viscosity

Every scattering calculation needs two properties of the solvent. Its **refractive index** $n(\lambda,\ T)$ sets the scattering vector q (Eq. 5), and its **dynamic viscosity** $\eta(T)$ sets the Stokes–Einstein radius R_{h} (Eq. 6). The program's **solvent library** offers both for a chosen solvent, temperature and wavelength, as a convenience default rather than a constant built into any analysis (**User-Manual** §6.3 names the solvents and shows the proposed values). §1.1.1 gives the two functional forms and the rule by which a proposed value reaches an analysis, §1.1.2 the tolerance shown beside each value, and §1.1.3 the solvents whose tabulated values a bench sample can leave.

1.1.1 The two forms

One functional form per property serves every solvent. The refractive index is a three-term Cauchy dispersion [1] with a linear temperature correction to a reference temperature T_{ref} (λ in µm) [2]:

$$n(\lambda,T)\ =\ A_{\mathrm{n}}\ +\ \frac{B_{\mathrm{n}}}{\lambda^2}\ +\ \frac{C_{\mathrm{n}}}{\lambda^4}\ +\ \frac{\mathrm{d}n}{\mathrm{d}T}\,(T-T_{\mathrm{ref}}) \tag{1}$$

where A_{n}, B_{n} and C_{n} are the solvent's Cauchy coefficients and $\mathrm{d}n/\mathrm{d}T$ its temperature slope (K^{-1}). The viscosity is the Vogel–Fulcher–Tammann (VFT) form (η in Pa·s) [3]:

$$\eta(T)\ =\ A_{\eta}\exp\!\left(\frac{B_{\eta}}{T-T_0}\right) \tag{2}$$

where A_{η} is a prefactor in Pa·s, B_{η} a temperature in K, and T_0 the temperature (K) at which the form diverges. The coefficients are fitted to each property's authoritative correlation over its validity range, and a fitted form says nothing about conditions outside that range. Each solvent carries a confidence tier — *primary* (dispersion and reference-correlation data) or *estimate* (weaker, e.g. bulk-compilation data). The full per-solvent source list is in the Citation-Index that ships beside this guide.

In the program: the solvent library *proposes* n and η for the committed solvent, temperature and wavelength, tagged as proposed and shown with the solvent's confidence tier as a badge; a proposed value reaches an analysis only when you commit it with the Data tab's **Update**, and a condition outside a record's stored wavelength or temperature range is declined rather than extrapolated.

1.1.2 The tolerance: shown, not propagated

Each property carries a tolerance — an absolute n uncertainty, a relative η uncertainty — evaluated at the requested wavelength and temperature rather than as one flat figure. A proposed n or η is a calibration-class systematic, and the reported ± is statistical/regression uncertainty, which excludes that class (§6, the statistical-only rule), so the tolerance informs your judgement rather than a fitted standard error. The refractive-index tolerance combines five terms in quadrature:

$$\sigma_{\mathrm{n}}(\lambda,T)\ =\ \sqrt{u_{\mathrm{src}}^2\ +\ r_{\lambda}(\lambda)^2\ +\ r_{\mathrm{T}}(T)^2\ +\ \left(u_{\mathrm{slope}}\,(T-T_{\mathrm{ref}})\right)^2\ +\ u_{\mathrm{lab}}^2} \tag{3}$$

where u_{src} is the source's stated accuracy; $r_{\lambda}(\lambda)$ and $r_{\mathrm{T}}(T)$ are the unified form's fit residual, as upper envelopes over wavelength and over temperature; u_{slope} is the uncertainty of the $\mathrm{d}n/\mathrm{d}T$ slope; and u_{lab} is an inter-laboratory allowance, defined below. The viscosity tolerance is built the

same way, and its $u_{\mathrm{lab},\eta}$ term applies only where the source is not a critically evaluated reference correlation (an evaluated correlation has already done the inter-laboratory analysis):

$$\sigma_{\eta,\mathrm{rel}}(T) \;=\; \sqrt{u_{\mathrm{src,rel}}^2 \;+\; r_{\mathrm{T},\eta}(T)^2 \;+\; u_{\mathrm{lab},\eta}^2} \tag{4}$$

with $u_{\mathrm{src,rel}}$ the source's stated relative accuracy of η and $r_{\mathrm{T},\eta}(T)$ the viscosity correlation's fit residual at T — the relative counterparts of the absolute terms above. The u_{lab} terms are fixed allowances for how much independent laboratories' published measurements of the same property typically disagree — the systematic a single source's stated accuracy cannot contain. They are measured, not chosen: $u_{\mathrm{lab}} = 1.2 \times 10^{-3}$ is the median, over wavelength nodes, of the full spread between the published $n(\lambda,\ T)$ curves covering that node, and $u_{\mathrm{lab},\eta} = 3.2\,\%$ is half the median 10th-to-90th-percentile spread of published viscosity measurements (≈ 1.28 standard deviations, adopted as-is). The band is flat in λ apart from the fit-residual envelope, so at the blue edge of a record's wavelength range — where dispersion sources are typically weakest — it can still be somewhat optimistic.

The one notable case is water, whose $\mathrm{d}n/\mathrm{d}T$ is genuinely temperature-dependent (the near-20 °C slope steepens by roughly two-thirds by 45 °C). Its stored coefficients are fitted directly to the IAPWS refractive-index formulation [4] evaluated densely across 400–800 nm and 15–60 °C, with the 0.1 MPa liquid density from the IAPWS-95 equation of state [5], so the drift is carried per condition by the residual envelope r_{T} rather than by an inflated slope allowance. One disclosure: [4] states its accuracy as 1.5×10^{-5} in the visible but only 1×10^{-3} between 700 and 800 nm, so the red edge of the water band rests on weaker truth there — the inter-laboratory allowance (1.2×10^{-3}) covers it.

In the program: each proposed n and η is shown with its tolerance (Eq. 3, Eq. 4); the tolerance is display-only and is never folded into a reported ±.

1.1.3 Hygroscopic solvents

Some error sources sit outside what the program can measure. The clearest is glycerol, which is strongly hygroscopic: its viscosity rises steeply as adsorbed water is excluded, so the tabulated η is for dry glycerol and a real, water-equilibrated sample can differ by tens of percent [3]; its refractive index shifts with water uptake as well [2]. DMSO and DMF are hygroscopic too, and their viscosity sources measured dried, water-quantified samples — a bench-equilibrated sample can sit outside the tabulated band, in a direction and by an amount the band cannot know.

1.2 Scattering Vector

$$q = \frac{4\pi n}{\lambda}\,\sin\!\left(\frac{\theta}{2}\right) \tag{5}$$

where λ is the vacuum wavelength of the laser, n is the solvent refractive index at that wavelength and temperature, and θ is the scattering angle. This is the standard definition for solution scattering [6].

In the program: n and λ are always explicit inputs, never assumed: n is the committed solvent value of §1.1 at the laser's wavelength and the measurement's temperature.

1.3 Stokes–Einstein Relation

The hydrodynamic radius follows from the translational diffusion coefficient $D = \Gamma/q^2$ (Γ the decay rate of the field autocorrelation g_1, §2.2.1) via the **Stokes–Einstein relation** [7]:

$$R_\mathrm{h} = \frac{k_\mathrm{B} T}{6\pi\eta D} \tag{6}$$

where k_B is the Boltzmann constant, T the absolute temperature, and η the solvent dynamic viscosity (§1.1), so the radius is a property of the solution at that temperature. A single-angle, single-concentration measurement gives the **apparent** R_h; the extrapolations to $q \to 0$ and $c \to 0$ are §3.5's.

In the program: T and η are required inputs, never assumed — a measurement without a committed viscosity reports its decay rate and diffusion coefficient but no radius. The $\pm$ on an R_h that follows from a fitted D is the reciprocal's delta-method pass-through (§6.2).

1.4 Refractive-Index Correction

The excess Rayleigh ratio of a sample in a solvent other than the calibration standard carries the factor

$$f_\mathrm{n} = \left(\frac{n_\mathrm{solvent}}{n_\mathrm{standard}} \right)^2 \tag{7}$$

which multiplies the solvent-subtracted intensity in the excess Rayleigh ratio ΔR_θ of §1.7, where it is the last factor of Eq. 12. When solvent = standard, $f_\mathrm{n} = 1$ exactly. The fixed squared form follows from refraction of the observed scattering volume and solid angle at the cell wall, and holds for rectangular and cylindrical cells alike [8].

In the program: the correction is applied whenever the solvent differs from the calibration liquid. If n_standard is not supplied it cannot be formed, and the measurement is treated as uncalibrated (§1.7.1) rather than computed with $f_\mathrm{n} = 1$.

1.5 Optical Constant and the Blank $\mathrm{d}n/\mathrm{d}c$

The prefactor in the Debye, Zimm and Berry equations of §4.3–§4.4 (vertically polarized geometry):

$$K = \frac{4\pi^2 n^2 \left(\mathrm{d}n/\mathrm{d}c\right)^2}{N_\mathrm{A}\, \lambda^4} \tag{8}$$

with N_A the Avogadro number. Units are mol·mL·g^{-2}·cm^{-1}, consistent with c in g/mL and R_θ in cm^{-1}.

> **Important.** $K \propto (\mathrm{d}n/\mathrm{d}c)^2$, so the refractive-index increment enters the molecular weight squared. For a low-contrast system a small $\mathrm{d}n/\mathrm{d}c$ makes the scattering weak and amplifies its error, which is why the program leaves the value to you (§1.5.1).

1.5.1 When no dn/dc exists

Some samples have none to give: no increment has been published for the solute, or — for a particle dispersion — the quantity is not defined at all. A **blank** dn/dc need not stop the analysis: K multiplies the *whole* ordinate $Kc/\Delta R$ (the Debye ordinate of §4.3), so a blank increment is the same class of absence as an uncalibrated measurement — an unknown positive factor on the ordinate — with the same survivors, which §1.7.1 argues once and lists (the Guinier R_g of §4.6 is the strongest case, since dn/dc does not enter ΔR at all). The two absences are independent: a fully calibrated measurement with no dn/dc still yields a molar mass M_w (§4.3) wrong by orders of magnitude, so calibrating does not repair a blank increment.

A dn/dc of zero is a different thing from a blank. It is not an "unknown" — it asserts that $K = 0$, which collapses the entire ordinate to zero and makes every fit meaningless. A negative increment is physical: K depends on the square, so the sign cancels, and a solute less refractive than its solvent genuinely has one.

One further case behaves like the blank one. Because K goes as the square of dn/dc (and as n^2 and λ^{-4}), a value that is legal, finite and non-zero can still leave K unusable — too small to represent, or too large — so whether a value is usable is a property of K itself rather than of the increment.

In the program: dn/dc is hand-entered, never proposed or defaulted. A blank increment sets $K = 1$, an arbitrary optical scale recorded on the result, disclosed beside every value that depends on it and flagged independently of the calibration; a zero is refused; a negative value is accepted; an increment that leaves K outside the usable range, checked on K itself, takes the $K = 1$ path with a note saying so.

1.6 Polarization Geometry and the Toluene Rayleigh Ratio

The calibrant's Rayleigh ratio must be taken in the geometry the instrument measures, so the convention comes first: §1.6.1 fixes the polarization subscripts and the depolarization ratio, §1.6.2 the toluene ratio and its conversion to that geometry.

1.6.1 The polarization convention

Two-letter subscripts are **(incident, analyzer)**: I_{VV} = vertical incident + vertical analyzer (the ordinary polarized intensity); I_{VH} = vertical incident + horizontal analyzer (the depolarized intensity). The **depolarization ratio** and its natural-light equivalent are

$$\rho_v = \frac{I_{VH}}{I_{VV}}, \qquad \rho_u = \frac{2\rho_v}{1+\rho_v}, \tag{9}$$

where ρ_u is the ratio for unpolarized incident light used by the classical literature (the conversion is exact). Physically $0 \le \rho_v \le 3/4$: $\rho_v = 0$ is an isotropic scatterer, $\rho_v = 3/4$ the fully anisotropic limit. A value outside this range signals stray depolarized light (polarizer/analyzer leakage, reflections, or an over-subtracted dark count). An instrument with a vertical laser and no analyzer is **VU** geometry: it collects $I_{VV} + I_{VH}$, which is why the calibrant's Rayleigh ratio of §1.6.2 must be taken in the same geometry. Figure 1 shows the convention from above, and also the length of beam the detector views, which sets the sin θ volume correction of §1.7.

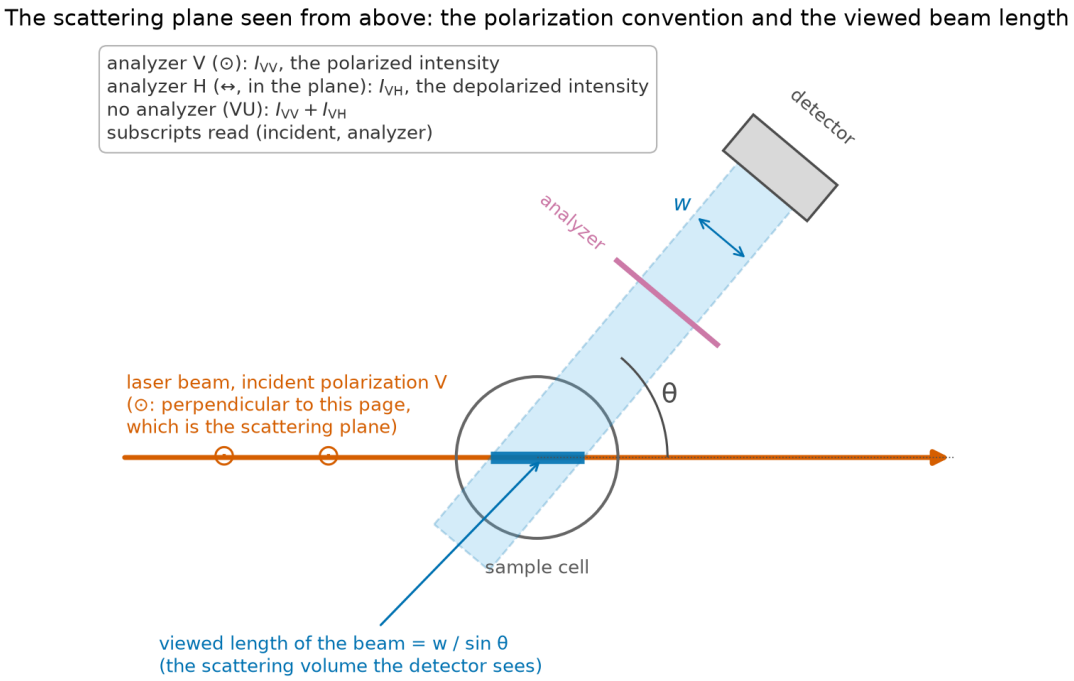


Figure 1: The scattering plane seen from above: the laser beam crosses the cell with its polarization V perpendicular to the plane; the detector at angle θ views a strip of width w, whose intersection with the beam — the scattering volume — has length $w/\sin\theta$; and the analyzer in front of the detector passes V (I_{VV}), H (I_{VH}) or, absent, both (VU, their sum). The subscripts read (incident, analyzer).

In the program: a ρ_v outside $0 \le \rho_v \le 3/4$ is flagged and not used.

1.6.2 The toluene Rayleigh ratio

All SLS calibration rests on the toluene Rayleigh ratio, corrected for temperature and for geometry (§1.7 says why a value stored by instrument software is no substitute). The base quantity is the polarized ratio R_{VV}, temperature-corrected with Sivokhin and Kazantsev's fitted coefficient [9], Eq. 9, p. 9501 and then converted to the measurement geometry:

$$R_{VV}(T) = R_{VV}(25^\circ\mathrm{C})\,[\,1 + a_T\,(T - 25)\,], \qquad a_T = +0.00421\ {}^\circ\mathrm{C}^{-1} \tag{10}$$

$$R_{VU} = R_{VV}\,(1 + \rho_v), \qquad R_{VH} = R_{VV}\,\rho_v \tag{11}$$

where T is in °C and a_T is positive: the Rayleigh ratio *rises* with temperature. ρ_v is toluene's own depolarization ratio (Eq. 9): 0.346 at 25 °C [9], Table 1, p. 9500, falling by 0.00135 per °C, the slope between that table's 10 °C and 50 °C values.

Wavelength	R_{VV} at 25 °C (cm^{-1})	Source
532 nm	2.34×10^{-5}	[10]
660 nm	8.456×10^{-6}	[9], Table 2, p. 9501

At 532 nm and 25 °C, for example, an instrument without an analyzer calibrates against $R_{VU} = 2.34 \times 10^{-5} \times (1 + 0.346) \approx 3.15 \times 10^{-5}$ cm^{-1}.

> **Important.** The geometry (VV / VU / VH) must match how the calibrant was measured (§1.6.1 says which geometry an instrument without an analyzer is), or the absolute scale is wrong by $\approx (1 + \rho_v) \approx 1.35$.

Other wavelengths need a separately sourced R_{VV}, not an interpolation: the wavelength dependence is steep — toluene's Rayleigh ratio scales as $\approx \lambda^{-4.17}$, steeper than the λ^{-4} Rayleigh law, reducing to $\approx \lambda^{-4}$ only once the $n(\lambda)$ dispersion is separated out [11] — so on that law a 2 nm offset from a tabulated wavelength already moves the ratio by ≈ 1.6 % at 532 nm.

In the program: the calibrant's Rayleigh ratio is always computed from this table at the standard's temperature and in the geometry you set (VU, VV or VH), with toluene's $\rho_v(T) = 0.346 - 0.00135\ (T - 25)$ at that temperature and no control to change it. A laser wavelength within 2 nm of a tabulated one is served that entry unscaled, with a note beside the calibration constant k_c (§1.7) giving the approximate size of the substitution on the $\lambda^{-4.17}$ law; the 2 nm tolerance is a convenience, not a physical criterion. A Rayleigh ratio stored in an instrument file follows §1.7's rule for stored constants.

1.7 Calibration to an Excess Rayleigh Ratio

Raw intensities are placed on an absolute scale in steps: subtract the detector dark count, subtract the pure-solvent scattering (to isolate the excess from the solute), correct for the scattering volume via $\sin\theta$ (the viewed length in Figure 1) [12], [13, p. 5], apply the refractive-index correction Eq. 7, [8], and scale by a calibration constant k_c:

$$\Delta R_\theta = k_c \sin\theta \left(I_{sample} - I_{solvent}\right) \left(\frac{n_{solvent}}{n_{standard}}\right)^2 \tag{12}$$

where I_{sample} and $I_{solvent}$ are the solution's and the pure solvent's intensities at θ, each with the dark count already subtracted (the first step above), in the units of the loaded intensity column. The calibration constant follows from a single calibrant measurement — the intensity of a standard liquid (usually toluene), the angle it was measured at, and that standard's Rayleigh ratio (§1.6, in the matching geometry) [6]:

$$k_c = \frac{R_{std}}{I_{cal,net} \sin\theta_{cal}} \tag{13}$$

where R_{std} is the standard's Rayleigh ratio (cm^{-1}), $I_{cal,net}$ the calibrant's intensity net of the dark count, and θ_{cal} the angle it was measured at, usually 90°. A constant stored with an instrument's files is no substitute: it goes stale after a re-alignment, and vendor Rayleigh ratios are often outdated.

In the program: k_c is computed from the calibrant values you enter and is shown and editable; a k_c or a Rayleigh ratio read from a data file is kept for reference and never used.

1.7.1 The uncalibrated fallback: what survives

Without a k_c — none can be computed or supplied, or the standard's refractive index is blank so that §1.4's correction cannot be formed — the intensities sit on an **arbitrary intensity scale**, and a measurement on it is uncalibrated. What that costs depends on how a quantity is formed from the ordinate. An unknown calibration scale multiplies every point of a series by the same positive factor, so it cancels in any quantity extracted as a **ratio of fitted coefficients** — a slope over an intercept, an amplitude over an amplitude — and survives in any quantity read from an absolute level. Hence these survive an uncalibrated measurement:

- every R_g (Guinier, Debye, Zimm, Berry, form-factor) and every form-factor radius;
- the calibration-free product $2{\cdot}A_2{\cdot}M_w$ (Eq. 82);
- every DLS quantity, which never sees the intensity scale;
- a per-mode apparent R_g (Eq. 75).

The absolute M_w, the absolute A_2, the single-angle apparent M_w and the Zimm two-route weights (§6.3) are read from an absolute level and carry the arbitrary scale.

Two qualifiers apply: the branching factors g and h survive only when the M_w scale is consistent across the samples compared (§5.3), and a blank that is unsubtracted or mis-shaped does not cancel (§1.7.2 says why). A blank $\mathrm{d}n/\mathrm{d}c$ scales the whole ordinate the same way, with the same survivors (§1.5.1). What a ± means on each side of that split is §6's statistical-only rule.

In the program: an uncalibrated measurement still runs, every result of it is marked uncalibrated, and it reports every value but the Zimm two-route weights (§6.3's rule). The apparent and absolute M_w and the absolute A_2 are flagged unreliable, with their arbitrary scale named in the export's Comments cells; R_g, the form-factor radii and $2A_2M_w$ carry no flag from it.

1.7.2 When no solvent blank was measured

I_{solvent} in Eq. 12 normally comes from a measured $c = 0$ series, subtracted angle by angle — the classical procedure, and the one Seery, Shorter and Amis use [12], Eq. 2, p. 1200. Where a series arrives without one, a stand-in can be computed from a single hand-entered number: the solvent's intensity at 90°, I_{90}, in the units of the loaded intensity column. With d the committed dark count rate,

$$I_{\text{solvent}}(\theta) = d + \frac{I_{90} - d}{\sin\theta} \tag{14}$$

so that after the pipeline's dark subtraction the net solvent term is $(I_{90} - d)/\sin\theta$. The dark count is additive and is not scaled by the scattering volume, which is why it sits outside the $1/\sin\theta$.

The $1/\sin\theta$ is the same scattering-volume factor as in Eq. 12 — the detection optics view a length of the beam that goes as $1/\sin\theta$ (the viewed length in Figure 1) [12], [13, p. 5] — combined with the scattering of a simple liquid being independent of angle once that volume is normalized out [14, p. 34] and p. 580, [13, p. 27], [15, pp. 061503–5]. Note that the entered value is therefore not subtracted uniformly: it is the 90° value, and the blank at 30° is twice as large.

Important. This is a flagged convenience simplification, not a published method: no

source consulted here prints or endorses a blank computed from one 90° value, which is why the program flags what is computed on it (the marked line below).

The computed blank is a two-step inference from the two facts cited above. It assumes vertically polarized incident light, detection in the horizontal plane, a cylindrical beam viewed through slit-type optics, and a solvent that scatters like a simple liquid. It is weakest at low angles, where flare and dust live and where a computed blank cannot cancel them, and for a solvent that is not a simple liquid — supercooled glass-forming organic liquids, for instance, show a strongly q-dependent isotropic excess of Ornstein–Zernike form [15], Eq. 7, p. 061503-5.

R_g loses the immunity §1.7.1 gives it because, unlike an unknown calibration scale, an unsubtracted or mis-shaped blank is *additive*, so it does not cancel. That is true whether the leftover varies with angle (a mis-shaped computed blank) or is constant (no subtraction at all, where the volume correction cancels the solvent's own $1/\sin\theta$). The same holds for the calibration-free product $2A_2M_w$ (§4.5).

A computed blank enters as a fixed offset before the fit, so it shifts the points without adding to their scatter. It is a *systematic*, like the calibration and $\mathrm{d}n/\mathrm{d}c$, which a regression ± cannot see, and a flag discloses it because no honest number for it exists; the route to a real ± is a measured blank.

In the program: a blank computed from I_{90} is used only where no measured $c = 0$ series exists, and every result computed on it is flagged unreliable, R_g included, with no term added to its ± (§6's statistical-only rule); leaving I_{90} empty or zero means no solvent subtraction, with the dark count still removed.

1.8 What This Chapter Hands On

Chapter 3 takes q (Eq. 5) and the Stokes–Einstein relation (Eq. 6) — with n, λ, θ, T and η as committed inputs — and nothing else from here: DLS never sees the intensity scale. Chapter 4 takes the excess Rayleigh ratio (Eq. 12) with its calibration constant, the refractive-index correction (Eq. 7), the optical constant (Eq. 8) and the polarization convention of §1.6, and §1.7.1 is the passage every later section means by "survives an uncalibrated measurement". Chapter 5 inherits the same split through the R_g, R_h and M_w it combines.

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2. The Measurement and Its Quality

A light-scattering instrument records two things: a **count-rate trace**, the scattered intensity sampled in time, and a **correlogram**, the autocorrelation of that intensity over a bank of delay times. This chapter is about the measurement itself, before any model is fitted to it. §2.1 gives the diagnostics that judge a trace — whether single samples stand out, whether the whole trace fluctuates more than photon statistics permit, how long it may be averaged, and whether anything in it is periodic. §2.2 defines the correlogram's two data forms and the Siegert relation that links them, and describes the multi-tau lag axis a real correlator produces, because the sample time of each channel enters the fits of chapter 3. §2.3 states the noise a photon-counting correlator puts on each channel of a correlogram, which is what those fits weight their residuals by. The chapter closes with what chapter 3 reads from here (§2.4).

2.1 The Count-Rate Trace and Its Diagnostics

The trace diagnostics answer two different questions, and keeping them apart matters when you read the plot. The outlier band asks whether any single *sample* stands out from the rest of the trace; the Fano factor asks whether the *whole trace* fluctuates more than photon statistics permit.

2.1.1 The outlier band

A sample is flagged when it lies outside a band centered on the trace's median and scaled by the trace's own scatter [16]:

$$\text{flag if } |I - \tilde{I}| > k_\sigma\, \hat{\sigma}_{\mathrm{R}}, \qquad \hat{\sigma}_{\mathrm{R}} = 1.4826\,\mathrm{MAD}, \qquad \mathrm{MAD} = \mathrm{median}_i |I_i - \tilde{I}|, \qquad k_\sigma = 3.5 \tag{15}$$

Here $\tilde{I}$ is the median count rate and k_σ the band's half-width in units of the robust scale $\hat{\sigma}_{\mathrm{R}}$. Both halves are chosen for resistance:

- The median center has a breakdown point of ≈50 %, so it is not dragged by the outliers being looked for — an arithmetic mean is, and a handful of dust spikes can shift it far enough that the *clean* samples fall outside the band.
- The factor 1.4826 makes $\hat{\sigma}_{\mathrm{R}}$ consistent with σ for Gaussian data [17].

The same identifier recurs in the literature under other names, all consistent with the default above. Iglewicz and Hoaglin write it as a modified z-score, $M_i = 0.6745(I_i - \tilde{I})/\mathrm{MAD}$ flagged at $|M_i| > D$, and recommend $D = 3.5$ from a simulation study; since $1/0.6745 = 1.4826$, k_σ here *is* their D. Davies and Gather give the identifier its general treatment and prove its breakdown optimality [18], on Hampel's influence-function framework [19], and report an asymptotic form, $|I - \tilde{I}| > 5.2\,\mathrm{MAD}$ [18, p. 790], which they attribute to a later paper of Hampel's (1985) and which agrees with the default, since $3.5 \times 1.4826 = 5.19$.

If more than half the samples share one value — a flat, stalled or heavily quantized trace — the MAD is exactly zero and the band has no width. Pearson and co-workers call this an *implosion window* and characterize it for the windowed form of the same filter [20]; their treatment is to identify such windows and set them aside, and the rest of that paper's theory is stated for *implosion-free* sequences.

Setting such a trace aside has a cost: a single dust spike on a perfectly flat trace goes unjudged too. The limit is inherent rather than a weakness of the MAD in particular. With over half the samples tied there is no scale to judge against, and the alternatives proposed precisely as improvements on the MAD, Rousseeuw and Croux's S_n and Q_n [17], carry the same 50 % implosion breakdown point and collapse on the same input.

On a shot-noise-limited trace the MAD scale is *not* the naive $\sqrt{\bar{r}}$ Poisson width of the mean count rate $\bar{r}$, and the intuition that it should be is wrong. The MAD measures the trace's actual scatter, so at sub-second sampling the robust scale comes out *wider* than a $\bar{r} \pm k_\sigma\sqrt{\bar{r}}$ band — about five-fold at 40 ms bins — because $\sqrt{\bar{r}}$ is the Poisson width only for raw counts binned at one second; with Δt the bin width, the true rate width is $\sqrt{\bar{r}/\Delta t}$. A band built on $\sqrt{\bar{r}}$ is therefore too narrow for sub-second count rates and flags nearly every sample.

One further limit is worth knowing. The MAD of a Poisson sample can only land on the integer count lattice, so below roughly fifty counts per bin it becomes granular and estimates σ to within

about a factor of two rather than a few percent — a very dim sample measured with short bins gets a coarse band.

The flagged fraction is a compact quality metric. The band is a *global* one: it flags departures from the trace's baseline of any duration without localizing them, so a slow drift trips it much as a single dust spike does. Localizing them is what the windowed companion of §2.1.2 adds.

In the program: the band multiplier k_σ is the **k** control (default 3.5), shared with the windowed layer; a flat or heavily quantized trace on which the MAD is zero is reported as not judged rather than flagged everywhere; the flagged fraction is shown in the summary line and exported.

2.1.2 The windowed Hampel decomposition

The same identifier run against a *moving* baseline [20]. Over a window of $w = 2H + 1$ samples centered on each point k (H the half-width, Pearson's K), take the median m_k and the robust scale S_k, and flag the center sample when it leaves the band:

$$m_k = \text{median}\{I_{k-H}, \ldots, I_{k+H}\}, \qquad S_k = 1.4826 \operatorname*{median}_{j\in[-H,H]} |I_{k-j} - m_k|, \qquad \text{flag if } |I_k - m_k| > k_\sigma S_k \tag{16}$$

where k_σ is the same band multiplier as Eq. 15. Pearson's filter *replaces* a flagged sample with m_k, which suits a filter; a diagnostic is better served by the flags and the curves m_k and S_k themselves.

The window sets what the layer can see. Pearson defines an impulse as a run of at most H samples, so an event shorter than about half the window can be flagged and anything longer is tracked as baseline — a persistent offset appears as a step in m_k rather than as flags. That blind spot is structural, not a defect, and showing m_k is what makes it visible.

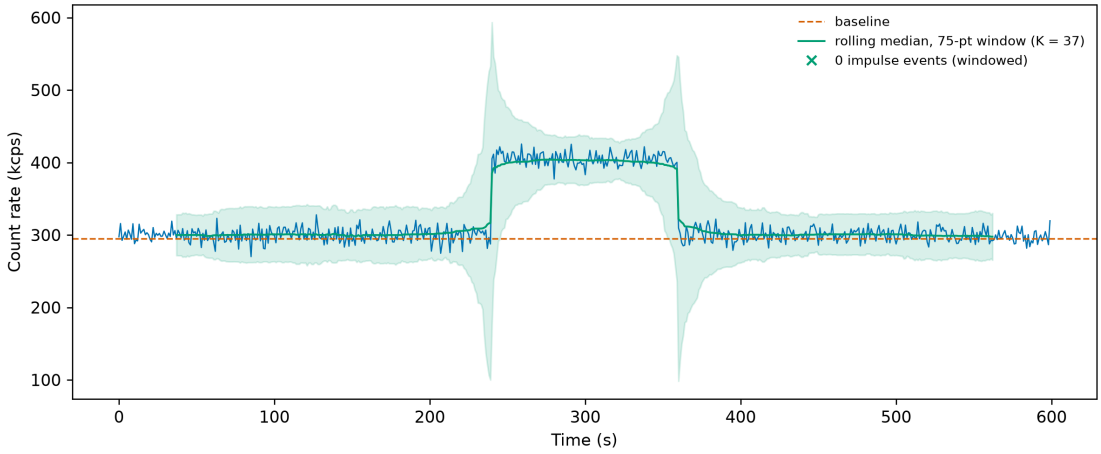


Figure 2: The windowed Hampel layer on a trace carrying a persistent offset (the **Traces** sub-tab, **Windowed Hampel** on): the rolling median m_k steps up with the offset instead of flagging it, the shaded S_k band widens where the window straddles a step, and the first and last H samples carry no band because no full window fits there. The legend's K is the half-width H, in Pearson's symbol.

No verdict is possible in two cases. Where no full window fits — the first and last H samples — there is nothing to judge against, and a window shrunk to fit there would be exactly the biased, implosion-prone regime a minimum width exists to exclude. And where more than H of a window's samples share one value the MAD is zero, Pearson's *implosion window*: no scale exists, exactly as for the global band above.

Two notes on the width. No one default suits every trace, since traces differ in length by orders of magnitude. And a minimum width can stand in for a finite-sample MAD correction: the uncorrected bias — $b_n = n/(n-0.8)$ for $n > 9$ [21], Eq. 4, p. 414, about 7.8 % at 11 points — stays below one step of the k_σ control, which moves in steps of 0.5, and at the default 3.5 it narrows the band as a k_σ of about 3.25 would. A floor keeps the regime needing a correction out of reach rather than correcting for it.

In the program: the band multiplier is §2.1.1's k control; no flagged sample is ever replaced; the curves m_k and S_k are drawn and exported with the flags; the window width is the **Hampel window (pts)** control, an odd number of points no smaller than 11 and with no finite-sample MAD correction (resolved from the trace length on *auto*, the value used shown and exported); and the stretches the layer did not judge — the edge half-windows, which are not shrunk to fit, and any implosion window, where nothing is flagged and no scale is substituted — are a gap in the band, with their fractions in the summary line.

2.1.3 The Fano factor

The count-rate histogram is fitted with a Gaussian and/or Poisson distribution; the key diagnostic is the Fano factor (variance/mean in count space):

$$F = \frac{\sigma^2}{\mu} \quad \begin{cases} \approx 1 & \text{shot-noise limited (ideal Poisson)} \\ > 1 & \text{excess fluctuations (slow modes, dust, drift)} \end{cases} \tag{17}$$

A trace can be far from shot-noise-limited and still flag no outliers at all, because the Fano factor totals the excess of the *whole* trace where the band judges single samples.

In the program: F is computed in count space from the committed sampling interval and shown in the summary line beside the mean and the coefficient of variation.

2.1.4 Blocking analysis

Cutting the trace into contiguous blocks of b samples and taking each block's mean gives a sequence $\bar{y}_k$ from which the standard error of the overall mean follows, with N the number of samples in the trace — Flyvbjerg and Petersen's blocking estimator [22], Eq. (26), p. 463:

$$\mathrm{SE}(b) = \frac{\mathrm{std}(\bar{y}_k)}{\sqrt{m}}, \qquad m = \lfloor N/b \rfloor \tag{18}$$

where std is the sample standard deviation of the m block means, normalized by $m-1$. At every block size this is a lower bound on the true standard error. For uncorrelated data it is flat in b. A curve that rises and plateaus is the signature of positive correlation — slow modes — and the plateau, not the $b = 1$ value, is the honest standard error: the bound grows with blocking until it reaches a fixed point and then stays constant within fluctuations, and that constant value is the estimate [22, p. 463]. Their blocking transformation halves the data at each step, so its block

sizes double; the program evaluates the same estimator at geometrically spaced block sizes instead, which samples the same curve more densely — a flagged convenience simplification of the doubling sequence.

In the program: the blocking curve is one of the views of the **Diagnostic** panel on **Utilities ▶ Traces** (**User-Manual** §11.1.5), and it is the correlation test that §2.1.8's periodicity rule reads.

2.1.5 The Allan deviation

The same partition, read for *stability* rather than precision. Where the blocking curve asks how precise the mean is at a given block size, the Allan variance asks how reproducible consecutive blocks are, which is what says how long a trace may usefully be averaged. It is the mean squared difference of adjacent block means, halved [23], Eq. (11), p. 109:

$$\sigma_{\mathrm{y}}^2(T_{\mathrm{a}}) = \left\langle \frac{(\bar{y}_{k+1} - \bar{y}_k)^2}{2} \right\rangle, \qquad \hat{\sigma}_{\mathrm{y}}^2(T_{\mathrm{a}}) = \frac{1}{2m} \sum_{k=1}^{m} (\bar{y}_{k+1} - \bar{y}_k)^2 \tag{19}$$

with $T_{\mathrm{a}} = b\,\Delta t$ the averaging time (Allan's and Barnes's τ) and m the number of differences averaged. The measure originates with Allan [24]; the compact difference form above, and the name, are Barnes and co-workers' [23]. The definition holds for contiguous gates — no dead time between measurements, Barnes's repetition time T equal to his τ — which a uniformly sampled trace satisfies exactly. In *fractional* form — the block means divided by the trace mean — the published slope mappings apply as printed [23], Eqs. (28)–(29), p. 111: white noise falls as $T_{\mathrm{a}}^{-1/2}$, a random walk rises as $T_{\mathrm{a}}^{+1/2}$, and a flicker floor is flat. The flat region is where averaging has stopped paying. One caution: a discrete spectral line — mains hum, a stirrer — makes $\sigma_{\mathrm{y}}^2(T_{\mathrm{a}})$ vary rapidly with T_{a}, and at an averaging time that happens to be a whole number of modulation periods the value can look far better than the trace deserves [23, p. 115]. Read this view beside the spectrum of §2.1.6, not alone.

In the program: σ_{y} is reported in fractional form.

2.1.6 The power spectral density

The diagnostics above describe how a trace's *amplitudes* are distributed; none of them can see whether anything in it is *periodic*. The program estimates the spectrum by Welch's method [25]: cut the record into overlapping segments of L_{s} points (his L), window and transform each, and average the resulting periodograms.

$$I_k(f) = \frac{L_{\mathrm{s}}}{U} \left| A_k(f) \right|^2, \qquad U = \frac{1}{L_{\mathrm{s}}} \sum_{j=0}^{L_{\mathrm{s}}-1} W^2(j), \qquad \hat{P}(f) = \frac{1}{N_{\mathrm{s}}} \sum_{k=1}^{N_{\mathrm{s}}} I_k(f),$$

$$S(f) = 2\,\Delta t\,\hat{P}(f) \quad [\mathrm{cps}^2/\mathrm{Hz}] \tag{20}$$

where $A_k(f)$ is the Fourier transform of the windowed segment k, $W(j)$ the window's j-th coefficient, U the window's mean square and N_{s} the number of segments averaged (Welch's K). Mind the second line, which is not Welch's. He works throughout in units where the sampling interval is one and reports $\hat{P}$ alone; converting to a one-sided spectral density in $\mathrm{cps}^2/\mathrm{Hz}$ costs a factor $2\,\Delta t$ — the 2 for folding the negative frequencies onto the positive ones, the Δt for going from power per bin to power per hertz. Without it the numbers would be a factor $2\,\Delta t$ from the axis label, and

— the reason it matters here rather than being pedantry — they could not be compared with the shot-noise floor of §2.1.7, which is the comparison this whole view exists to support.

Averaging is not a refinement here but the point: a single periodogram has 100 % relative scatter at every frequency however long the record, so an unaveraged spectrum cannot be read. How much scatter survives is set by the equivalent degrees of freedom (E.D.F.) of the approximating chi-square distribution, which depends on how strongly the overlapping segments are correlated [25, pp. 71–72]:

$$C_{\mathrm{ov}}(j) = \frac{\left[\sum_k W(k)W(k+j\Delta_{\mathrm{s}})\right]^2}{\left[\sum_k W^2(k)\right]^2}, \qquad \text{E.D.F.} = \frac{2N_{\mathrm{s}}}{1 + 2\sum_{j=1}^{N_{\mathrm{s}}-1} \frac{N_{\mathrm{s}}-j}{N_{\mathrm{s}}} C_{\mathrm{ov}}(j)} \tag{21}$$

where $C_{\mathrm{ov}}(j)$ (Welch's $\rho(j)$) is the correlation between segments j starts apart and Δ_{s} the offset between segment starts (his D). With no overlap C_{ov} vanishes and this reduces to Welch's E.D.F. $= 2N_{\mathrm{s}}$. The window enters only through C_{ov}, so the same expression serves a modern Hann window as well as Welch's own two; the E.D.F. halves at zero frequency and at Nyquist. Welch treats the segment length purely as a resolution-versus-variance trade and prescribes no value; for the overlap he recommends half a segment, for a record of fixed length.

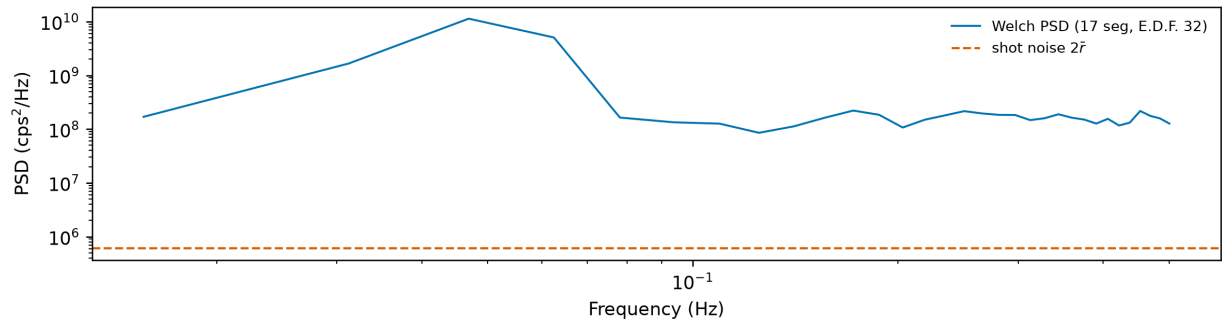


Figure 3: The Diagnostic panel's Welch spectrum (Utilities ▶ Traces) of a synthetic trace carrying a 0.05 Hz modulation, the peak at half the frequency of the axis's one labelled tick: the line stands two decades above the broadband level, and the shot-noise floor $2\bar{r}$ of §2.1.7 lies far below both — the comparison the one-sided cps^2/Hz scale exists for. The legend gives the segment count and the E.D.F.; the chi-square band of Eq. 22 is not part of the plot.

That number is what an error bar on the spectrum is built from. Welch names the approximating chi-square distribution and gives its degrees of freedom but prints no interval, so the band below is the standard one constructed from that distribution, not his:

$$\frac{\nu_{\mathrm{e}}\,\hat{P}(f)}{\chi^2_{\nu_{\mathrm{e}},\,1-\ell/2}} \;\le\; P(f) \;\le\; \frac{\nu_{\mathrm{e}}\,\hat{P}(f)}{\chi^2_{\nu_{\mathrm{e}},\,\ell/2}}, \qquad \nu_{\mathrm{e}} = \text{E.D.F.} \tag{22}$$

with $1-\ell$ the two-sided confidence level ($\ell = 0.05$ gives a 95 % interval); as a pair of multipliers it can be applied to whichever points a reader cares about. Being a chi-square interval it is asymmetric — the upper reach is always the longer one — and it is a statement about the *estimator's* scatter alone: it says nothing about whether the detector was linear or the trace stationary.

In the program: the spectrum is plotted and exported as a one-sided density in $\mathrm{cps}^2/\mathrm{Hz}$; the segment length and window are yours, the overlap is fixed at half a segment with no control to change it; the E.D.F. (Eq. 21) is computed from the window in use and shown in the plot's legend; and the E.D.F. and the two chi-square band multipliers of Eq. 22, for the 95 % band, are exported, while the band itself is neither drawn nor exported.

2.1.7 The shot-noise floor

A spectrum needs a reference, and for photon counting there is a natural one. Counts in a bin of width Δt have variance equal to their mean $\bar{c} = \bar{r}\,\Delta t$ (the mean count per bin), so the count *rate* $r = c/\Delta t$ has variance $\bar{r}/\Delta t$; a white sequence of variance V sampled at $f_\mathrm{s} = 1/\Delta t$ has one-sided spectral density $2V/f_\mathrm{s}$. Hence

$$S_\mathrm{shot} = 2\,\frac{\bar{r}}{\Delta t}\,\Delta t = 2\bar{r} \quad [\mathrm{cps}^2/\mathrm{Hz}] \tag{23}$$

flat at twice the mean count rate and independent of the sampling interval. This assumes dead-time-free Poisson detection; a detector with appreciable dead time is sub-Poissonian and sits below it. Broadband power above the floor is the same excess the Fano factor totals, resolved by frequency; a narrow peak above it is a candidate periodicity — whose significance is a separate question, answered in §2.1.8.

In the program: the floor is drawn on the spectrum at $2\bar{r}$ from the trace's own mean count rate, with no dead-time correction (§4.1's rule).

2.1.8 Periodicity significance

A spectrum always has a highest point, so "there is a peak" means nothing without asking how often noise alone would produce one that tall. For Gaussian noise the power at a given frequency is exponentially distributed *when measured in units of the data variance* σ_0^2, and the largest of N_f independent powers then has false-alarm probability [26], Eq. (14), p. 839

$$\Pr\{Z > z\} = 1 - \left[1 - e^{-z}\right]^{N_\mathrm{f}}, \qquad Z = \max_n P(\omega_n), \qquad N_\mathrm{f} = N/2 \tag{24}$$

where $P(\omega_n)$ is the periodogram's power at frequency ω_n in units of σ_0^2, Z the largest of them, z the level it is tested against, and $N_\mathrm{f} = N/2$ the independent frequencies of an evenly sampled record of N samples (Scargle's N_0) [26], Eq. (5), p. 836. Two details decide whether this number means anything. First, the null belongs to the plain, unaveraged periodogram normalized by σ_0^2 — Welch's averaged estimate is chi-square with the degrees of freedom above, not exponential, so this level means nothing drawn against the averaged spectrum. Second, the null assumes noise that is white as well as Gaussian. A drifting or correlated trace breaks that assumption in the direction that manufactures convincing low-frequency peaks.

In the program: the test runs on the plain periodogram and its level is never drawn against the averaged spectrum; when the blocking curve (§2.1.4) rises, reporting correlations, the verdict is shown marked unreliable rather than withheld or believed.

2.2 The Correlogram and Its Lag Axis

A correlogram arrives in one of two data forms on a lag axis a real correlator lays out in blocks: §2.2.1 fixes the forms and the Siegert relation between them, §2.2.2 the axis and the sample time each channel carries.

2.2.1 The two data forms and the Siegert relation

For the g_2 form, the baseline is removed by

$$g_2(\tau) - 1 = \frac{G(\tau) - G_\infty}{G_\infty} \tag{25}$$

where $G(\tau)$ is the raw intensity autocorrelation the correlator accumulated at delay τ and G_∞ its long-delay level — the normalization level Koppel writes B and ISO 22412 writes A (§3.1.2) — so that $g_2(\tau) - 1$ is the normalized excess correlation and decays to zero. For the g_1 form, the **Siegert relation** [6] converts the field autocorrelation to the measured quantity:

$$g_2(\tau) - 1 = \beta\,|g_1(\tau)|^2 \tag{26}$$

where $g_1(\tau)$ is the normalized field autocorrelation, the quantity every model of chapter 3 is written for, and β is the instrument coherence factor (an optical property, typically 0.01–1.0 for standard goniometers). Because the intensity correlogram is the *square* of the field one, a decay rate Γ of g_1 appears as 2Γ in $g_2 - 1$, a factor a fit of $g_2 - 1$ must carry to report the rate of g_1.

In the program: an instrument parser knows its file's form; a generic plain-text file is loaded with the form you state, and the fits then work on $g_2(\tau) - 1$ whichever form the file held. Every DLS fit determines β rather than assuming it, carries the Siegert factor of 2 explicitly, and reports the g_1 decay rate Γ.

2.2.2 The multi-tau lag axis

A real correlator does not sample delay time on an evenly log-spaced grid. It runs a bank of linear correlators whose sample time doubles from block to block, giving a **quasi-logarithmic staircase** that reaches nine decades of lag in a few hundred channels [27, Sec. 3.2], p. 300. With L channels per octave and a finest sample time Δt_0, block j has sample time Δt_j and covers the lags

$$\Delta t_j = 2^j\,\Delta t_0, \qquad \tau_{j,k} = k\,\Delta t_j, \qquad k = L,\,\ldots,\,2L-1 \quad (j \geq 1), \tag{27}$$

with block 0 running $k = 1$ … $2L{-}1$. The blocks are contiguous: block $j{-}1$ ends at $(2L{-}1)\Delta t_{j-1}$ and block j begins one fine step later. The $\tau = 0$ channel is omitted, since none of the formats the program reads carries one. The lag spacing and the sample time are therefore different quantities on such an axis: within a block every channel accumulates over the same Δt_j, while the lags themselves are that step's multiples, and a correlator whose sections run at constant-ratio spacing can have a lag spacing many times its sample time.

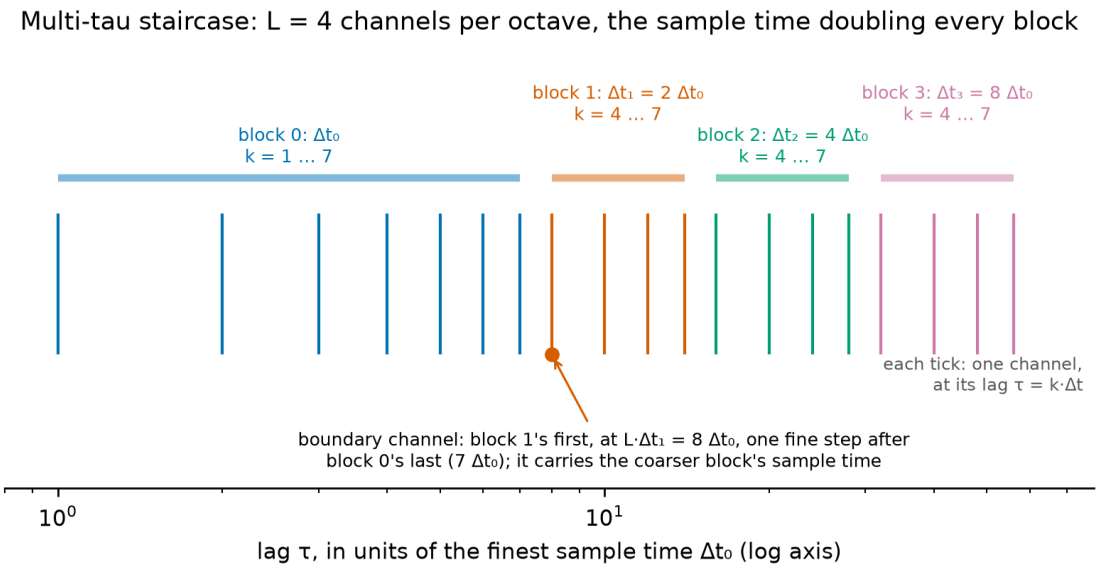


Figure 4: The multi-tau staircase of Eq. 27 for $L = 4$, its channel positions computed by the program's own axis builder: block 0 holds $2L - 1$ channels at Δt_0, every later block L channels at twice the previous sample time, and the blocks are contiguous — block 1's first channel sits one fine step after block 0's last and carries the coarser block's sample time.

The sample time a channel was accumulated over is written Δt_k from here on, with k the channel's index within its block as in Eq. 27, so a channel of block j has $\Delta t_k = \Delta t_j$. It is an input of the noise model of §2.3 and of the residual weights in §3.2.8 and §3.4.2. Under the layout above it is each channel's step to the next [27, Sec. 3.2], p. 300, and no lag is shared between blocks — block j begins one fine step after block $j-1$ ends, at $L\ \Delta t_j$ — so the boundary channel is the *coarser* block's first and carries that block's sample time. A real correlator's firmware may compute a lag at that position in either block, which the axis alone cannot reveal; only a format that lists the boundary lag twice, once from each block, shows that the finer block computed it. A file that states the correlator's sampling structure supplies Δt_k per channel; one that states none leaves it to be read off the lag axis. Which formats list the boundary lag twice, which state a sample time, a run duration or a count rate, and where in the file, is the **User-Manual** §5's.

In the program: the layout above is taken for every staircase axis; how a fit obtains Δt_k — as stated, or read off the axis — and what it records of that are §3.2.9's rules.

2.3 The Noise on a Correlogram

Correlogram noise is not one number. The variance of $g_2(\tau)-1$ varies channel by channel, and it is a property of the correlator, not of the sample: a photon-counting correlator estimates each channel from a finite number of products of a finite number of photons, so the channel's variance follows from the decay rate, the sample time, the run duration and the count rate. Both the cumulant fits (§3.2) and the distribution fits (§3.4) weight their residuals by a model of that variation, and the generator (§7.2) draws from it. Taking Schätzel's per-lag variance in its **classical limit** — the mean photon count per sample time $\bar{N}_k \to \infty$ (his $\bar{n}$), so the shot and cross terms vanish and only the intensity-noise group survives [27], Eq. (49), p. 296 — gives, with $\Theta_k = \bar{\Gamma}\,\Delta t_k$ the decay per sample time (Schätzel's Γ, dimensionless; $\bar{\Gamma}$ the mean decay rate of §3.2.1), k the channel index

within its block — the same k that subscripts Θ, Δt and M below, marking that channel's own quantities — and $M_k = T_{\text{run}}/\Delta t_k$ the number of products averaged over the run duration T_{run}:

$$\begin{aligned}\operatorname{var}\left(g_2(\tau_k) - 1\right) = \frac{\beta^2}{M_k}\Big[&\coth 2\Theta_k + e^{-4\Theta_k k}(2k + \coth 2\Theta_k) \\ &+ 2\beta e^{-4\Theta_k k}(-2k + \coth 2\Theta_k - 2\coth\Theta_k) + 4\beta e^{-2\Theta_k k}\coth\Theta_k\Big].\end{aligned} \tag{28}$$

The resulting noise field is not monotonic: it falls through the correlogram's decay, then turns up in the last stage, where the lag times become comparable with the run and the products $M - k$ left to average are few [28, p. 29382]. What these sources corroborate is the shape — the non-monotonic profile and the last-stage upturn — not a magnitude: this model predicts only a *relative* span (roughly a factor of twenty over a 22-octave axis), and its absolute scale in the synthetic generator is whatever its **Noise level** is set to (§7.2), so the absolute per-channel range measured on real data [29, p. 152] cannot confirm or contradict it. The span is what carries the physics: weighting the accumulation count alone would ramp *monotonically* by a factor of ~1400, burying the longest lags under noise larger than the signal.

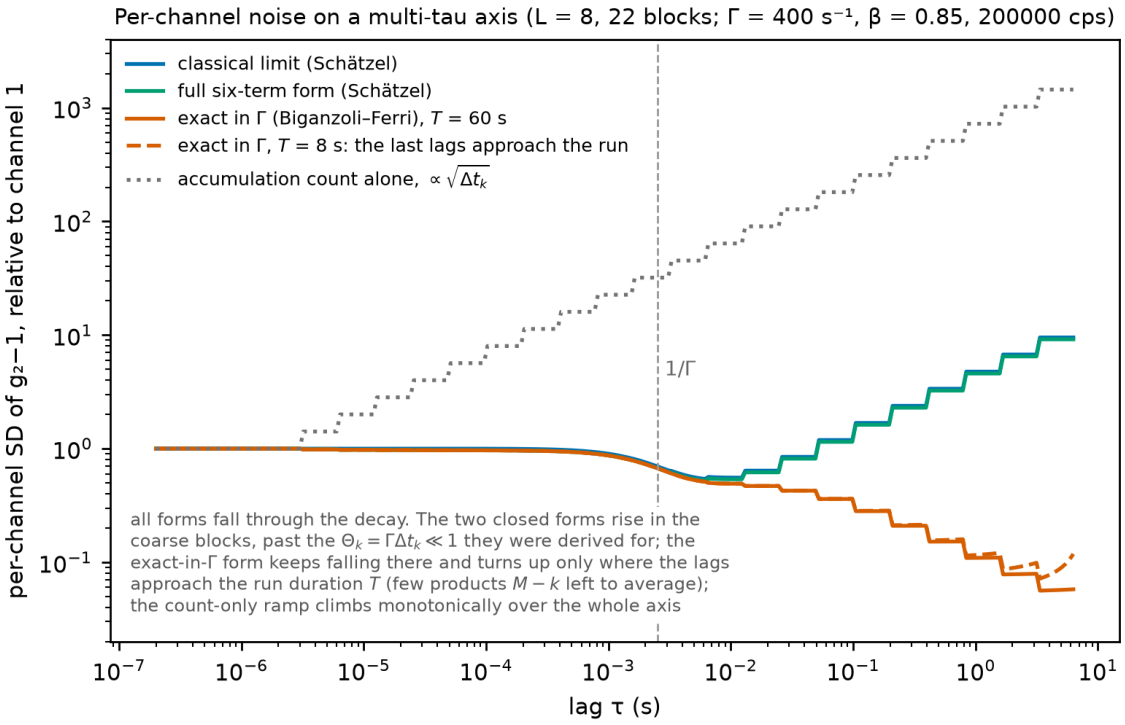


Figure 5: The per-channel SD of $g_2{-}1$ across a multi-tau axis of 22 blocks, for a decay rate whose $1/\Gamma$ sits at block 11, each form relative to its first channel. The classical limit (Eq. 28), the full six-term form (Eq. 29) and the exact-in-Γ form (Eq. 40) all fall through the decay; the two closed forms then rise in the coarse blocks, past the $\Theta_k \ll 1$ they hold for, while the exact form keeps falling and turns up only where the lags approach the run duration (dashed: an 8 s run against lags reaching 6 s). The accumulation count alone (dotted) would ramp monotonically by a factor of ~1400.

With the count rate known, the two dropped groups return. Writing $\bar{N}_k = \bar{r}\,\Delta t_k$ for the mean photon count per sample time of channel k, with $\bar{r}$ the mean count rate (§2.1.7) — the per-block substitution that carries Schätzel's single-sample-time result onto a multi-tau axis [28, p. 29381] — the full six-term form [27], Eq. (49), p. 296 adds to the bracket of Eq. 28

$$\operatorname{var}\left(g_2(\tau_k)-1\right) = \frac{\beta^2}{M_k}\Big[\ \cdots\ + \frac{2}{\beta\,\bar{N}_k}\left(1+e^{-4\Theta_k k}+2\beta e^{-2\Theta_k k}\right) + \frac{1}{\beta^2\,\bar{N}_k^2}\left(1+\beta e^{-2\Theta_k k}\right)\Big], \tag{29}$$

where the ellipsis stands for the four classical groups. In both closed forms the run duration enters only through $M_k = T_{\mathrm{run}}/\Delta t_k$, a factor common to every channel, so it cancels from the *relative* shape a fit weights by; the form exact in Γ (§3.2.8) keeps it, and §3.2.9 says what a fit does when a file states none. The two groups matter exactly where a fit looks: they fall as $\bar{N}_k^{-1}$ and $\bar{N}_k^{-2}$, so they dominate the finest blocks and vanish in the coarse ones, which is what widens the per-channel span at short lag. The full form is the **canonical** variance for a photon-counting correlator's channel (benchmarked against: the per-channel sample SD of real replicate measurements on multi-tau axes).

Both forms are closed forms derived for $\Theta_k \ll 1$, which the coarsest blocks of a multi-tau axis violate by construction. The form exact in Γ, the Biganzoli–Ferri per-channel variance, is stated with the residual weight it was derived for (Eq. 40, §3.2.8); it tends to Schätzel's forms above as $\Theta_k \to 0$ and departs from them where Θ_k is not small, which is why it is the default weight of the Nonlinear and distribution fits (§3.2.12, §3.4.2) and the generator's noise (§7.2).

In the program: no noise model is fitted to a measurement; these forms are evaluated on a fitted decay rate to weight a fit's residuals (§3.2.8, §3.4.2), and what a fit records of the weighting it ran under is §3.2.12's rule.

2.4 What This Chapter Hands On

Chapter 3 reads $g_2(\tau) - 1$ in one data form and fits models written for g_1 (Eq. 26); its shared apparatus (§3.1) selects channels on the lag axis of §2.2.2, and its residual weights (§3.2.8, §3.4.2) are the per-channel variance of §2.3 evaluated on a first-pass fit, with the sample time Δt_k of §2.2.2 as their input. The trace diagnostics of §2.1 feed no fit; they are the measurement's own quality record, and §4.1 reads the trace's mean as a static intensity.

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3. Dynamic Light Scattering

Dynamic light scattering extracts particle dynamics — decay rates, diffusion coefficients and hydrodynamic radii — from a measured $g_2(\tau)-1$ (§2.2). Every method fits $g_2(\tau)-1$ directly, under §2.2.1's convention for the Siegert factor, so Γ throughout is the decay rate of g_1; then $D = \Gamma/q^2$, and R_{h} follows from Stokes–Einstein, Eq. 6, the one step of the three that needs the viscosity (§1.3).

The chapter opens with what every DLS fit shares (§3.1): the delay window, the determined baseline, the decay-completeness check and the size range, stated once so that no method has to restate them. §3.2 is the cumulant analysis, the standard first fit, in its **Linear** and **Nonlinear** forms (the Settings labels), with the higher moments and the residual weight; §3.3 the single, double and stretched exponentials (the last the Kohlrausch–Williams–Watts form, KWW); §3.4

the distribution methods and the choice of regularization; §3.5 the fits that combine angles or concentrations, and the mode map; §3.6 depolarized DLS and the shape models built on it; §3.7 the reweighting of an intensity distribution to volume, surface or number, which borrows two results from chapter 4. No DLS fit reports a ± from a single correlogram (§6, the single-correlogram rule); the honest dynamic ± is the replicate average of §6.4, and the chapter closes with what chapters 4 and 5 take from a DLS result (§3.8).

3.1 What Every DLS Fit Shares

Four rules govern every DLS fit the program offers — the cumulants, the exponential fits and the distribution methods alike — and they are stated here once. A fit reads the correlogram through one **delay window** (§3.1.1); it subtracts, or floats about, a baseline that is determined rather than fitted (§3.1.2 and §3.1.3); it is asked whether the measurement actually observed the decay it reports (§3.1.4); and its size is compared against the range the technique can establish (§3.1.5). Where one of these checks, or a precondition of a method, fails, a computed value is usually better flagged than withheld: the number is real output a reader may need, and the flag names what failed. Each section's marked line says which it does.

3.1.1 The delay window and the leading-channel skip

Every fit (cumulant, exponentials, KWW, NNLS/CONTIN/lognormal) selects which correlogram points it uses through one shared window with two complementary restrictions:

1. a **leading-channel skip** — drop the first N correlator channels *by index* (the shortest, finest-spaced lags). The first few channels of a multi-τ correlator carry detector **afterpulsing** and dead-time artefacts that are not diffusion: a spuriously high first channel mimics a fast decay and biases the fit toward a falsely small R_h;
2. a physical delay window $[\tau_{\min}, \tau_{\max}]$ in time.

A channel that collected no coincidence ($g_2 = 0$) has nothing to fit. Taking the skip before the window keeps a corrupted leading channel from distorting the amplitude cutoff/intercept (cumulant) or the β-to-$\tau\to 0$ extrapolation (distributions). Use the skip sparingly with distribution methods: the short lags constrain the *fast / small* end of the size distribution, so an aggressive skip that a cumulant *mean* merely nudges can erase a genuine small-particle population entirely.

In the program: the skip is one integer for every DLS fit, set in Settings (default 0, keep all), and is applied first; the two compose as an intersection, so the first fitted point is the later of channel N and the first channel with $\tau \geq \tau_{\min}$. The delay window is the **Correlogram** sub-tab's shared region, and until you move it each measurement is fitted over its own full lag range — a window proposed from one measurement never reaches one on another lag axis (**User-Manual** §8.1.2). Inside the window every fit drops a channel with $g_2 = 0$ and counts it in a note, so the cumulants, the exponential fits and the NNLS, CONTIN and lognormal inversions all read one **live-channel mask**.

3.1.2 The baseline is determined, not fitted

In the cumulant method's own source the baseline is a normalization computed from the count rate, never a free parameter [30]: the normalized "signal" — his $S(\tau)$, this guide's $g_2(\tau) - 1$ — is formed from the correlator sum $C(\tau)$ and the accidental correlation level (his B, written G_∞ here as in Eq. 25), which "is calculated from the single channel counting rates",

$$g_2(\tau) - 1 \equiv \frac{C(\tau)}{G_\infty} - 1, \qquad G_\infty = M^{-1}\Big[\sum_{i=1}^{M} N_i\Big]^2 \tag{30}$$

where N_i is the photon count in the i-th sample interval and M the number of intervals (Koppel's $n(t_i)$ and N), so that after a correct normalization the baseline of $g_2(\tau) - 1$ is zero by construction, and a residual tail is a normalization error or an unresolved slow mode. [31] says the same independently — the baseline (its A, this guide's G_∞) is "a time-independent constant proportional to the square of the time-averaged scattered intensity" — and prescribes determining it two ways (from the total photon count, and from $G(\tau)$ at $\tau \gg 25/\bar{\Gamma}$), retaining the larger, and discarding the measurement when the two differ, relative to the smaller, by more than 10^{-3} (Annex A.1.2). Its linearization then subtracts that determined baseline before the logarithm:

$$y(\tau) = \tfrac{1}{2}\ln\big[G(\tau) - G_\infty\big] \cong \tfrac{1}{2}\ln(G_\infty\beta) - \bar{\Gamma}\,\tau + \tfrac{1}{2}\mu_2\,\tau^2 \tag{31}$$

where $y(\tau)$ is the half-logarithm of the baseline-subtracted correlation, the ordinate the fit regresses on τ, and ISO writes the baseline A and the intercept B; this guide keeps G_∞ and β, with $\bar{\Gamma}$ and μ_2 the mean decay rate and second cumulant of §3.2.1. Divided by G_∞ and doubled, this is the cumulant expansion of §3.2.1 (Eq. 35) applied to $(G - G_\infty)/G_\infty$ — the Linear fit with the baseline *pinned* at a value determined beforehand; the program's **baseline region** (§3.1.3) plays the part of that determined value. Because that value is subtracted from the already-normalized $g_2{-}1$, the Linear fit's β differs from ISO's exact form by a factor $1{+}\varepsilon$, with ε the pinned offset of §3.1.3; $\bar{\Gamma}$ and μ_2 are unaffected. The Nonlinear fit's floating B is not a contradiction of this — [32] floats it to *detect* the small deviations from unity that noise leaves after a good correlator's own normalization, which is why its bound is kept narrow (§3.2.5).

In the program: which fits subtract the determined baseline, and which float one, is §3.1.3's rule. The two-baseline comparison ISO prescribes is not a refusal here; which of a file's two stated baselines forms its $g_2 - 1$ is a loading choice (**User-Manual** §5.2, §12.3).

3.1.3 How each fit applies the determined baseline

The same determined baseline is subtracted before the single-exponential, double-exponential and KWW fits, so each of Eq. 42, Eq. 43 and Eq. 44 is fitted in the form

$$g_2(\tau) - 1 = \varepsilon + \beta\,|g_1(\tau)|^2 \tag{32}$$

with ε the determined baseline. The same fork is stated for the single-exponential fit by [33]: "depending *inter alia* on whether the value of the baseline is reliably known or not" — and when it is, fit $\ln(G/G_\infty - 1) = \ln\beta - 2\Gamma\tau$ (his G_2 and B; Eqs. 4.4–4.5, p. 178); the same chapter prints the stretched exponential with an additive baseline term (Eq. 4.41, p. 196), so a baseline term on these models is not the program's invention ([34] itself is the decay function only). What *is* the program's own choice is that ε is pinned to the determined value rather than fitted, a flagged convenience simplification of those forms: a floating baseline recovers the offset only where the correlogram has already decayed and drifts as the window shortens, which is the regime a baseline term matters in most. The offset is not optional in effect: a flat offset that is *not* removed is absorbed differently by each model — a small monotone size bias in the single-exponential and KWW fits, and in the double exponential a spurious, infinitely slow second mode that also biases the surviving fast mode.

In the program: the determined baseline is the baseline region on the Correlogram sub-tab — the mean of g_2-1 over the channels the fit itself reads: past any skipped leading channels, inside the delay window, and live ($g_2 > 0$), so a region with none of those counts as not placed. The Linear fit (Eq. 31) and the exponential fits subtract it and never fit a baseline of their own; the Nonlinear fit floats its own B (§3.2.5). A region you have not moved supplies nothing: until then the Linear fit and the exponential fits subtract nothing, the Nonlinear fit measures its floating baseline over its own index tail (§3.2.5), and the distribution methods estimate theirs automatically (§3.4.1). A region you have placed is applied as given — the offset ε recorded with the result and named in the export — and one reading of it serves every fit that reads it, so the fit shown and the fit pooled, averaged or mapped are one fit of the correlogram. The controls, the region's reach into the other sub-tabs and the reset are the **User-Manual** §8.1.2.

3.1.4 Did the mode actually decay?

A parametric fit will happily report a rate for a mode whose decay the measurement never saw. The correlogram then constrains only the *start* of that decay, and the fitted Γ — with the R_h that follows from it — is an extrapolation past the end of the data rather than a measurement. It is a common condition, not an exotic one: a static or aggregated sample is the usual cause, and nothing else in the fit remarks on it, since the model fits the data it was given perfectly well.

The measurable question is how much of the decay the lag range covered. ISO's criterion has two equivalent expressions; a mode has not decayed when

$$\Gamma\,\Delta\tau < \tfrac{1}{2}\ln 50 \approx 1.96 \qquad \text{or} \qquad \frac{\min_j[G(\tau_j)-G_\infty]}{G(\tau_1)-G_\infty} > \tfrac{1}{50} \tag{33}$$

The left form is a **rate** test: for a mode of decay rate Γ measured across a lag span $\Delta\tau = \tau_\mathrm{max} - \tau_\mathrm{min}$, the dimensionless product $\Gamma{\cdot}\Delta\tau$ counts the relaxation times observed. The right form is the same requirement asked of the **data** — did the measured correlogram ever fall to a fiftieth of where it started?

The threshold is [31]'s, not an invention here. Annex A.1.2 fixes the range of points a cumulant fit may retain and requires at least one of them to fall below 1/50 of the initial amplitude, and states no consequence for a range that does not; §10.2.1 a) states the same condition qualitatively, as a delay-time range that "covers the set of decay times". Since the measured amplitude $[G(\tau) - G_\infty]$ follows $|g_1|^2 = \mathrm{e}^{-2\Gamma\tau}$, the 1/50 requirement *is* $2\Gamma{\cdot}\Delta\tau \geq \ln 50$ — which is why the two forms above are one criterion. Xu [35] states the same quantity directly as a delay-range rule of thumb, $\tau_\mathrm{max} = 2.5/\bar{\Gamma}$, and gives an independent, stricter version of the data form.

The rate form assumes the mode decays exponentially, so it holds only where that is true, and it can ask the question *per mode*. A stretched exponential's rate parameter cannot answer it: a KWW (§3.3.3) absorbs a decay it never observed into its *stretch exponent* rather than into τ_c, so a τ_c-based test passes correlograms that plainly never decayed, and its two reported sizes are two readings of one decay. A cumulant's mean rate cannot answer it for the same reason (§3.2.3): it absorbs an undecayed slow component into the higher cumulants, and the Nonlinear fit into its floating baseline as well. Note the data form takes the minimum over the delay window, not the last point: a decayed correlogram's tail is noise about zero, and whichever channel lands last may sit back above the threshold. It also assumes the correlogram's baseline is already removed — a residual drift offset larger than the threshold reads as an incomplete decay.

What the standard states for a whole correlogram and one mean rate, a fit that resolves modes needs per mode — a two-mode fit routinely pairs a measured fast mode with an undecayed slow one, and a single verdict cannot say that. The per-mode step is this program's extension; the criterion and its threshold are the standard's.

Judged per mode, the healthy mode beside an undecayed one keeps a clean number. The check is deliberately independent of the size range of Eq. 34 — a mode can lie past that range and still have decayed, in which case a real, very large scatterer was genuinely measured and only the size range speaks. The two criteria answer different questions: one asks whether sedimentation has overtaken diffusion, the other whether the decay was observed at all — and the size of an unobserved decay is not a size.

In the program: the check is the **decay-completeness check**, applied per mode, in its rate form, wherever the fit resolves modes (the single and double exponentials, each mode judged independently), and as one verdict, in its data form, for the cumulant and KWW fits, before the cumulant's retained range is fitted; a mode that fails it is flagged with that reason, its rate and size reported rather than withheld, and the size range of §3.1.5 is asked only of a mode that passed. A pooled fit inherits these verdicts by §3.5.3's rule.

3.1.5 The size range

A fit's own diagnostics say nothing about whether the size it returns lies where DLS can measure at all. Sedimentation sets the upper bound — [31] puts it "much less than 10 µm", stated as a diameter, so a radius is bounded by half that, and a fitted size lies past the range when

$$R_\mathrm{h} > R_\mathrm{h}^\mathrm{max} \tag{34}$$

A genuinely large scatterer is real data, so a size past this bound is a number the technique does not establish rather than a failed fit. The limit is the technique's, not the cumulant's. The companion lower bound is the **size floor**, below which a fit has chased a short-lag artifact (detector afterpulsing in the first channels) rather than a particle, and skipping the leading channels (§3.1.1) removes it. A free floating-baseline fit that lands there has no measurement behind it at all, which is why the Nonlinear fit is treated differently from the others.

In the program: every DLS method — the cumulants, the single/double-exponential modes and both KWW sizes — compares its size against both bounds, the size range's 5 µm and the size floor's 0.2 nm; a value outside them is reported with its flag, and below the floor the flag names the leading-channel skip as the remedy — the one exception is a Nonlinear fit that lands below the size floor, which is rejected — and the flag travels with it into the Summary table, the exports and the Cross-Sample candidates. A replicate average whose fits include such a size is not written to the sample (§6.4).

3.2 The Cumulant Methods

The cumulant analysis is the standard first fit of a correlogram: a mean decay rate, a width and, from them, one size and one polydispersity index. The program offers it in two forms, the Linear log-polynomial fit of Koppel (§3.2.3) and the Nonlinear moment-expansion fit of Frisken (§3.2.5) — the Settings labels — and both report the same quantities from the expansion of §3.2.1: $\bar{\Gamma}$, μ_2, the PDI and R_h, with μ_3 and μ_4 at the higher orders of §3.2.6. §3.2.2 says which average the size is, §3.2.4 what the Linear fit's truncation looks like on screen, §3.2.7 what a moment resting on its

bound means, and §3.2.8–§3.2.12 how the nonlinear fit weights its residuals — the Biganzoli–Ferri variance (§3.2.8), its inputs (§3.2.9), its named divergences (§3.2.10), the other two weightings and the fallback (§3.2.11) and the default (§3.2.12).

3.2.1 The cumulant expansion and the polydispersity index

The standard first analysis [30] takes the log of the baseline-subtracted signal and fits a polynomial in τ:

$$\ln[g_2(\tau) - 1] = \ln \beta - 2\bar{\Gamma}\,\tau + \mu_2\,\tau^2 - \tfrac{1}{3}\,\mu_3\,\tau^3 + \cdots \tag{35}$$

The mean decay rate $\bar{\Gamma}$ gives $\bar{D} = \bar{\Gamma}/q^2$ and R_h; the second cumulant gives the **polydispersity index**

$$\mathrm{PDI} = \frac{\mu_2}{\bar{\Gamma}^2} \tag{36}$$

and the third (order-3 fit) the third central moment μ_3 — a measure of the distribution's asymmetry, though not its skewness, which is $\mu_3/\mu_2^{3/2}$ and is not reported; an order-4 fit adds the fourth term, whose coefficient and meaning are given below at Eq. 39. The cumulant result is reliable only for a moderately narrow distribution; past that, a distribution method is the right tool. The limit is not a precondition the standard states — the only figure ISO 22412 gives for the polydispersity index is that a monodisperse spherical test sample typically reads below 0.07 ([31], 3.3 Note 1, p. 2). It is a precondition of the expansion, not the accuracy of a particular order, so one limit serves every order and both methods; how the accuracy depends on the order is §3.2.6's question.

In the program: Eq. 35 fitted directly is the Linear fit (the Settings label), on the baseline of §3.1.3, by weighted linear least squares under the weight of §3.2.12. A PDI above 0.3, at every order and for both methods, flags the R_h as well as the PDI cell, and the verdict travels with the value into the Summary tables, the exports and the Cross-Sample candidates. A cumulant result carries no ± (§6's single-correlogram rule).

3.2.2 Which average the cumulant size is

$\bar{\Gamma}$ is the mean decay rate over the scattered-intensity distribution of decay rates $G(\Gamma)$ (ISO's $C(\Gamma)$; Eq. 46 fits the same function), and the Stokes–Einstein diameter is inversely proportional to the decay rate — so averaging the rate averages the *reciprocal* diameter. [31] therefore defines the DLS diameter $\bar{d}_\mathrm{DLS}$ as

$$\frac{1}{\bar{d}_\mathrm{DLS}} = \int_0^\infty \frac{1}{d}\,G[\Gamma(d)]\,\mathrm{d}\left(\frac{1}{d}\right) \tag{37}$$

where d is the particle diameter (ISO's x; its $\bar{x}_\mathrm{DLS}$ is $\bar{d}_\mathrm{DLS}$ here) and $G[\Gamma(d)]\,\mathrm{d}(1/d)$ is the fraction of the scattered intensity from particles of diameter between d and $d + \mathrm{d}d$. The cumulant size is thus the **scattered-intensity-weighted harmonic mean** — sometimes also referred to as the z-average ([31], Formula (A.14), printed p. 19, and Note 2 to entry 3.2, printed p. 2). This program reports the radius $R_\mathrm{h} = \bar{d}_\mathrm{DLS}/2$, reached by exactly that route: $\bar{D} = \bar{\Gamma}/q^2$ into Eq. 6, whose reciprocal proportionality to the size is what makes the resulting mean harmonic. It is not

the same average as any distribution method's peak size, which is why the program names each size by its method rather than by its average.

In the program: the cumulant radius is named *cumulant* on screen and in every export, and a distribution method's peak size is named by its method; the two are never shown under one label.

3.2.3 The Linear fit: its fitted range and the cutoff

The log-polynomial fit is taken over a contiguous interval of the correlogram, bounded below in amplitude: the run of channels that contains the peak and reaches out in each direction until the signal first falls to a cutoff fraction of that peak. The rule follows [31] Annex A.1.2, which states the retained values as *a range in* $[G(\tau_j) - G_\infty]$ (ISO's $[G_2(\tau_j) - A]$), bounded above by the first channel's amplitude $[G(\tau_1) - G_\infty]$ and below by a fixed fraction of it. Three departures from the Annex's letter are deliberate, and none is the standard restated:

- The reference amplitude is the windowed peak, not the first channel. On a clean correlogram these are the same channel and the readings coincide. They part company on a noisy head, and the peak is the safer anchor: it is the one channel guaranteed to clear a threshold measured from it, so the interval is never empty for a positive peak, where a run anchored at τ_1 can be emptied by a single depressed leading channel and refuse a measurement the rest of which is sound.
- The Annex's sentence does not by itself settle contiguity. Read strictly it states a band in *amplitude*, which an individual channel either satisfies or does not; reading it as an interval in τ is an inference. What the interval reading rests on is the two treatments below — not on the Annex alone.
- The fraction is a tenth, not a hundredth. A.1.2 prescribes a hundredth of the initial amplitude; [36] truncate their linear cumulant analysis at exactly 10 % of it, in a comparison of linear against nonlinear cumulants on distributions of known width. The tenth is a **benchmarked deviation** from the standard (benchmarked against: real multi-tau correlograms): a hundredth reaches into channels where the decay is comparable to the noise, where the parabola flattens and the reported PDI falls, which can silence the polydispersity warning a measurement was carrying. A deeper cutoff is not a uniformly finer reading.

Two independent treatments describe the amplitude rule as a truncation rather than a filter, which is the shape the program takes from them: [27] takes the retained set to end at the first channel whose normalised correlation goes negative — stated there for *noise estimation* on the amplitude estimator, so it is a precedent for the shape and not a prescription for a cumulant fit's range — and [36] truncate their own linear cumulant analysis when $g_2(\tau) - 1$ *drops to* 10 % of its initial value.

The distinction matters because a correlogram decays monotonically only *in expectation*. Once the underlying decay approaches the threshold, noise carries individual channels above and below it; a rule applied channel by channel keeps only the ones that landed above, and the regression fits that rectified tail as a slow decay mode — $\bar{\Gamma}$ and μ_2 both fall and R_h rises without bound. Terminating at the first crossing cannot do this, because there is nothing past the crossing to select from.

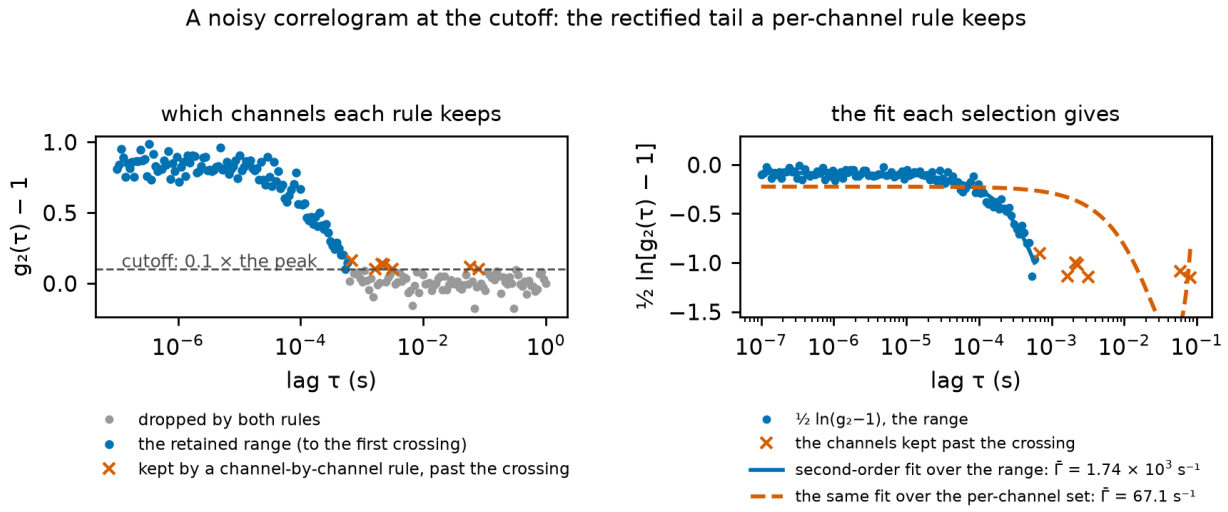


Figure 6: One synthetic correlogram with constant noise at 6 % of its intercept on a 200-channel log grid, at the 10 % cutoff. Left: the retained range ends at the first crossing (blue), where a rule applied channel by channel would also keep the seven tail channels that noise carried above the threshold (crosses). Right: the second-order fit of Eq. 35 over each selection — over the range $\Gamma = 1.74 \times 10^3$ s^{-1}, the value the correlogram was built with; over the per-channel set the rectified tail reads as a slow decay and Γ collapses to 67 s^{-1}.

One limit is worth stating, because the rule does not reach it. A.1.2 states the range on $[G - G_\infty]$ with the baseline G_∞ determined beforehand; where a residual baseline is left unsubtracted and sits *above* the cutoff fraction times the peak, the plateau never crosses the threshold, nothing terminates, and every channel is retained. That regime is a separate mechanism, and the range rule neither causes nor cures it.

The Annex states four requirements. Two are the retained range, with the departures above, and the requirement that at least one retained value fall below a fiftieth of the initial amplitude — which is the decay-completeness check of §3.1.4 in its data form, and its equivalent rate criterion $\bar{\Gamma} \cdot \Delta\tau$. The data form is the one [31] itself states for the cumulant's retained range (Annex A.1.2), and §3.1.4 says why the rate form is the wrong instrument for it. The other two — at least twenty retained points, and every retained value positive — carry the Annex's remedy of *discarding the measurement*. That is also why the Annex's own hundredth is safe there and is not the default here: A.1.2 reaches that depth only on measurements it has already refused to analyze.

In the program: the cutoff is a tenth of the windowed peak by default and is settable as **Cumulant fit cutoff** (**User-Manual** §12.1.5), so a measurement that warrants the Annex's own hundredth can be given it; the range a size was fitted over travels with it in the export. The decay-completeness check is asked under §3.1.4's rule. Every retained value is positive by construction, the range holding only values above the cutoff; the twenty-point minimum is not applied, the fit needing one point more than its order and refusing a shorter range; and no measurement is discarded for failing either Annex requirement.

3.2.4 Where the truncated expansion turns around

Eq. 35 truncated at second order is a parabola in τ with positive curvature μ_2, so it has a minimum and rises afterwards. Setting its derivative to zero puts that minimum at $\tau^* = \bar{\Gamma}/\mu_2 = 1/(\mathrm{PDI}\cdot\bar{\Gamma})$ — that is, at 1/PDI mean decay times. Below the PDI ≈ 0.3 limit above, τ^* lies at more than three decay times, out where the correlogram is already noise; above it the minimum moves inside the measured lags, where the data fall away from the parabola's continuation; on a very broad sample it moves inside the retained range itself, and the fitted curve drawn over the data visibly bends back upward at its right-hand end. That upturn is the truncation leaving its range of validity, not a failure of the regression, and it is the same condition the PDI flag reports — read together, they say the sample needs a distribution method. The Nonlinear fit (§3.2.5) carries its polynomial as a prefactor rather than in the exponent, and turns upward only for $\mu_2 \geq 2\bar{\Gamma}^2$, i.e. PDI ≥ 2.

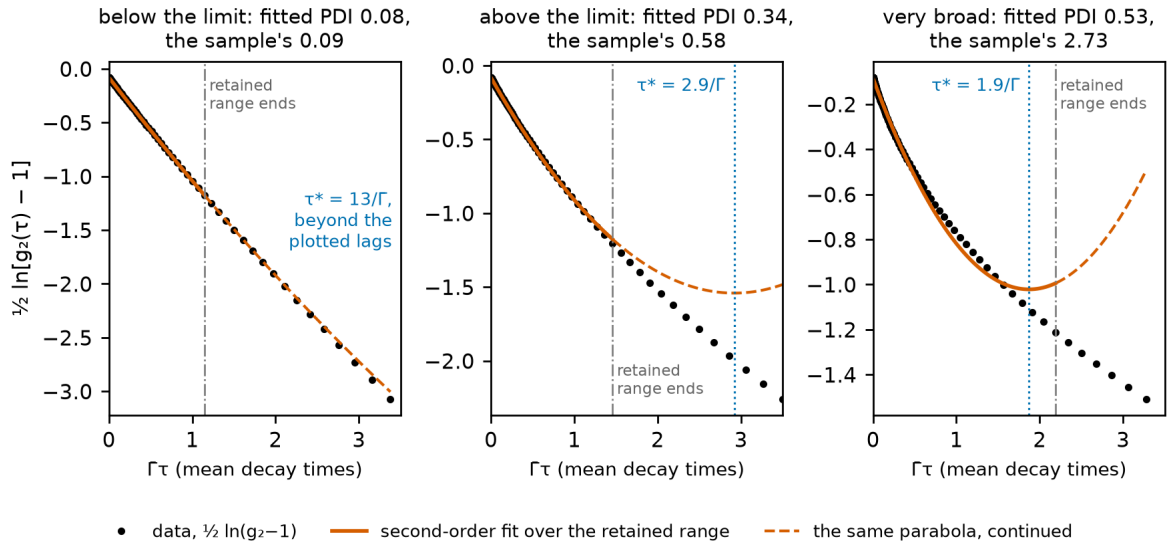


Figure 7: The second-order fit of Eq. 35 (solid, over the retained range) and the same parabola continued (dashed) on three synthetic lognormal samples. Below the limit the minimum τ^* lies far beyond the plotted lags; above it the parabola turns around inside the measured lags but beyond the retained range, so the drawn fit stays monotone while the data fall away from it; on a very broad sample τ^* moves inside the retained range and the fitted curve itself bends upward at its right-hand end. In both broad cases the fitted PDI is well below the sample's true one, as the truncation implies.

In the program: the fitted curve is drawn over the retained range on the Correlogram sub-tab, so the bend is visible there when τ^* falls inside it.

3.2.5 The Nonlinear fit

The Nonlinear fit [32] fits $g_2 - 1$ *directly* by nonlinear least squares, expanding about the mean rate (Frisken's moments-about-the-mean form) with a **floating baseline** B:

$$g_2(\tau) - 1 = B + \beta\left[e^{-\bar{\Gamma}\tau}\left(1 + \tfrac{1}{2}\mu_2\tau^2 - \tfrac{1}{6}\mu_3\tau^3 + \cdots\right)\right]^2 \tag{38}$$

It yields the same $\bar{\Gamma}$, μ_2, PDI and R_h as the Linear fit but is more robust to baseline drift and noisy/low-count data, because B and the coherence β are fit rather than assumed and every live channel of the windowed correlogram is used (no amplitude cutoff; the live-channel mask of §3.1.1). Expanding about $\bar{\Gamma}$ (rather than about $\tau = 0$, as the log-polynomial does) keeps it stable at large τ.

The baseline is constrained to small drift about the long-time tail so the fit cannot absorb the decay into B, by a bound sized to the data: a fixed width is wide enough on a low-amplitude correlogram to swallow a real baseline error and shift $\bar{\Gamma}$ with it.

Where the tail lies is a choice. [31] (Annex A.1.2) defines the tail-estimated baseline at $\tau \gg 25/\bar{\Gamma}$, which needs the very rate the fit is being run to obtain, so a tail taken by index is a flagged convenience simplification of the standard, not the standard itself; and [32] prescribes no tail region, so a tail you place is a flagged convenience simplification too. Placing it does not fix B, which still *floats*: a region says where the baseline is measured, never what it is. But the bound's centre and the fit's starting guess are built from the tail's mean B_0, which is why a region placed away from the correlogram's own tail can move the reported size.

When the fitted B comes to rest on an edge of its bound, $\bar{\Gamma}$ is set by that edge rather than by an interior optimum. A boundary-active baseline is a statement about where the tail was measured, not a failed precondition of the cumulant expansion, which is why it earns a note rather than a flag.

In the program: the Nonlinear fit (the Settings label *Nonlinear*) is the default, chosen once for every cumulant-based step (**User-Manual** §12.1.1), and runs under the residual weight of §3.2.12. Its tail is the last quarter of the windowed lags that collected coincidence, or the channels of a baseline region you have placed (§3.1.3), from which nothing is subtracted; the tail's mean B_0 is the centre of the baseline's bound and the starting guess, for the fit and for the first pass at which its residual variance is evaluated, and the floating B may move from it by the larger of five times the tail's own scatter and 10^{-3} of the correlogram's peak, and never by more than the peak. A placed region holding fewer than two delay channels is refused by name, not widened back to the index tail; a fitted B on an edge of its bound is a passive note that leaves the reliability flag untouched; a fit that does not converge reports no values and is labeled *(failed)*. The fit carries no ± (§6's single-correlogram rule); a replicate set gives one (§6.4).

3.2.6 Orders 3 and 4: the higher moments

Both methods fit up to the fourth order. The two expansions agree term by term up to the third, and part company at the fourth, where the log-polynomial's coefficient is the fourth *cumulant* and the moment expansion's is the fourth *central moment*:

$$\ln[g_2(\tau) - 1] = \cdots + \tfrac{1}{12}(\mu_4 - 3\mu_2^2)\tau^4, \qquad g_1(\tau) = e^{-\bar{\Gamma}\tau}(1 + \cdots + \tfrac{1}{24}\,\mu_4\,\tau^4) \tag{39}$$

The factor of two between the 1/24 and the 1/12 is the Siegert square, and $\varkappa_4 = \mu_4 - 3\mu_2^2$ is the relation between the two fourth-order quantities ([36], Eqs. 12–13 and the terminology note, preprint pp. 5–6: *the cumulants only differ from the central moments at order 4 and higher*). Two fits that report different fourth-order quantities under one name disagree through a hidden convention — a fit that reported $\varkappa_4$ under the name of μ_4 would read only the fraction $2\sigma^2/(1+2\sigma^2)$ of μ_4 on a Schulz distribution, between 7 % and 24 % at $\sigma = 0.2$–0.4 ([36], Eqs. 18–19, pp. 6–7). μ_4 is an even central moment and so non-negative; the Nonlinear fit bounds it at zero. The Linear fit's converted

μ_4 is not bounded and, under realistic noise, is not a usable number: the quartic coefficient of a log-polynomial amplifies exactly the tail noise the Nonlinear fit's exponential prefactor damps, which is the Linear fit's general disadvantage at every order ([36], §4.2, p. 11). Read μ_4 from the Nonlinear fit only, and read the Linear fit's as a sign that its fourth order has nothing to say about the sample.

They are diagnostics, not measurements: [36] fit simulated correlograms of known width and find, for a fourth-order fit at $\sigma = 0.4$, an uncertainty of ± 0.045 in the recovered $\mu_3/\bar{\Gamma}^3$ against a true value of 0.0512 — "DLS can do little more than hint at the sign of μ_3" (their §4.3, preprint p. 12) — and up to $\sigma \approx 0.4$ an uncertainty in μ_4 "comparable to its actual value" (§4.4, p. 12–13); they are "not aware of any experiment where μ_3 or μ_4 have been convincingly measured" (p. 14). That is a constant property of the method, true on every measurement, and a per-measurement flag that is always down is not a flag, which is why the program does not flag them. Their value is diagnostic — the extra parameters decorrelate μ_2 from $\bar{\Gamma}$ in the fit (p. 14) — so compare them between fits of the same sample, and do not read them as the skewness and kurtosis of the distribution. A fitted μ_4 of 0 is the bound, not a measured fourth moment (§3.2.7), and the fitted μ_3 is reliable in sign only for broad distributions.

Which order for which sample? The accuracy of the width is order-dependent: [36] find nonlinear fits accurate in μ_2 up to a Schulz width $\sigma \approx 0.3$ (their width parameter), i.e. PDI ≈ 0.09, at orders 2–3, and up to $\sigma \approx 0.6$, PDI ≈ 0.36, at order 4 (§4.2, p. 11; steps (iii)–(iv), p. 14), so the PDI flag of §3.2.1 is more permissive than that domain at orders 2–3 and more conservative at order 4. Their recommendations, by sample:

Sample	Order	Why ([36])
broad	4	it reduces the systematic error in μ_2 that a second-order fit carries there (§5, p. 13)
narrow, $\mu_2/\bar{\Gamma}^2$ below ≈ 0.1	2	the second-order fit is the *more* precise there (step (iv), p. 14)
noisy, at a noise level of 10^{-2}	2	fourth-order fits "were not useful"; second-order fits still gave $\bar{\Gamma}$ to about 2 % (§4.7, p. 13)
any	not 3	the τ^4 term is comparable to or larger than the τ^3 term even at small $\bar{\Gamma}\tau$, so a third-order fit absorbs the quartic content into μ_3 and μ_2 (§5, p. 13)

Taking order 2 as the default is a benchmarked deviation from that recommendation (benchmarked against: synthetic correlograms of known moments under the generator's noise models). The recommendation still holds for a sample whose width you know: order 4 on a broad one and on clean data, and not order 3.

In the program: order 2 is the default (Settings, with a per-measurement box on the Correlogram sub-tab whose value wins); both methods report the central moment μ_4, the Linear fit's converted from its fitted coefficient; μ_3 and μ_4 are shown bare, with no reliability glyph of their own.

3.2.7 Two passive notes on a fitted moment

When the fitted μ_2 comes to rest on its lower bound of zero (the bound's source is [32, p. 4089]: "the first two cumulants must be positive"), the PDI shown is that bound, not a measurement; the same holds of μ_4 at order 4, whose bound leaves the PDI untouched.

No cause can be named, because more than one thing drives a moment onto its bound and the fit cannot tell them apart. The reading that is right *inside* the method's domain: the polydispersity is below what this correlogram's noise resolves, so the sample is not distinguishable from monodisperse at this order — never "zero". The reading that is right *outside* it: the truncated expansion does not describe the sample at this order, and a wide or multimodal distribution fitted at order 3 can stop with μ_2 on the wall on every draw, exactly as a narrow one does. Between the two readings, compare the orders, look at the Linear fit's PDI, and prefer a distribution method for a sample that may be wide; a μ_4 at its bound is read only as a diagnostic for comparing fits of one sample.

Read on the *dimensionless* moment, four orders below the smallest polydispersity a correlogram resolves, a value at the bound is the wall and nothing a reader could take for a measured width; and it marks no failed precondition of the expansion, which is why it earns a note rather than a flag.

In the program: a μ_2, or at order 4 a μ_4, whose dimensionless value — the PDI $\mu_2/\bar{\Gamma}^2$ the result shows, or $\mu_4/\bar{\Gamma}^4$ — rests on its zero bound (below 10^{-6}) carries a passive note naming the bound, each note firing on its own moment only; the notes leave the reliability flag untouched and name no cause.

3.2.8 The residual weight: the Biganzoli–Ferri variance

The Nonlinear fit minimises $\sum_k[(g_2-1)_k - \text{model}_k]^2/\sigma_k^2$, and *which* σ_k is a choice that moves every number the fit produces — which is why the weighting a fit ran under is recorded (§3.2.12), as the SE estimator is (§6.1.1). Three weightings exist for the Nonlinear fit: the Biganzoli–Ferri variance (the Settings label *Sourced variance*), the instrument's own standard deviations, and the unweighted form.

In the **Biganzoli–Ferri variance**, σ_k^2 is the per-channel variance of a photon-correlation estimate for a single exponential, exact in the decay per sample time $\Theta_k = \bar{\Gamma}\,\Delta t_k$ (the dimensionless product Biganzoli and Ferri write Γ) [28], Eq. (9), p. 29378. With $S_n = \sinh^n\Theta_k/\Theta_k^n$, k the channel index within its block ($\tau_k = k\,\Delta t_k$, the same index as in §2.3), M the number of sample times in the measurement — M_k of §2.3 on a multi-tau axis, by the per-block substitution of §3.2.9 — and $\bar{N}$ the mean photon count per sample time ($\bar{N}_k$ of §2.3; Biganzoli and Ferri's $\langle n\rangle$),

$$
\begin{aligned}
\sigma^2_{g_2}(k) = \frac{\beta^2}{M-k}\Big\{ & 2S_2\hat{\chi}_0^2 e^{-4\Theta_k k} + 4\beta S_2\hat{\chi}_0^2 e^{-2\Theta_k k} + 8\beta S_3\hat{\chi}_0 e^{-4\Theta_k k} \\
& + S_4\big[\coth 2\Theta_k - 1 + 4\beta e^{-2\Theta_k k}(\coth\Theta_k - 1) + e^{-4\Theta_k k}(a\coth 2\Theta_k + c)\big] \\
& + 4\beta S_6 e^{-4\Theta_k k}(\coth\Theta_k - 1) - 8\beta S_5 e^{-4\Theta_k k}(k - 2 + \coth\Theta_k) + \hat{\chi}_0^4 \\
& + \frac{2}{\beta\bar{N}}\big[\hat{\chi}_0^2 + S_2 e^{-2\Theta_k k}(e^{-2\Theta_k k} + 2\beta\hat{\chi}_0) + 2\beta S_3 e^{-4\Theta_k k} - 2\beta S_4 e^{-4\Theta_k k}\big] \\
& + \frac{1}{\beta^2\bar{N}^2}\big[1 + \beta S_2 e^{-2\Theta_k k}\big]\Big\},
\end{aligned}
\tag{40}
$$

with, from their Eq. (10), $\hat{\chi}_0^2 = (2\Theta_k - 1 + e^{-2\Theta_k})/2\Theta_k^2$, $a = 1 + 2\beta$ and $c = 2k + 4\beta k - 2 - 8\beta + 4\beta\hat{\chi}_0^2 - 16\beta\hat{\chi}_0$. As $\Theta_k \ll 1$ it tends to Schätzel's closed form, Eq. 29 [27], Eq. (49), p. 296, as the same page states, and departs from it where Θ_k is not small — the coarse blocks §2.3 names, where it keeps falling while the closed forms rise, and turns up only where the lags approach the run duration, the $1/(M-k)$ of its prefactor [28, p. 29382] (Figure 5). It is evaluated on the best-fit single exponential rather than on the noisy channel it weights — the prescription of [30, p. 4818] (the prose after his Eq. 26: the variance is approximated "with the best fit single exponential").

A first pass fits the model of Eq. 38 at order 1 under uniform weights, and its $\bar{\Gamma}_1$ and β_1 are the evaluation point:

$$w_k = \frac{1}{\sigma_k^2}, \qquad \sigma_k^2 = \operatorname{var}\left(g_2(\tau_k) - 1\right)\Big|_{\bar{\Gamma}=\bar{\Gamma}_1,\ \beta=\beta_1,\ \Delta t_k,\ \bar{N}_k=\bar{r}\,\Delta t_k} \tag{41}$$

with Δt_k the sample time each channel was accumulated over (§2.2.2).

In the program: the Biganzoli–Ferri variance is evaluated on the first-pass single exponential of Eq. 41; which weighting is the default and what a fit records of it are §3.2.12's rules, and the fallback when this weight cannot be formed is §3.2.11's.

3.2.9 The weight's inputs: sample time, count rate, run duration

A file that states the correlator's sampling structure supplies Δt_k per channel. A file that states none leaves Δt_k to be read off the delay axis: each channel's step to the next where the lags are whole multiples of their block's sample time, the block-boundary channel carrying the coarser block's sample time (§2.2.2). On most channels of a real file the step to the next channel does not divide the lag a whole number of times — a written lag carries rounding, sections meet at boundaries, a block's lags sit offset from its spacing's multiples, or a uniform log or linear grid has no blocks at all. Eq. 41 is written on a whole lag index, $\tau_k = k\,\Delta t$ [28], Eq. (2), p. 29376, so a stand-in Δt_k must give one. No source says what sample time to use when a file states none; the nearest sample time that gives a whole lag index keeps the variance on its own premise rather than claiming a more accurate Δt_k. A *stated* sample time that gives no whole lag index has no such premise to keep.

$\bar{N}_k$ comes from the measurement's committed mean count rate $\bar{r}$ by the per-block substitution of [28, p. 29381]; the two shot-noise groups need it, and the classical infinite-count-rate limit does not — an instrument-dependent difference, since some formats carry a count rate in their header and some do not (**User-Manual** §5).

Only *ratios* of σ_k reach the fit, so no absolute noise scale is needed — but the run duration T_{run} does not cancel from the exact form: $1/(M-k) = \Delta t_k/(T_{\text{run}} - \tau_k)$ keeps it in the relative shape unless the run is long against the last lag. Which files state the run duration of the correlogram is the **User-Manual** §5's. Both passes run under every weighting in the same scaled parameter units [37, Sec. 6], Eq. (6.3), p. 112, [38, Sec. 7.1], pp. 155–157, so two weightings differ in the weight alone.

In the program: a stated per-channel sample time is used as stated, and one that does not divide its lag a whole number of times is refused, the fit falling back to the unweighted form (§3.2.11) with a note naming the channel. Without a stated one, $\Delta t_k = \tau_k/\ell_k$, with ℓ_k the ratio of the lag to its step to the next channel rounded to the nearest whole number — ties round up, on every channel alike, and the ratio is at least 1, so a lag shorter than its step takes its own lag; such a sample time is a quantity the program **inferred**, and every scalar the fit produced says so in its export's Comments. With a committed count rate the full form is used (recorded `biganzoli-full`), without one the classical limit (`biganzoli-classical`). A run duration stated in the file is used, and one stated nowhere or ambiguously takes the $T_{\text{run}} \to \infty$ form — read from the header or honestly absent, never inferred, so its form is recorded on the result and travels no further. The result records which sample time, which count-rate form and which run-duration form it was evaluated on.

3.2.10 The four named divergences, and where the exact form matters

The form is the canonical one; four practicality divergences are named:

1. The weight is diagonal — no whitening: the same source restricts its covariance to channels of one sample time ([28], pp. 29381 and 29387), and it names the variance as the form "more commonly used in the data fitting procedures" (p. 29387), so whitening across a multi-tau axis has no in-library source.
2. The classical limit of Eq. 40, its two shot-noise groups dropped, standing in for the full form when no count rate is committed (§3.2.9).
3. The $T_{\text{run}} \to \infty$ form standing in for a missing run duration (§3.2.9).
4. At large Θ_k — a slow sample's coarsest blocks — Eq. 40 is evaluated in its asymptotic form, which agrees with it to working precision at the switch and stays finite where the hyperbolic terms do not.

The equation is defined on integer k and, near $k = 1$ at large Θ_k, is not a variance anywhere else, which is why a lag index is held to an integer. One premise is disclosed rather than chosen: the equation assumes symmetric normalization (their Eqs. (2)–(3), p. 29376), and the normalization an instrument used is not something a file states.

Where the exact form matters is set by a yardstick, not by the source: the closed form's condition $\Theta_k \ll 1$ carries no number [28, p. 29378], so this guide borrows the 0.1 at which [27, Sec. 3.1], p. 300 bounds a *different* approximation, the triangular-averaging amplitude distortion. It is a reading yardstick only: no switch in the program depends on it. The two variances weight the far tail in opposite directions, so the channels the closed form discounts are the ones the exact form leans on, and on a real multi-tau axis — where a slow sample decays across a section's long sampling time — a large share of a fit's weight sits on channels past that yardstick, which is the reason for the exact form.

The check of the weight itself is whiteness: a weight that matches the noise leaves the weighted residuals (residual $\div\ \sigma_k$) of an in-domain fit white, as in the whitened residuals of [32, Fig. 1], p. 4089.

In the program: the weight is diagonal; the asymptotic form takes over above $\Theta_k = 30$; a lag index within a rounding of an integer is evaluated at that integer, and any other is refused, with §3.2.11's fallback; the count-rate and run-duration forms are §3.2.9's rule.

3.2.11 The other two weightings, and the fallback

The second weighting is the instrument's own reported per-channel standard deviations, the route of [32, p. 4089] ("weights were calculated from standard deviations provided by the ALV-Laser software"); no file this program reads carries them, and the synthetic generator's noise model supplies its own.

The third is the **unweighted** form: $\sigma_k \equiv 1$, plain least squares [39] on the method's own residual — $g_2 - 1$ here, $\ln(g_2 - 1)$ on the Linear fit. It needs no equation of its own. It switches off the prescription each method would otherwise apply: the Biganzoli–Ferri variance of [28], Eq. (9), p. 29378 on this method, and on the Linear fit the weight ISO 22412 prescribes for its logarithm (§3.2.12). It is not a candidate for the default, because it is the absence of a noise model, and no source licenses it on the multi-tau data this program reads. As an opt-in it has three honest uses:

- comparing like-for-like with a result fitted without a noise model (the role the classical-OLS covariance plays for a ±, §6.1.1);

- a sensitivity readout, where the gap between the two fits on one measurement shows how much the noise model drives R_h and the PDI on *that* measurement;
- a measurement whose weight inputs the program itself labels as stand-ins — an axis-derived sample time, or no committed count rate.

What it costs is not ranked: real data carry no ground truth, and a score drawn under one noise model cannot rank "no model" against it. The same form is also the fallback when Eq. 40 refuses its inputs, for the reason §3.1 gives for flagging rather than withholding.

In the program: the instrument weighting runs only on data that state per-channel standard deviations; the unweighted form is a labelled opt-in, chosen in **Settings ▶ Cumulant residual weighting** for either cumulant method, and is the fallback when the Biganzoli–Ferri weight refuses its inputs — a decay per sample time, count rate, run duration or lag index it cannot use, or a non-finite SD: the fit records the unweighted form, flags the size and names the refused input in its note; too few live channels, or a first pass that did not fit, still withhold the values, since no fit can run; in a Γ vs q^2 or D vs c fit a point falls back on its own, and the result and its export name the points that did.

3.2.12 Which weight is the default

The Biganzoli–Ferri variance is the canonical variance of a photon-correlation estimate, in its exact-in-Γ form (benchmarked against: synthetic correlograms of known moments; the real replicate groups of §2.3 score the full form, not this one). An amplitude weight, evaluated on the noisy observation it weights and printed by no source, under-reports μ_2 inside the method's own domain (an attenuation bias). The unweighted opt-in weights no channel by its own noisy value, so it carries no such bias, only the cost of ignoring a variance that is not flat. For the Linear fit, ISO 22412 prescribes a normalized weighting factor that "accounts for the nonlinear transformation of the raw data" into the logarithm, and prints no form for it [31], Annex A.1.2, p. 20. The form the program uses is its own derivation of that prescription: each squared residual of $\ln(g_2 - 1)$ weighted by $(g_2 - 1)^2$, the inverse of the logarithm's variance when the noise on $g_2 - 1$ is the same on every channel.

In the program: the Biganzoli–Ferri variance is the Nonlinear fit's default, and an amplitude weight is not offered for it; the Linear fit's is the program's form of ISO's weight, each retained point's squared residual weighted by the square of its own baseline-subtracted $g_2 - 1$ (the unweighted opt-in is §3.2.11's); whichever weighting ran is recorded on the result and named in the exports wherever it is *not* the default — which exports carry the record, and which surfaces do not, is the **User-Manual** §13.2 — and every place the program fits a cumulant (the DLS tab, the per-angle points behind Γ vs q^2, replicate averaging, the Cross-Sample size candidates and DDLS) uses the same weighting and the same committed count rate for a measurement.

3.3 Single, Double and Stretched Exponentials

Where the cumulant expansion describes a narrow distribution about one mean rate, three parametric models describe a correlogram whose shape is known: one discrete mode, two, or a broad continuum with a single characteristic time. Each is fitted to $g_2(\tau)-1$ through the Siegert relation with β floating, over the delay window of §3.1.1, in the pinned-offset form of Eq. 32 (§3.1.3), by unweighted least squares on $g_2(\tau)-1$ — a flagged convenience simplification: the per-channel variance of §2.3 that the cumulant fits weight by (Eq. 41) is not applied to these three models, so their residuals are equal-weighted across the axis, and a size from one of them is compared with a

cumulant's with that in mind.

In the program: the three exponential fits run unweighted; the weighting choice of §3.2.12 does not reach them.

3.3.1 Single exponential

Is the sample monodisperse? For a single population of one decay rate Γ the field correlation is a single exponential, so

$$g_2(\tau) - 1 = \beta\, e^{-2\Gamma\tau} \tag{42}$$

which is equivalent to the first cumulant for clean data. It is valid where one decay carries the whole correlogram — a monodisperse sample, or one whose polydispersity the correlogram cannot resolve — and the size it returns is the Stokes–Einstein radius of that one rate.

In the program: the fit carries the verdicts of §3.1.4's decay-completeness check and §3.1.5's size range with its result. It carries no ± (§6's single-correlogram rule); a replicate set gives one (§6.4).

3.3.2 Double exponential

Do two distinct populations relax on separate time scales — a polymer and a slow aggregate, say? Two discrete modes enter through the full Siegert relation,

$$g_2(\tau) - 1 = \beta \left[\, A_1\, e^{-\Gamma_1\tau} + (1 - A_1)\, e^{-\Gamma_2\tau} \,\right]^2 \tag{43}$$

where A_1 is the amplitude fraction of the faster mode. The model is identifiable only where the two rates are well separated — roughly a decade apart — and both decays lie inside the delay window; two modes closer than that trade amplitude and rate against each other, and a mode the window did not observe is an extrapolation, not a measurement.

In the program: the fit reports whether it converged, and each mode carries its own verdicts of §3.1.4 and §3.1.5. Across angles, the modes are the points of the mode map (§3.5.4). The fit carries no ± (§6's single-correlogram rule).

3.3.3 Stretched exponential (Kohlrausch–Williams–Watts)

Is the relaxation broadly distributed rather than discrete — glassy dynamics, or a continuum of modes? The Kohlrausch–Williams–Watts form [34] stretches a single exponential with an exponent $s \in (0,\, 1]$, where $s = 1$ recovers §3.3.1:

$$g_2(\tau) - 1 = \beta\, \exp[-2\,(\tau/\tau_\mathrm{c})^{\,s}] \tag{44}$$

where τ_c is the characteristic time. The mean relaxation time is the first moment of the relaxation-time distribution the stretched exponential implies, which [40] derive from the decay function alone (their Eq. (19), p. 3350, with τ_ww and β_ww for this guide's τ_c and s):

$$\langle\tau\rangle = \frac{\tau_\mathrm{c}}{s}\,\Gamma_\mathrm{E}\!\left(\frac{1}{s}\right) \tag{45}$$

with $\Gamma_E(1/s)$ Euler's gamma function (the subscript keeps it apart from a decay rate). The model describes a distribution's *envelope*, not its modes, so it is the model to prefer over the double exponential when the correlogram is broad but not bimodal.

In the program: both τ_c and $\langle\tau\rangle$ are reported, each with its hydrodynamic radius; the two carry §3.1.4's one verdict, and each size its own verdict of §3.1.5. They carry no ± (§6's single-correlogram rule).

3.4 Distribution Methods

These recover a full distribution of decay rates without assuming a functional form. §3.4.1 states the inversion every method here solves, §3.4.2 the residual weight it runs under and §3.4.3 that weight's scale anchor and fallback; §3.4.4 is the unregularized NNLS, §3.4.5 CONTIN with its Tikhonov penalty and the three ways the program chooses its strength — robust GCV (§3.4.7, with classical GCV as its limiting case, §3.4.6), the L-curve corner (§3.4.8) and Provencher's F-test (§3.4.9) — and §3.4.10 the single-mode lognormal form.

> **Important.** For all distribution methods: in the multi-modal inversions (NNLS and CONTIN), fine structure — small secondary peaks especially — is sensitive to noise and the regularization choice and is not always reproducible, so a ± across repeated measurements of it is a reproducibility spread, not a calibrated error. Treat the main peaks as reliable; cross-check width against the cumulant PDI and the parametric fits. (Lognormal sidesteps this by assuming a single peak, at the cost of not resolving more than one.)

3.4.1 The inversion

The field autocorrelation is a weighted sum of exponentials, so through the Siegert relation the measured correlogram is

$$g_2(\tau) - 1 = \beta \left[\int G(\Gamma)\, e^{-\Gamma\tau}\, \mathrm{d}\Gamma \right]^2 , \tag{46}$$

discretized on a log-spaced grid of hydrodynamic radii. The program inverts the field autocorrelation g_1 directly [7, Sec. 8.11]: the single-Γ kernel is $A[i,j] = \exp(-\Gamma_j\ \tau_i)$ and the data vector is g_1 recovered from the Siegert relation, $\mathbf{y} = \mathrm{sign}(u)\sqrt{|u|}$ with $u = (g_2{-}1{-}\text{baseline})/\beta$ — a sign-preserving square root, so long-delay noise stays zero-mean rather than rectifying into a positive pedestal that a non-negative fit would read as spurious large-R_h weight. The reconstruction back to measured space is $g_2{-}1 = \beta(\mathbf{Ax})^2$ + baseline. Every method here needs β and a baseline.

In the program: the grid is log-spaced over 1–1000 nm by default and yours to set. β is estimated by extrapolating to zero delay and the baseline from the last quarter of the live channels inside the delay window; both can be overridden — the baseline by the Correlogram sub-tab's baseline region once you place it (§3.1.3) — and the values used are always reported. No distribution method carries a ± (§6's single-correlogram rule); what a replicate average of its peaks (§6.4) means is §3.4's caution.

3.4.2 The residual weight

The residual is statistically weighted: each channel's residual carries a weight w_k, written as the diagonal matrix $\mathbf{M} = \mathrm{diag}(w)$ in the equations below, and, as for the cumulant (§3.2.8), the

choice of weight moves every number the fit returns: here the recovered distribution, the selected regularization strength α (§3.4.5) and every array its sweep returns. Both forms available here are inverse-variance weights on the recovered g_1. Propagating a g_2-1 noise of per-channel SD σ_k through the square root gives $\mathrm{Var}(g_1) \approx {\sigma_k}^2/(4\beta^2 g_1(\tau_k)^2)$, so the per-lag weight is the delta-method inverse variance

$$w_k \;\propto\; \frac{g_1(\tau_k)^2}{\sigma_k^2}, \tag{47}$$

which down-weights the noise-dominated long-delay channels. The two forms differ in what they put in σ_k, and with it in where the factor $g_1(\tau_k)^2$ is read.

In the Biganzoli–Ferri variance (§3.2.8), σ_k is the per-channel SD of g_2-1 of Eq. 40, exact in Γ. Both factors of Eq. 47 are evaluated on a fitted model, neither on the noisy observation:

- σ_k at the $\bar{\Gamma}$ and β of an order-1 single-exponential fit under uniform weights — Koppel's prescription [30, p. 4818], the prose after his Eq. 26, and the cumulant arm's own evaluation point (Eq. 41), so both fitters read one variance at one point. It stays at a single exponential because Eq. 40 is a closed form for one.
- $g_1(\tau_k)$ as the fitted field correlation $(\mathbf{A}\mathbf{x}_0)_k$ of a first pass by the same method under the ${g_1}^2$ form below, at the same α selector or the same fixed α. That is where [41], prose prescription, p. 4674 evaluates an error model: an intermediate spectrum is computed "assuming that the statistical errors are independent and have the same variance", the correlation function is calculated from it, and the error model is evaluated there — one pass, because iterating further "hardly changed" the estimates [41, p. 4675]. The first pass's own result is not reported, and the inversion runs twice, so a fit under this form takes about twice as long.

Only the evaluation point is taken from that work, and the two divergences from it are named rather than implied: it fits g_2 directly and carries no square root, so the delta-method factor ${g_1}^2$ is this program's, from its g_1-space inversion; and it evaluates the variance on the spectrum as well, where the variance here is a closed form for one exponential and so stays at Koppel's point. The two shot-noise groups of Eq. 40 need a committed mean count rate; the classical limit needs none, so this form runs on every instrument.

The second form is the ${g_1}^2$ form. Taking $\mathrm{Var}(g_2-1)$ as constant across channels leaves σ_k a constant, so Eq. 47 collapses to $w \propto {g_1}^2 = |u|$ — the recovered g_1, squared, read off the observation itself. It needs nothing beyond the correlogram, and it is the form to select to reproduce a result reported under it. It is a flagged simplification of the Biganzoli–Ferri variance: its premise, a constant variance across channels, is not what a correlator produces (§2.3).

In the program: the Biganzoli–Ferri variance is the default and the ${g_1}^2$ form a labelled choice (Settings, and per measurement on the Distribution sub-tab); the form a fit ran under is recorded on the result and written into the per-fit and mode-map exports. The count-rate forms, and what the result records of them, follow §3.2.9's rule; a fit without a committed count rate also carries a passive note naming the absent input and what committing it would buy.

3.4.3 The scale anchor, the fallback and the diagonal simplification

This section reasons with the regularization strength α, its sweep and the selectors that pick from it, which §3.4.5–§3.4.7 introduce; a reader new to them may take those first.

The weight's overall scale cancels in the NNLS argmin, in the lognormal arm's objective, and in the fit at any fixed α. It does not cancel in the automatic α selection: rescaling w by a factor a_{w} reparametrizes the Tikhonov objective of §3.4.5 (Eq. 49) as $\alpha \to \alpha/\sqrt{a_{\mathrm{w}}}$, so the selected α slides by $\sqrt{a_{\mathrm{w}}}$ across the fixed sweep — and the per-measurement β enters the ${g_1}^2$ form as exactly such a rescale ($w \propto |u| = |g_2{-}1{-}\text{baseline}|/\beta$). The selectors re-minimize and track the shift, leaving the recovered distribution nearly unchanged unless the tracked optimum reaches an end of the sweep (§3.4.5); a user-supplied fixed α does not compensate. A rescale is therefore a slide of the sweep window along the solution path, and the Biganzoli–Ferri variance returns a *relative* shape in arbitrary units — so the Biganzoli–Ferri weight is multiplied by the one scalar that matches the squared Frobenius norm of the weighted design to the ${g_1}^2$ form's over the same channels:

$$a_{\mathrm{w}} = \frac{\mathrm{tr}(\mathbf{A}^\top \mathbf{M}_{\mathrm{ref}} \mathbf{A})}{\mathrm{tr}(\mathbf{A}^\top \mathbf{M} \mathbf{A})}, \qquad \mathrm{tr}(\mathbf{A}^\top \mathbf{M} \mathbf{A}) = \sum_k w_k \sum_j A_{kj}^2, \tag{48}$$

with $\mathbf{M}$ the Biganzoli–Ferri weight before anchoring and $\mathbf{M}_{\mathrm{ref}}$ the ${g_1}^2$ form's. The construction is the program's own, not something read out of a source. The effective degrees of freedom (Eq. 50) are computed from the weighted design $\mathbf{M}^{1/2}\mathbf{A}$ against $\alpha^2 \mathbf{L}^\top \mathbf{L}$, so matching that trace holds the NDF-against-α curve at the high-α end of the sweep and leaves the sweep bounds where they were derived. What no scalar can hold is the rest of the curve: the weight's *shape* — not its scale alone — sets the curve's slope in NDF per decade of α, so the two curves part toward the low-α end. For the same reason no anchor makes a hand-set α mean the same amount of smoothing under a weight of a different shape — one reason the ${g_1}^2$ form stays selectable.

A diagonal weight corrects the *precision map* and ignores the correlations between channels. The canonical method for a regularized fit under correlated errors is **generalized least squares**, with the inverse covariance in the fidelity term rather than a diagonal: the penalized objective of [42, Sec. 3.2], Eq. (8), p. 140, the GLS normal equations of [43, Sec. 4.3.2], Eq. (4.3.15), p. 163, and the covariance-weighted DLS inversion of [41], Eq. (15), p. 4675. The diagonal weight is a flagged convenience simplification on practicality grounds alone: it runs on every instrument whether or not a count rate exists, and it needs no matrix factorization. The premise §3.2.10 discloses — symmetric normalization, which no file states — holds here as well. The weight is a fit-only precision map and adds no ± (§3.4.1).

In the program: the anchor is applied on every fit under the Biganzoli–Ferri variance. Three things refuse the weight: an input the variance will not be evaluated on (a stated sample time that does not divide its lag, a lag at or past the run's end, or a non-finite result), an order-1 first pass that does not fit, and a first-pass spectrum that failed or is zero on every channel, which leaves the anchor nothing to match. A refused weight falls back to the ${g_1}^2$ form, records that form, and carries a passive note naming the refused input rather than a proxy, the distribution reported rather than withheld; a solver that fails after the weight was formed is an ordinary fit failure instead. The distribution methods carry no reliability flag, so the recorded form and note are the whole disclosure. The diagonal weight is the form every instrument can run.

3.4.4 NNLS

Non-negative least squares (NNLS) solves $\min \|\mathbf{M}^{1/2}(\mathbf{A}\mathbf{x} - \mathbf{y})\|^2$ with $\mathbf{x} \geq 0$ and no smoothing — Eq. 49 at $\alpha = 0$. It resolves well-separated peaks but produces spiky, noise-sensitive distributions.

In the program: NNLS is one of the three methods the Distribution sub-tab offers, on the same grid, weight and baseline as the others.

3.4.5 CONTIN and the choice of α

CONTIN [44] adds Tikhonov regularization [45]:

$$\min_{\mathbf{x}\geq 0} \|\mathbf{M}^{1/2}(\mathbf{A}\mathbf{x}-\mathbf{y})\|^2 + \alpha^2 \|\mathbf{L}\mathbf{x}\|^2 \tag{49}$$

with $\mathbf{L}$ the second-difference operator. The regularized fit's **effective degrees of freedom**, which every selector below reads, are the trace of the Tikhonov influence ("hat") matrix $\mathbf{A}_\alpha$ [46] — Provencher's NDF (his Eqs. 3.15–3.16) [45]:

$$\mathrm{NDF}(\alpha) = \mathrm{tr}[\mathbf{A}(\mathbf{A}^\top\mathbf{A} + \alpha^2\mathbf{L}^\top\mathbf{L})^{-1}\mathbf{A}^\top] = \sum_j \frac{s_j^2}{s_j^2+\alpha^2}, \tag{50}$$

with s_j the singular values of the weighted design. It is evaluated at every α from one generalized singular value decomposition of the pair $(\mathbf{M}^{1/2}\mathbf{A}, \mathbf{L})$ — the route CONTIN itself takes [45], Appendix, Eqs. A.15 and A.21–A.25, pp. 225–226 — through Hansen's general-form filter factors $f_i = \tilde{s}_i^2/(\tilde{s}_i^2+\alpha^2)$ (with $\tilde{s}_i$ the generalized singular values, Hansen's γ_i) [46, Sec. 4.2], p. 73, so the trace is monotone in α by construction. The constrained problem is solved as an **augmented non-negative least-squares** stack — αL is stacked beneath the weighted design and the whole system solved with NNLS, so non-negativity and smoothing are imposed together (rather than a closed-form Tikhonov solve). The regularization strength α is chosen by one of three selectors over the same sweep — robust GCV (§3.4.7), the L-curve corner (§3.4.8) or the F-test (§3.4.9). CONTIN is generally better for broad distributions and real data.

CONTIN itself sweeps α relative to the problem's largest singular value, from machine precision up to that value [47, Sec. 4.1.3]; the fixed sweep here is a flagged convenience simplification of that grid, and its ceiling diverges from CONTIN's: CONTIN trims the top of its sweep so the regularized solution keeps "approximately at least DFMIN degrees of freedom", and DFMIN is 2 by default [47, Sec. 4.1.3], [44], Table 4, p. 233, where the ceiling here is a fixed absolute α that can hold solutions with fewer. Reading the CONTIN manual's selection criterion (its Eqs 1–2, §4.1.6.5), the smallest-α solution is the F-test's *reference*, so CONTIN itself is unlikely to choose it — but the F-test here can pin to the floor on a fine R_h grid, so a pick at either edge is read with care, whichever selector made it:

- A pick pinned to the high-α end of the sweep means the selector wanted a larger α than the range allows, and the fit shown is the smoothest the range holds (raise the ceiling, or inspect the L-curve).
- A pick pinned to the low-α end means the selector wanted a smaller α than the range allows. At a floor low enough that regularization has not yet acted, the pick is then essentially the unregularized NNLS fit, and extra peaks in it may be inversion noise rather than populations — the "solutions with extra peaks" of an unregularized inversion [47, Sec. 3.2.3]; at a user-raised floor it is simply the least-smoothed answer the range permitted.

In the program: the sweep is a fixed α range you can set, its floor 10^{-4} by default; the three selectors are yours to choose, and the full sweep is returned for inspection; a fixed α can be supplied instead. Each result records which selector chose its α, and a pick at either end of the range is flagged with a caution rather than trusted, whichever selector made it.

3.4.6 Generalized cross-validation (GCV)

Generalized cross-validation ([48]; the robust GCV of §3.4.7 at $\gamma_r = 1$) estimates the leave-one-out prediction error without knowing the noise level. For the weighted Tikhonov fit, with $\mathrm{tr}(\mathbf{I} - \mathbf{A}_\alpha) = N_y - \mathrm{NDF}(\alpha)$,

$$\mathrm{GCV}(\alpha) = \frac{N_y\, V(\alpha)}{\left(N_y - \mathrm{NDF}(\alpha)\right)^2}, \qquad V(\alpha) = \|\mathbf{M}^{1/2}(\mathbf{A}\mathbf{x}_\alpha - \mathbf{y})\|^2, \tag{51}$$

where $V(\alpha)$ is the weighted residual, $\mathrm{NDF}(\alpha)$ the Tikhonov effective degrees of freedom (Eq. 50), and N_y the number of fitted lag channels (inside the delay window, after any skip); the chosen α is the minimizer. GCV is σ^2-free and asymptotically optimal for independent errors. Beside the correlated-error failure §3.4.7 explains, its documented failure mode is a flat minimum [46]; the edge cautions of §3.4.5 apply to it and to robust GCV alike.

In the program: classical GCV is the $\gamma_r = 1$ setting of the robust selector (§3.4.7).

3.4.7 Robust GCV

Classical GCV (Eq. 51) assumes the errors are independent, and under positively correlated errors it under-smooths: cross-validation and GCV "tend to underestimate smoothing parameters when data are correlated" [42, p. 141], the more so the stronger the correlation (their Table 1, p. 135). A correlogram's lag channels are correlated [27], and the Biganzoli–Ferri variance with a committed count rate is the weight that exposes it here, so classical GCV can split one broad mode into two where the correlation is strong. Robust GCV (Eq. 52) is the literature's repair for exactly this failure, and classical GCV stays one setting away.

Lukas's robust GCV, RGCV [49], Eq. 1.4, p. 521, first proposed in [50], weights the solution's variance into the criterion, for a correlation that is not known: the selector minimises

$$\bar{V}(\alpha) = \left[\gamma_r + (1 - \gamma_r)\, t_2(\alpha)\right] \mathrm{GCV}(\alpha), \qquad t_2 = \frac{\mathrm{tr}\, \mathbf{A}_\alpha^2}{N_y}, \tag{52}$$

where $\mathbf{A}_\alpha$ is the influence matrix of §3.4.5, t_2 the normalized trace of its square (Lukas's μ_2, a different quantity from the second cumulant) and $\gamma_r \in (0, 1]$ the robustness parameter (Lukas's γ): $\gamma_r = 1$ is classical GCV (Eq. 51; "for $\gamma = 1$, the RGCV method is just GCV" [51, Sec. 4.15.2], p. 1829), and the smaller γ_r the more weight the solution's variance carries, so the selector smooths more. t_2 is evaluated exactly from the same decomposition that gives NDF (Eq. 50) — the sum of the squared filter factors over N_y. Each of its sources uses $\gamma_r = 0.1$ [49, p. 543], [50, pp. 1885–1886], [51], Eq. 33, p. 1829. Because t_2 is a trace divided by N_y it is dimensionless: no scaling of the penalty multiplier (Lukas's λ, this guide's α^2) enters it, so the printed value needs no conversion to this problem, and its verdict does not depend on the R_h grid. Being divided by N_y, the variance term's weight does depend, modestly, on the number of fitted channels — the delay window, skipped channels and the instrument's layout. On Lukas's own correlated-error test problem, at a first-order autoregressive (AR(1)) correlation of 0.4 between neighboring errors, the median inefficiency of the α it picks is 1.32 against classical GCV's 8.11 [49], Table 1, p. 545; at a correlation of 0.8 it rises to 3.43, so its protection weakens as the correlation grows.

Robust GCV is the canonical method for errors correlated in an unknown way: its robustness parameter is printed, and that parameter needs no normalization of its own. Lukas also defines a

strong robust form, R_1GCV [49], Eq. 1.5, p. 521, whose robustness parameter carries the scale of the penalty multiplier; its two published values, from [52, Sec. 5] and [51], Eq. 34, p. 1831, belong to two different normalizations [52], Eq. 4.13, p. 11, neither of which this problem can reproduce, which is why the program does not offer the strong form. Smoothing more has a price that grows with the noise: on a noisy sample expected to be multimodal, a run at $\gamma_r = 1$ is the comparison to make, since the robust selectors can return two modes as one where classical GCV still separates them.

In the program: robust GCV at $\gamma_r = 0.1$ is the default selector; γ_r is set per measurement on the Distribution sub-tab, seeded from Settings, and recorded with the result and in the export beside the selector; the strong form R_1GCV is not offered.

3.4.8 The L-curve corner

The L-curve corner ([46]; located on the sampled sweep by the adaptive pruning algorithm of [53]) is read off the trade-off curve of the weighted residual against the **size of the distribution**, $(\log\|\mathbf{M}^{1/2}(\mathbf{A}\mathbf{x}_\alpha - \mathbf{y})\|,\ \log\|\mathbf{x}_\alpha\|)$, as the α at the corner of the "L", defined as the point of **maximum curvature** [46], Eq. 7.24:

$$\kappa(\alpha) = \frac{\zeta'\Lambda'' - \zeta''\Lambda'}{\left((\zeta')^2 + (\Lambda')^2\right)^{3/2}}, \qquad \zeta = \log\|\mathbf{M}^{1/2}(\mathbf{A}\mathbf{x}_\alpha - \mathbf{y})\|, \quad \Lambda = \log\|\mathbf{x}_\alpha\|, \tag{53}$$

primes denoting derivatives along the curve (Hansen writes the size axis η; Λ here keeps η for the viscosity). Below the corner (small α) the fit is noise-dominated and the size of the distribution grows fast for little gain in fit; above it (large α) the fit degrades while the size hardly changes. Log axes are essential: they are what make the two legs distinguishable [46, Sec. 7.5.1].

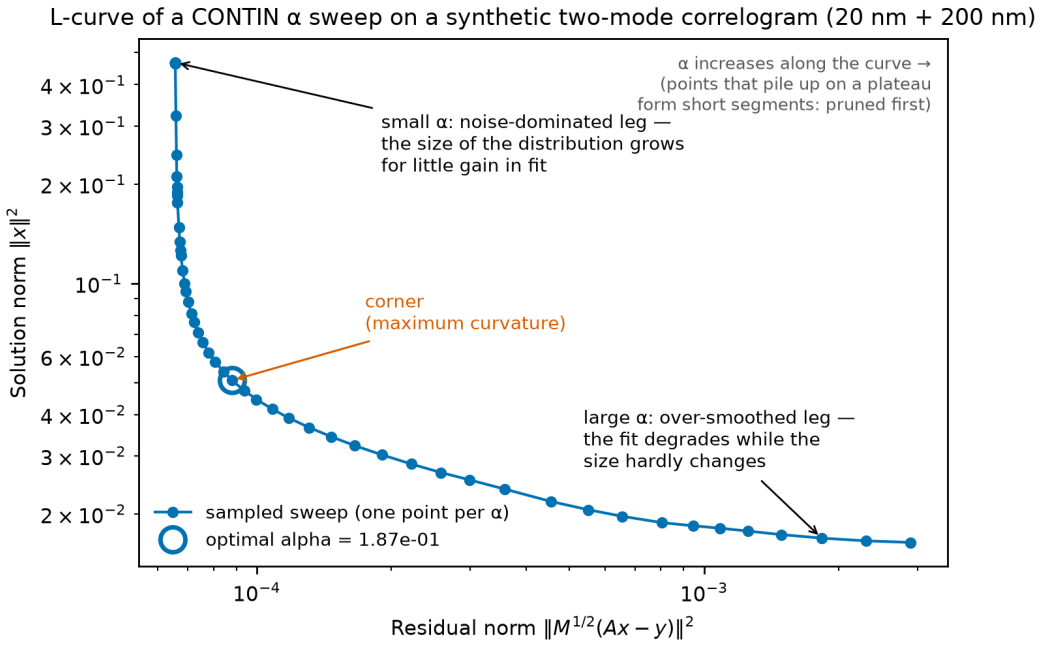


Figure 8: The L-curve of a CONTIN α sweep on a synthetic two-mode correlogram (20 nm and 200 nm, intensity fractions 0.6 and 0.4): the size of the distribution against the weighted residual, one point per α, on log axes. The program's axes plot the two norms squared, which on log axes puts the corner where Eq. 53 does. The vertical leg is the noise-dominated regime and the horizontal leg the over-smoothed one; the ring marks the corner the adaptive pruning algorithm picks, the point of maximum curvature, and the points that pile up along each leg form the short segments pruning drops first.

The solution axis is a benchmarked deviation. Hansen's L-curve for a general-form penalty plots the penalty's own seminorm, $\|\mathbf{L}\mathbf{x}_\alpha\|$ [46, Sec. 7.5]. On this kernel that axis has no floor: as α grows the non-negative distribution collapses toward the constant-plus-linear null space of the second-difference operator, which the penalty leaves untouched [54, p. 175], so $\log\|\mathbf{L}\mathbf{x}_\alpha\|$ keeps falling as fast as the residual rises, the curve never turns horizontal, and every corner rule prefers an over-regularized bump that merges two resolvable modes (benchmarked against: synthetic two-mode cases). The 2-norm of a non-negative distribution whose amplitude the data fix is bounded below, so its curve has the horizontal leg the corner needs. Any norm of the solution vector is an admissible size measure for the L-curve [55, p. 1495]; the locator's own source defines the discrete L-curve it searches on exactly this axis [53], Eq. 1; and the DLS literature plots it under CONTIN's second-derivative penalty [56, Sec. II.A], [57], Eq. 15, [58].

The curve is known only at the sweep's points, so the corner is located by the adaptive pruning algorithm [53, Fig. 2]: pruned copies of the curve keeping its 5, 10, 20, … longest segments are each searched for a corner by two routines — the point whose adjacent segments turn most nearly through a right angle, and the point nearest the meeting of the most horizontal and the most vertical segment — and the candidates are resolved by taking the last one before the curve turns vertical. It has no tuned parameter, and pruning is what makes the pick insensitive to the sweep's extent: the points that pile up on the two plateaus form short segments and are dropped first. The cubic-spline corner-finder of [55] is not used: its authors' later verdict is that it "has an undesired

tendency to track the unwanted local corners" [53, p. 485]. Two details the paper leaves open — the fallback when no candidate passes both conditions of the final test, and the order in which the second routine tries segment pairs — are the program's own choices.

A sweep window need not hold both legs of the L: with no horizontal leg every solution in it is under-smoothed, with no vertical leg every one is over-smoothed, and a sweep with fewer than three distinct points, or on which no pruned copy shows a convex corner, has no corner to find. Which α such a window reports is a choice made here rather than in the source, and widening the range past the end its note names is the remedy. Hansen's own caveat stands, that "the heuristic L-curve algorithm can fail no matter how it is implemented" [53, Sec. 5]; the no-corner note is where such a failure is disclosed rather than hidden. The L-curve's documented weaknesses run both ways: it under-smooths very smooth solutions at low noise, where it is "likely to fail" [46, p. 190], [54, Sec. 5.3], and it consistently over-smoothed Hansen's own second-derivative test problem [46, Sec. 7.7.1].

In the program: a window with no horizontal leg reports the **ceiling** note and returns the largest α of the range, one with no vertical leg the **floor** note and the smallest, and a sweep with no corner the **no corner** note and the least-regularized α, so the plot's ring marks the α the distribution was computed at.

3.4.9 The Provencher F-test

The third selector is Provencher's F-test ("probability to reject"; [45]; [56]). Among the sweep of increasingly-smoothed solutions, each is compared to the least-squares reference by the F-statistic

$$F(\alpha) = \frac{V(\alpha) - V_0}{V_0} \frac{N_y - \mathrm{NDF}_0}{\mathrm{NDF}_0}, \qquad V(\alpha) = \|\mathbf{M}^{1/2}(\mathbf{A}\mathbf{x}_\alpha - \mathbf{y})\|^2, \tag{54}$$

where V_0 is the minimum residual over the sweep — at the reference α_0, the least-regularized (least-squares) end — N_y the number of fitted lag channels (§3.4.6), and $\mathrm{NDF}_0 = \mathrm{NDF}(\alpha_0)$ the effective degrees of freedom at that reference, Eq. 50 of §3.4.5 — Provencher's NDF (his Eqs. 3.15–3.16), the trace of the influence matrix, *not* the raw grid size. Crucially NDF_0 is a single fixed value (evaluated at α_0), used in both the scaling factor and the F-distribution — not the per-α NDF. For an ill-posed Laplace kernel NDF is small (typically a handful to a couple dozen for a correlogram). $F(\alpha)$ is F-distributed with $(\mathrm{NDF}_0, N_y - \mathrm{NDF}_0)$ degrees of freedom, so its cumulative value $P_{\mathrm{rej}}(\alpha) = \mathrm{CDF}_F(F(\alpha); \mathrm{NDF}_0, N_y - \mathrm{NDF}_0)$ is the "probability to reject": ~0 where the regularizer barely changes the fit (under-smoothed) and ~1 where it changes it a lot. The chosen α is the one whose P_{rej} is closest to a chosen level; a higher level tolerates more fit degradation and selects a smoother, more-parsimonious distribution, a lower level a rougher one.

The F-test also carries the two checks the CONTIN manual prescribes for the solution table [47, Sec. 3.6.5], points 1 and 3:

1. The reference solution's residual must "agree with OBJ. FCTN. in the first few figures": its penalty $\alpha_0^2\|\mathbf{L}\mathbf{x}_0\|^2$ must be negligible beside its residual $V(\alpha_0)$; otherwise the sweep floor is not small enough to serve as a reference.
2. The chosen solution's probability to reject must lie "between about 0.05 and 0.95"; otherwise the sweep does not bracket the test's transition.

The second check reads on the chosen solution, as the CONTIN manual states it. The F-test picks the α whose probability to reject is nearest the requested level, so a level at or beyond a band edge

places the pick outside the band by the user's own construction. At CONTIN's own level of 0.5 the check is exactly "some sweep point's probability to reject lies inside the band"; at any other level it is not, which is why the chosen solution is consulted. The F-test treats the N_y residuals as independent, but a correlogram's lag channels are correlated (§6, the single-correlogram rule), so N_y overstates the independent information and the level is a guide, not an exact test. It is offered for comparison with legacy CONTIN output. No single criterion is best in every case [46], which is why the program offers all three (§3.4.5).

In the program: the level is 0.5 by default and yours to set; the first check's tolerance is a relative 10^{-3} by default, a documented parameter; the second check applies only when the requested level lies strictly inside the 0.05–0.95 band, and a level at or beyond a band edge is recorded on the result rather than flagged; either check's failure flags the result with a note naming which failed, and the distribution is still reported.

3.4.10 Lognormal

The lognormal method is a *single-mode parametric* distribution [59]: it assumes the intensity-weighted size distribution is lognormal in R_h and fits its median and log-width to the same $\mathbf{Ax} = \mathbf{y}$ model. The discrete weight is the lognormal density sampled at each node of the log-spaced grid — a flagged convenience simplification of integrating the density over each bin, which is the exact discretization:

$$w_i \propto \exp\left[-\frac{(\ln R_{h,i} - \mu)^2}{2\sigma^2}\right], \qquad \mu = \ln(\text{median } R_h) \tag{55}$$

with σ the log-width (polydispersity); the $1/R_h$ of the lognormal pdf cancels the log-grid spacing, so the discrete weight carries no extra Jacobian. It cannot resolve multiple modes, but it is robust to noise and always returns a smooth, single-peaked distribution — a good default for a known-unimodal sample.

In the program: a lognormal fit that fails to converge says so, and no peaks, Summary rows or sources for ρ (§5.1) are offered from it.

3.5 Combining Measurements

Two fits pool several measurements of one sample — §3.5.1 the decay rates across angles and §3.5.2 the diffusion coefficients across concentrations — and §3.5.3 says what each pooled point is, what weight the pooled lines run under and what a pooled fit refuses. §3.5.4 is the mode map, which pools every extracted mode without pairing them.

3.5.1 Γ vs q^2: multiple angles

For pure translational diffusion the decay rate is proportional to q^2:

$$\bar{\Gamma} = D\,q^2 \tag{56}$$

A straight line through the origin gives D (and R_h). Three linearity diagnostics test whether the motion is genuinely diffusive: the fit R^2; the intercept of an unconstrained fit (should be ≈ 0 — a nonzero value indicates a non-diffusive contribution); and the trend of $D_{app} = \Gamma/q^2$ with q^2 (should be flat — an upturn signals internal/segmental modes).

In the program: the line is fitted through the origin, unweighted (§3.5.3), over the angles you tick; the three diagnostics are reported and summarized as one diffusive / non-diffusive label, and the pooled result's reliability flag follows §3.5.3's rule. The ± on D is the through-origin regression SE of §6.1, statistical only, and passes to R_h by Eq. 116.

3.5.2 D vs c: concentration extrapolation

Extrapolating the apparent diffusion coefficient to zero concentration removes inter-particle interactions:

$$D_{app}(c) = D_0\,(1 + k_D\,c) \tag{57}$$

The intercept D_0 gives the infinite-dilution $R_{h,0}$; the slope gives the diffusion interaction parameter k_D (positive in good solvents, negative in poor ones).

In the program: the line is fitted unweighted (§3.5.3) over the concentrations you tick; a concentration whose fit returns no usable decay rate is dropped by §3.5.3's rule, and fewer than two remaining concentrations withhold D_0, $R_{h,0}$ and k_D. The ± on the intercept and slope is the regression SE of §6.1, statistical only.

3.5.3 The per-point fits

Each point's Γ in both fits is a cumulant fit of that correlogram, so the table of points and the line are one fit of the sample.

Each point of a pooled line comes from one correlogram, which carries no ± to weight it by (§6's single-correlogram rule), so the points' variances are unknown and need not be equal — the case §6.1's pairing of unweighted least squares with the HC3 covariance is for.

A set that mixes polarizations (VV with VH) cannot be pooled: a sample may hold both — that pairing is what DDLS is for (§3.6) — but $\Gamma_{VH} = \Gamma_{VV} + 6D_r$ carries a rotational term, so a line through both has the wrong slope and an intercept that is not physical.

In the program: every pooled line in this chapter — Γ vs q^2, D vs c, a mode family's fits (§3.5.4) and the Γ_{VV} line of DDLS (§3.6.1) — is fitted by unweighted least squares, its ± from the HC3 covariance of §6.1. Each point is fitted at the Settings cumulant order, method, cutoff and residual weighting, the settings every fit without a per-measurement control of its own reads. When the **Use the Correlogram tab's delay window and baseline** box is ticked, each point is fitted over the Correlogram sub-tab's delay window and baseline, read for that correlogram (§3.1.3). A passive note a point's own fit carries is relayed to the pooled result once, with the points that carried it. The export records the weighting each point ran under and names any point whose sample time was derived from the delay axis. A concentration dropped for returning no usable decay rate (a non-positive Γ, which a weak scatterer on a noisy baseline can produce) is named beside the result, on the plot and in the export. A set that mixes polarizations is refused, naming the tags found. The Γ vs q^2 and D vs c results carry a reliability flag of their own that is false when any per-point fit they rest on was flagged (§3.1.4) or when the pooled R_h itself falls outside the size range (§3.1.5), so a flagged single-angle value is never laundered into an unflagged pooled one. The **User-Manual** §8.4 and §8.6 describe the plot marks and the export cells.

3.5.4 Mode-resolved dynamics: the mode map

When a sample relaxes on more than one time scale — the fast + slow two-mode structure of salt-free polyelectrolytes [60], or the slow modes of neutral semidilute θ-solutions [61] — a single pooled Γ per angle averages physically distinct processes. The **Mode Map** (DLS tab) instead plots *every* extracted mode of *every* measurement in (q^2, Γ) space — the two-mode double-exponential's modes, or a CONTIN/NNLS distribution's resolved peaks. Pairing those modes into families by rank breaks exactly where the analysis is interesting: where a diffusive mode's Γ sweeps past a q-independent mode's across the angle range. Each family is then fitted separately with Eq. 56 (through-origin D, the same linearity diagnostics), and with the slow-mode deviation form of Sedlák & Amis [60]:

$$\tau_s^{-1} = a\, q^2 + b\, q^4 \tag{58}$$

where τ_s is the mode's relaxation time ($\tau_s{}^{-1}$ its decay rate), a the mode's $q \to 0$ apparent diffusion coefficient and b the coefficient of the q^4 term, which captures the apparent (non-diffusive) character of a slow mode.

The deviation is the angular fingerprint that Burchard's first-cumulant expansion, $\Gamma/q^2 = D_z(1 + C_B R_g{}^2 q^2 - \ldots)$, assigns to internal structure, with D_z the z-average diffusion coefficient and his $\langle S^2 \rangle_z$ equal to $R_g{}^2$. His Table 4 puts C_B at $13/75 \approx 0.17$ for a monodisperse coil, $1/5$ at $M_w/M_n = 2$ (M_n the number-average molar mass), and down to $2/15 \approx 0.13$ in its branched and star limits [62].

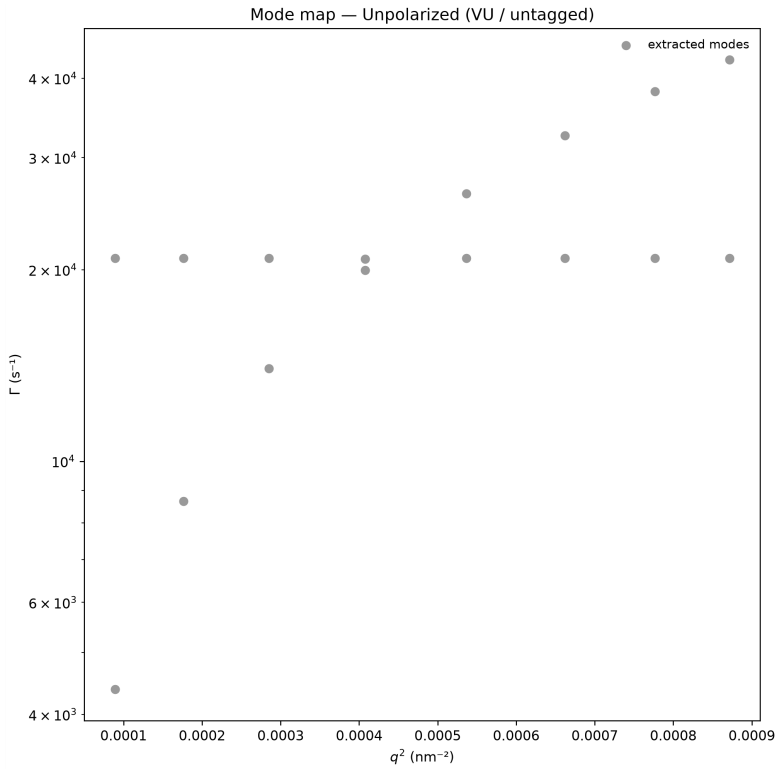


Figure 9: The Mode Map sub-tab on the two-mode synthetic set — every extracted mode of every angle in (q^2, Γ) space, before any family is picked: the fast mode's Γ rises with q^2 and sweeps past the slow mode's constant Γ near $q^2 = 4 \times 10^{-4}$ nm^{-2}, where a rank-based pairing would swap the two. That crossing is why the families are picked by hand.

Every family's R_h is computed by Stokes–Einstein (Eq. 6), and several diagnostics qualify it:

- For a slow mode that reflects collective or structural relaxation rather than a particle, Stokes–Einstein is not strictly applicable [63], so the number is an apparent size for comparison.
- A mode carrying less than ~2 % of the measurement's amplitude is below the practical resolvability floor of the decomposition [64].
- The fast mode of a salt-free polyelectrolyte is the coupled polyion–counterion mode (the classic treatment: [7] §13.7; a modern account: [65] §7.6–7.7), whose amplitude is expected q-flat.
- The map's **amplitude report** plots A_f/A_s against $\sin^2(\theta/2)$ — the axis of Sedlák & Amis's Fig. 4 [60] — for the two role-designated families, with an angle-flatness diagnostic whose relative-span threshold is a flagged convenience simplification with no source: no printed criterion exists in the sources cited here.

A family's fit combines independent angles, which a single correlogram's modes do not.

In the program: no automatic association is performed — you pick the families on the Mode Map; each family's lines are fitted unweighted (§3.5.3); each family's R_h is reported flagged apparent and mode-resolved, never gated away; a mode below that resolvability floor gets a note that its decay rate and amplitude are imprecise, not a reliability flag; whether a family passes through the origin is a label, never an eligibility test, with its stated tolerances ($R^2 \geq 0.98$, |intercept|/mean $\Gamma \leq 0.1$ by default); the angle-flatness diagnostic (relative span, default 20 %) gates a note, never a verdict

a result depends on; and a recorded family becomes the **mode set** the Sedlák partition of §4.2 reads. A mode's amplitude and its per-measurement Γ carry no $\pm$ (§6's single-correlogram rule); a family's D, R_h, a and b carry the regression SE of §6.1, statistical only; a single-sample dynamic $\pm$ comes from replicate averaging (§6.4).

3.6 Depolarized DLS: Rotational Diffusion and Shape

Depolarized DLS pairs a polarized (VV) with a depolarized (VH) correlogram at each angle to separate translational from rotational diffusion. §3.6.1 gives the two decay rates and what each channel supplies, §3.6.2 the validity limit and the $\pm$, §3.6.3 the case in which the depolarized channel decays no faster than the polarized one; §3.6.4–§3.6.6 turn the two coefficients into the dimensions of an assumed shape — a sphere, a rigid rod, an ellipsoid of revolution — and §3.6.7 gives those dimensions' $\pm$.

The VH signal is intrinsically weak (the matrix/solvent scatters polarized), so depolarized acquisition needs a clean analyzer and long counting; flexible polymer coils barely depolarize, so DDLS targets rods and anisotropic particles [66]. How to tag each correlogram's polarization, how same-angle repeats are averaged and held to the repeat-set check of §6.4, and how to pre-average weak VH correlograms are described in the **User-Manual** §8.7.

The translational and rotational diffusion coefficients, D_t and D_r (§3.6.1), are **shape-free observables**; turning them into particle *dimensions* requires assuming a geometry, so the dimensions of §3.6.4–§3.6.6 are **model-dependent** — dimensions of an *assumed* shape, never a direct measurement. Three models are offered — the sphere, the rigid rod and the ellipsoid of revolution — and read together they diagnose the shape. The program's rules for all three, including their flags, are stated once, in §3.6.7's marked line.

3.6.1 Rotational diffusion from the VV/VH pair

For an **optically anisotropic** scatterer (a rigid rod, or an anisotropic particle), the depolarized (VH) correlogram decays faster than the polarized (VV) one because molecular *reorientation* adds to the decay. In the small-qL regime (L the length of the rod or anisotropic particle, §3.6.5) the two field (g_1) decay rates are

$$\Gamma_{VV} = q^2 D_t, \qquad \Gamma_{VH} = q^2 D_t + 6 D_r, \tag{59}$$

where D_t is the translational and D_r the rotational diffusion coefficient [7], [67] (the dynamic-scattering-from-rotation framework originates with [68], which gives the translational and rotational broadening as separate limits). Subtracting isolates the rotation, and its inverse is the reorientation time:

$$D_r = \frac{\Gamma_{VH} - \Gamma_{VV}}{6}, \qquad \tau_{rot} = \frac{1}{6 D_r}. \tag{60}$$

Γ in Eqs. 59, 60 is the field rate (§2.2.1). Across angles the VV channel carries the whole translational signal, cleanly measured, while each angle's VH − VV difference carries the rotation. The $6D_r$ term is q-independent, so a plot of Γ_{VH} vs q^2 is a line of slope D_t and intercept $6D_r$ — the translational/rotational separation at a glance.

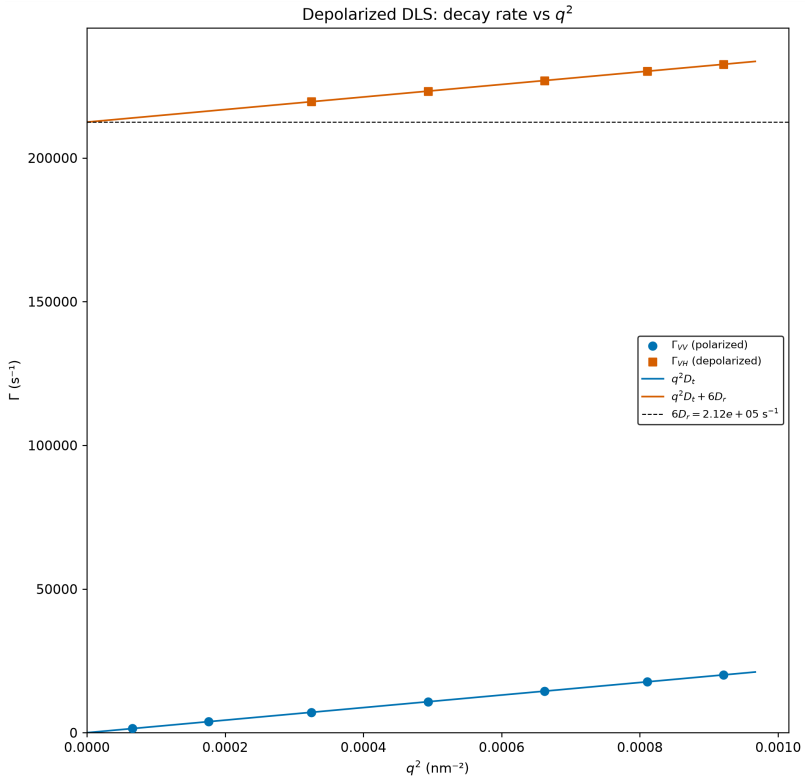


Figure 10: The **DDLS** sub-tab after a run on the synthetic rod's VV and VH sets at seven angles: Γ_{VV} against q^2 through the origin, slope D_t, and Γ_{VH} on the parallel line whose intercept is $6D_r$ (dashed) — the translational/rotational separation at a glance.

In the program: each correlogram is fitted for its Γ with the cumulant of §3.2; D_t is the through-origin slope of Γ_{VV} vs q^2, fitted unweighted (§3.5.3), and D_r the mean of the per-angle differences; the polarization tag of each correlogram is yours (**User-Manual** §8.7), and the two-letter convention is §1.6.1's.

3.6.2 Validity and uncertainty

Eq. 59 is a single exponential only for $qL \lesssim 3$ (above it, higher rotational modes and intramolecular interference appear; [67]).

A single angle gives one correlogram per channel; across angles the points are independent measurements.

In the program: with a rod length supplied, qL is evaluated per angle and the result is flagged past the limit; a single angle carries no ± (§6's single-correlogram rule), and its exported ± columns say why they are blank rather than keeping the multi-angle label; across angles D_t carries the through-origin SE of §6.1 and D_r the SD/$\sqrt{N}$ of the per-angle estimates, statistical only.

3.6.3 When $\Gamma_{VH} \leq \Gamma_{VV}$

The difference in Eq. 59 is then zero or negative, so D_r is unphysical for an anisotropic scatterer — check the VV/VH pairing, the signal-to-noise, or whether the particle is optically isotropic. A single bad angle can leave the *mean* D_r positive, and every value that is a *function* of D_r — the

mean, its ±, $\tau_{\mathrm{rot}} = 1/(6D_{\mathrm{r}})$ and every shape-model dimension, each inverted from the (D_{t}, D_{r}) pair — is in doubt with it, while the shape models still run and still report a length. D_{t}, the through-origin slope of Γ_{VV} vs q^2, is unaffected by VH noise or by an optically isotropic scatterer. That is not a claim of immunity: a swapped VV/VH pairing corrupts both channels at once, so separating the two families marks the number in doubt only for the causes a reader can act on.

In the program: the check runs per angle, so it fires even when the mean D_{r} is positive; its note names a swapped VV/VH pairing as the first cause and marks the D_{r} family and every dimension inverted from it, on the result and in the Comments cell of each such export column (**User-Manual** §8.7), and leaves the D_{t} family unmarked.

3.6.4 Shape models: the sphere

With one unknown, the rotational diffusion of a sphere of radius R is the Stokes–Einstein–Debye relation [66], whose inverse gives R:

$$D_{\mathrm{r}} = \frac{k_{\mathrm{B}}T}{8\pi\eta R^3}, \qquad R = \left(\frac{k_{\mathrm{B}}T}{8\pi\eta D_{\mathrm{r}}}\right)^{1/3}. \tag{61}$$

For a *true* sphere this equals the translational (Stokes) radius $R_{\mathrm{h}} = k_{\mathrm{B}}T/(6\pi\eta D_{\mathrm{t}})$, so the ratio $R(D_{\mathrm{r}})/R_{\mathrm{h}}$ tests the shape: ≈ 1 means a sphere is plausible; a ratio above 1 is the **rod signature** (rotation reports a larger effective size than translation, because a rod tumbles more slowly than a sphere of its translational radius), a sign that the sphere model does not apply.

3.6.5 The rigid rod

With two unknowns: for a cylinder of length L and diameter d (axial ratio $p = L/d$), the Tirado–García de la Torre [69] relations are

$$D_{\mathrm{t}} = \frac{k_{\mathrm{B}}T}{3\pi\eta L}\,(\ln p + \nu), \qquad D_{\mathrm{r}} = \frac{3k_{\mathrm{B}}T}{\pi\eta L^3}\,(\ln p + \delta_{\perp}), \tag{62}$$

$$\nu = 0.312 + \tfrac{0.565}{p} - \tfrac{0.100}{p^2}, \qquad \delta_{\perp} = -0.662 + \tfrac{0.917}{p} - \tfrac{0.050}{p^2}. \tag{63}$$

With two observables and two unknowns the rod is exactly determined. There is no closed-form inverse, so the program solves it numerically: at a trial p, D_{t} fixes L (Eq. 62, left); the predicted D_{r} is monotonic in p, so a one-dimensional root find recovers p, then L and $d = L/p$.

The end-effect polynomials (Eq. 63) are fitted only for $2 < p < 30$, and a pair of coefficients need not correspond to any rod at all.

3.6.6 The ellipsoid of revolution

For a prolate or oblate ellipsoid with polar semi-axis a and equatorial radius b (axial ratio $p = a/b$, so $p > 1$ is prolate, $p < 1$ oblate), both friction factors follow from Perrin's shape integral S.

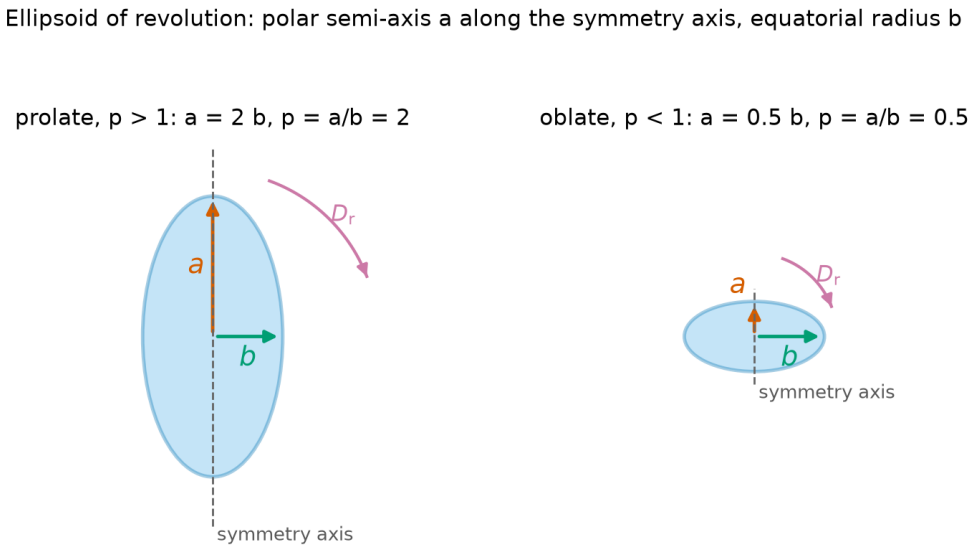


Figure 11: The ellipsoid of revolution: the polar semi-axis a lies along the symmetry axis and b is the equatorial radius, so $p = a/b > 1$ is prolate and $p < 1$ oblate; DDLS observes the end-over-end tumbling of the symmetry axis, D_r. §4.9 fits the same p from the angular form factor.

In the reciprocal axial ratio $\phi = b/a = 1/p$ (Perrin's and Koenig's ρ), the dimensionless product aS of the polar semi-axis and that integral is

$$aS = \begin{cases} 2\,(1-\phi^2)^{-1/2}\,\ln\dfrac{1+\sqrt{1-\phi^2}}{\phi}, & \phi < 1 \text{ (prolate)}, \\ 2\,(\phi^2-1)^{-1/2}\,\tan^{-1}\sqrt{\phi^2-1}, & \phi > 1 \text{ (oblate)}. \end{cases} \tag{64}$$

The translational coefficient is Perrin's friction factor as tabulated by Tanford [70]; the rotational coefficient — the end-over-end tumbling of the symmetry axis, which is what DDLS observes, written D_r as in §3.6.1 (Koenig's rotary friction coefficient C_2 about that axis) — is Koenig's corrected form [71] of Perrin's [72] expression:

$$D_t = \frac{k_B T}{6\pi\eta R_0}\cdot\frac{\phi^{2/3} aS}{2}, \qquad R_0 = (ab^2)^{1/3}; \tag{65}$$

$$D_r = \frac{k_B T}{C_2}, \qquad C_2 = \frac{4C_0\,(1-\phi^4)}{3\phi^2\,[\,aS\,(2-\phi^2)-2\,]}, \qquad C_0 = 8\pi\eta\,ab^2. \tag{66}$$

Both reduce to the sphere limits (Eq. 61 and Stokes–Einstein) as $p \to 1$. As with the rod, the 2-D inverse is reduced to a 1-D root find in p (at a trial p, D_t fixes the size, then D_r is matched). Unlike the rod, an ellipsoid inverse is not single-valued: a given (D_t, D_r) can be reproduced by both a prolate and an oblate ellipsoid, and the oblate branch is non-monotonic (a maximum near $p \approx 0.72$), so up to three ellipsoids fit, and these data alone cannot prefer one.

Koenig's factors are tabulated for $0.1 < p < 10$.

3.6.7 Uncertainty of the shape models

The sphere radius SE follows from D_r by the delta method ($R \propto D_\mathrm{r}^{-1/3} \Rightarrow \mathrm{SE}(R)/R = 1/3 \cdot \mathrm{SE}(D_\mathrm{r})/D_\mathrm{r}$). This is a single, smooth, monotonic power, so the delta method is accurate at realistic precision and degrades only for large $\mathrm{SE}(D_\mathrm{r})/D_\mathrm{r}$; it is Monte-Carlo-validated (§6) — sampling D_r and re-inverting matches the delta SE to ~1.5 % at a stressing 10 % spread.

The rod and ellipsoid inverses are nonlinear with no closed form, so their dimension SEs are propagated by Monte Carlo directly — sampling D_t, D_r from their SEs and inverting each (for the ellipsoid, each draw's solutions are matched to the base solution they belong to) — which is the sampling spread, with no linearization to validate. The rod ± is checked against the sampling SD of independent noise realizations (§6).

The inversion is solved over the model's whole domain (every $p > 1$ for the cylinder, every $p > 0$ for the ellipsoid), so a draw is lost only where no such shape reproduces it — a negative rate, or a D_r faster than the $p \to 1$ limit admits for that D_t — and not to a shape window. The census — draws reproduced of draws taken, and the seed — matters because the ± is the spread over exactly those draws.

The aspect ratio's distribution is heavy-tailed at large input uncertainty (a slightly slow D_r draw is a much longer, thinner rod), so its ± can be large next to the point estimate; L and d stay well behaved.

In the program: all three models are computed from the fit on screen after **Run** (the sphere needs the solvent viscosity); the sphere reports R, R_h and their ratio; a recovered p outside a model's tabulated range (the rod's $2 < p < 30$, Koenig's $0.1 < p < 10$) is flagged as an extrapolation; a pair of coefficients that no rod reproduces is reported as the rod model not fitting; every ellipsoid solution that fits is listed, none silently preferred, and the result is flagged degenerate when more than one does; the Monte Carlo census is reported beside every ±.

3.7 Weighted Size Distributions

This section consumes two results chapter 4 derives — the sphere form factor of §4.8 and the Mie amplitude of §4.10 — and states only which scattering law applies in which size regime.

The distribution methods of §3.4 return an **intensity-weighted** size distribution $w_\mathrm{I}(R)$: each bin's weight is proportional to how much light that size scatters. Because large particles scatter far more than small ones, the intensity view over-represents them. Re-expressing the distribution as **volume-, surface-, or number-weighted** requires an optical model to convert "how much this bin scatters" into "how many particles are there." Writing $N(R)$ for the number distribution and $I_1(R,\, q)$ for the scattering per particle, the intensity weight of a bin is

$$w_\mathrm{I}(R_i) \propto N(R_i)\, I_1(R_i, q) \;\Rightarrow\; N(R_i) \propto \frac{w_\mathrm{I}(R_i)}{I_1(R_i, q)}, \tag{67}$$

and the other weightings follow from the geometric moments of a sphere,

$$w_\mathrm{surface}(R_i) \propto N(R_i)\, R_i^2, \qquad w_\mathrm{volume/mass}(R_i) \propto N(R_i)\, R_i^3. \tag{68}$$

This inversion is well-posed only for homogeneous spheres, where the DLS-measured hydrodynamic radius equals the geometric radius ($R_\mathrm{h} = R$) and I_1 is a clean law. The scattering per particle

depends on the size regime:

The **Rayleigh** law, for a point scatterer, is valid for very small particles, $x = 2\pi nR/\lambda \lesssim 0.1$ (the size parameter, with n the solvent's refractive index and λ the vacuum wavelength):

$$I_1(R, q) \propto R^6. \tag{69}$$

The **Rayleigh–Debye–Gans (RDG)** sphere is valid at low contrast, $|m - 1| \ll 1$ and $2x|m - 1| \ll 1$, with $m = n_{\text{solute}}/n$ the relative refractive index [73]:

$$I_1(R, q) \propto V^2\, P(qR) \propto R^6\, P(qR), \qquad V = \tfrac{4}{3}\pi R^3, \tag{70}$$

with P the sphere form factor (Eq. 88). RDG reduces to the Rayleigh R^6 law as $qR \to 0$. The bulk optical-contrast factor — proportional to $(m^2 - 1)^2$ ($\approx (2(m - 1))^2$ in the RDG small-contrast limit) — is the same for every size bin (one material), so it cancels in Eq. 67 after normalization: m enters only the RDG validity gate, never the reweighting arithmetic itself.

The **full Mie** sphere is the *general* case (any x, any contrast — §4.10): the per-particle intensity is the full Mie result $I_1 \propto |S_1(x, m, \theta)|^2$ (Eq. 94). Unlike RDG, the contrast does not cancel size-independently, so each bin carries its own x, m dependence; m (from the solute refractive index) is required. Mie reduces to the RDG law as $m \to 1$ and to Rayleigh as $x \to 0$, so it is the top of the **Rayleigh → RDG → Mie** ladder. It is what makes the classic high-contrast case (polystyrene latex in water, $m \approx 1.19$) work where RDG and Rayleigh must refuse.

Each law holds only in its own regime, which a bin's size parameter $x = 2\pi nR/\lambda$, contrast $|m - 1|$ and product qR locate for Rayleigh and RDG, and its normalized angular intensity $P(\theta) = |S_1|^2/|S_1(0)|^2$ for Mie. The RDG conditions are Bohren and Huffman's [73], Eqs. (6.2)–(6.3), p. 159, $|m - 1| \ll 1$ and $kd|m - 1| \ll 1$ with d a characteristic dimension (the diameter here, so $kd = 2x$), and they state them only qualitatively. Rendering each "$\ll 1$" as a numeric bound is the program's own threshold, a flagged convenience simplification with no printed source. Mie has no upper contrast/size validity bound (it is the general case), but near a Mie form-factor minimum $N = w_{\text{I}}/I_1$ spikes on noise, exactly as it does past the RDG sphere's first minimum; a genuinely small particle ($P \approx 1$) is far from one.

A reweighted distribution derives from the ill-posed NNLS/CONTIN inversion (§6.5.2), and the division by I_1, which goes as R^6, amplifies small-R noise into large spurious small-particle populations; it is model-dependent, distinct from the measured intensity distribution it derives from. This is why ISO 22412:2017 §9.1 recommends against — and specifically deprecates — number distributions from DLS [31].

In the program: the reweighting offers sphere models only and refuses a non-spherical choice. Each bin is checked against default bounds — Rayleigh $x \leq 0.1$; RDG $|m - 1| \leq 0.1$, $2x|m - 1| \leq 0.1$ and qR below the sphere's first form-factor minimum 4.4934; Mie $P(\theta)$ above a documented floor (default 0.01), with a documented large-x convergence ceiling and an informational note where the cheaper RDG limit already agrees. Bins outside a model's range are flagged — a validity overlay marks them, and the number/volume view is still drawn — never silently reweighted with the wrong law, and the model can be set to refuse instead; switching the weighting redraws the existing fit and never re-runs it. A reweighted distribution is report-only and carries no ± (§6.5.2). The **Weighting** and **Sphere model** controls, and where the solute refractive index the RDG and Mie laws read is entered, are the **User-Manual** §8.3.3.

3.8 What This Chapter Hands On

Chapter 4 takes two things from a DLS result: the time-averaged count rate of a measurement, which §4.1 reads as a static intensity, and a recorded mode set from the mode map (§3.5.4), which §4.2 uses to partition a static series by mode. Chapter 5 takes R_h — the cumulant's harmonic mean of §3.2.2, or a pooled R_h from §3.5 — for ρ and for the hydrodynamic contraction factor, with the apparent / thermodynamic label of its inputs (§1.3, §3.5) attached. Chapter 6 states why none of this chapter's single-correlogram values carries a ± and where a DLS ± does come from (§6.4).

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4. Static Light Scattering

Static light scattering extracts the weight-average molar mass M_w, the radius of gyration R_g and the second virial coefficient A_2 from the angular and concentration dependence of the excess Rayleigh ratio of §1.7. The chapter begins with where a static intensity can come from besides a static scan — the time average of a DLS count rate (§4.1), and the share of it a recorded mode carries (§4.2) — and then follows the analyses from the least to the most complete: the single-concentration Debye plot (§4.3), the Zimm and Berry double extrapolation that gives the thermodynamic values (§4.4), the calibration-free $2A_2M_\mathrm{w}$ product (§4.5), the Guinier plot (§4.6) and the depolarization correction (§4.7). §4.8–§4.11 replace the small-qR_g expansion by the full form factor of a sphere or coil, an ellipsoid, a Mie sphere and a core–shell sphere. Every section closes on where its ± comes from, and the chapter closes with what chapters 5 and 6 take from it (§4.12).

Two rules are shared by the fits of this chapter — the Debye, Guinier, Zimm and Berry lines, the calibration-free product, the Amplitude Guinier fit and the nonlinear form-factor fits of §4.8–§4.11. No per-point weight exists for them that an independent check has confirmed (§4.1 gives the reason for a static intensity), so the points' variances are unknown — the case §6.1's pairing of unweighted least squares with the HC3 covariance is for. And an excess Rayleigh ratio that is **non-positive** — the solution scattering less than its solvent blank — is the ordinary signature of an over-subtracted reference at high angle.

In the program: every fit of this chapter is unweighted least squares on its ordinate, its ± from the HC3 covariance of §6.1. A linear fit — Debye, Guinier, Zimm and Berry alike — drops a non-positive point before the fit and counts it, refuses a set left with fewer than two usable points rather than fitting it, and, where the dropped points would otherwise leave no trace in an export, states the drop in the Comments cells of the affected columns.

4.1 Derived Intensities

I_sample in Eq. 12 need not come from a static scan. The scattered intensity a static experiment measures *is* the time average of the photon count rate, so a measurement recorded for DLS supplies it directly — Sedlák and Amis measured their static intensities exactly this way, on the same Brookhaven BI-200SM goniometer as their correlograms, "by counting the total number of counts per a defined time interval" [60]. The program will therefore build an SLS measurement from a set of DLS measurements, one intensity per angle:

$$\bar{I}(\theta) = \frac{1}{N}\sum_{k=1}^{N} I_k(\theta) \tag{71}$$

where I_k are the samples of that measurement's count-rate trace. Where an instrument file records the measurement's totals instead of a trace, the same average follows from the header — for a format that states the run's totals, the total counts divided by the number of correlator samples and the sampling time they were accumulated over [74] (the **User-Manual** §5.2 says which):

$$\bar{I} = \frac{N_{\text{counts}}}{N_{\text{samples}}\,\Delta t_{\text{sample}}} \tag{72}$$

Two conditions the program cannot check for you, and so only states:

- **Comparability.** Intensities may be compared across measurements only if the attenuation and the laser power were the same for all of them. An instrument that attenuates automatically between measurements breaks this silently.
- **Detector linearity.** At high count rates the detector's dead time makes the measured rate fall below the true one. The correction is instrument-specific — for the BI-9000AT, $I_{\text{true}} = I_{\text{meas}}/(1 - \tau_{\text{d}} I_{\text{meas}})$ with $\tau_{\text{d}} = 25$ ns [75] — so consult your instrument's manual for its working range rather than assuming one; the dead time depends on the detector, and it is not a number the program can know.

Where two or more measurements share an angle — repeats at one setting — a large difference between their mean count rates usually means a changed pinhole, attenuator or laser power rather than noise. The standard error of the repeats' mean (Eq. 118, §6.4), and the trace mean's own sampling error, would both make poor weights for the Zimm or Debye fit, for two reasons: the sampling error of a count-rate average understates the real uncertainty in a static intensity, which is dominated by alignment, cell cleanliness and the calibration, so a ± built from it would read as precision the number does not have; and a fit weighted by it would rest on weights no independent check has confirmed (§6). The honest route to a ± on these points is repeats, exactly as for any other static measurement.

Because the derived measurement is an ordinary SLS measurement, everything downstream (§4.3–§4.6) applies to it unchanged, including the uncalibrated fallback (§1.7.1).

In the program: a derived measurement records whether each intensity came from the trace or the header, and that record travels into the export. Repeats at one angle are averaged into one point through the repeat-set check of §6.4, every contributing measurement recorded so the result does not depend on the order they were loaded in; a repeat whose mean count rate sits more than 5 % from the group mean — the tolerance the Data tab applies between a committed rate and its own trace — is disclosed by name with the spread, and the averaged value is used as is. The point records the standard error of its repeats' mean (a single measurement carries none, and none is invented), which is not propagated into the Zimm or Debye weighting. No dead-time correction is applied.

4.2 Mode-Resolved Intensities: the Sedlák Partition

When a sample relaxes by more than one mode, its static intensity is a sum of the modes' shares — and the DLS decomposition says how big each share is. Sedlák's method [63] splits the standard-normalized excess intensity by the modes' relative amplitudes, taken from the spectrum of relaxation times (§4.2.1 gives the per-mode apparent size, §4.2.2 what a full SLS analysis on one mode's share

means):

$$A_{\mathrm{s}}(\theta) = \frac{I(\theta)/I_{\mathrm{B}}(\theta)}{1 + A_{\mathrm{f}}(\theta)/A_{\mathrm{s}}(\theta)}, \qquad A_{\mathrm{f}}(\theta) = \frac{I(\theta)/I_{\mathrm{B}}(\theta)}{1 + A_{\mathrm{s}}(\theta)/A_{\mathrm{f}}(\theta)} \tag{73}$$

where $I(\theta)$ is the solution's excess intensity, $I_{\mathrm{B}}(\theta)$ the intensity of the calibration standard at the same angle (Sedlák's subscript), so that I/I_{B} is the standard-normalized excess, and $A_{\mathrm{f}}(\theta)$, $A_{\mathrm{s}}(\theta)$ the fast and slow modes' shares of that excess, with the closure $I(\theta)/I_{\mathrm{B}}(\theta) = A_{\mathrm{f}}(\theta) + A_{\mathrm{s}}(\theta)$. The ratio $A_{\mathrm{f}}/A_{\mathrm{s}}$ on the right of each equation is the ratio of the two modes' amplitudes in the DLS spectrum, so each share follows from the measured excess and that ratio. The same closure holds for any number of modes [64]. Dividing each share by the excess gives the mode's **amplitude fraction** $\hat{A}_i$, the number the DLS spectrum supplies and a recorded mode set holds; the fractions sum to one when the recorded modes exhaust the spectrum. Since the excess Rayleigh ratio is the normalized excess intensity times a fixed optical factor, the partition carries over to the Rayleigh side, where the whole calibration pipeline (§1.7) is already behind it:

$$\frac{I(\theta)}{I_{\mathrm{B}}(\theta)} = \sum_i A_i(\theta), \qquad \hat{A}_i(\theta) = \frac{A_i(\theta)}{I(\theta)/I_{\mathrm{B}}(\theta)} \quad \Longrightarrow \quad \Delta R_i(\theta) = \hat{A}_i(\theta)\, \Delta R(\theta) \tag{74}$$

The partition belongs after solvent subtraction: the amplitudes decompose the solute's correlated (excess) scattering, and solvent scatter belongs to no mode — splitting the raw intensity would allocate a share of the solvent baseline to each mode and then over-subtract it angle-dependently.

The DLS side owns the decomposition: a mode set is recorded from the Mode Map (§3.5.4) — per-angle amplitude fractions with their source measurements, extraction method, regularization and integration bounds — and any intensity series of the sample (a static scan or a derived series, §4.1) is partitioned by it. The method assumes the modes are *widely separated* on the time axis — Sedlák's own stated condition [63] — and a manual split of the spectrum exists precisely because the peaks overlap, so a manual split always strains that condition.

In the program: the partition runs after solvent subtraction, on the Rayleigh side, in the SLS tab's Mode-Resolved sub-tab; the fractions of a recorded mode set are used as measured, never renormalized, and when the recorded modes do not exhaust the spectrum the sum below 1 is disclosed rather than reassigned; the separation is checked as a ratio of adjacent decay rates against a fixed default of 5, and a strained separation or a manually set bound is a note, never a refusal.

4.2.1 The per-mode apparent size

Each mode's share is an ordinary angular series, and Sedlák's per-mode static analysis is a Guinier-type fit of the *slow-mode amplitude alone* [63] — the total-intensity form is valid only when one mode carries nearly all of the intensity, while the amplitude form survives any mixture because the fast mode's amplitude is angle-flat:

$$\frac{A_{\mathrm{s}}(0)}{A_{\mathrm{s}}(q)} \cong 1 + \frac{1}{3} R_{\mathrm{g,app}}^2\, q^2, \qquad q \to 0 \tag{75}$$

Fitting the exact rearrangement $1/A_{\mathrm{s}}$ vs q^2 makes the normalization $A_{\mathrm{s}}(0)$ the fit's own intercept rather than a hand-picked lowest angle, and $R_{\mathrm{g,app}}^2 = 3\,(\mathrm{slope}/\mathrm{intercept})$, a size that survives an uncalibrated measurement and a missing $\mathrm{d}n/\mathrm{d}c$ alike (§1.7.1). The subscript is Sedlák's own (he writes $R_{\mathrm{G,app}}$): the number is apparent, a semiquantitative estimate whose *relative changes* are the

meaningful signal. Below a resolution floor computed from the fitted q-window (never copied from anyone's optics),

$$R_{\mathrm{g,floor}} = \sqrt{\frac{3\,\delta_{\mathrm{A}}}{q_{\max}^2 - q_{\min}^2}} \tag{76}$$

with δ_{A} the minimum relative amplitude change, the predicted variation across the window is inside the scatter that ill-posed amplitudes carry. The two-mode intensity-fraction decomposition itself ($\hat{A}_{\mathrm{f}} + \hat{A}_{\mathrm{s}} = 1$, per-mode contributions tracked against angle and temperature) is standard practice well beyond polyelectrolytes [76].

In the program: a size below the resolution floor of Eq. 76, with δ_{A} a fixed default of 0.05, is labeled relative-changes-only; the size carries no flag from a missing calibration or $\mathrm{d}n/\mathrm{d}c$ (§1.7.1).

4.2.2 SLS analyses on a mode's share

A partitioned series is an ordinary Rayleigh series, so Guinier, Debye and (across concentrations) Zimm/Berry run on it unchanged, computed live in the Mode-Resolved sub-tab. Every such result is apparent, and its M_{w}/A_2 rest on three assumptions the data cannot check, because nothing in them says which mode is the chains and which is solvent structure or domains:

- The $Kc/\Delta R$ ordinate uses the total concentration and the chain $\mathrm{d}n/\mathrm{d}c$, while a domain's own mass concentration and contrast are unknown — for a domain-type mode, scattering goes as the number, volume and contrast of the domains jointly [77], so an M_{w} computed with the chain increment is unanchored by an unverifiable contrast assumption, not just noisy.
- For charged systems, absolute M_{w} and A_2 additionally require dialysis-equilibrium (Donnan) conditions [64].
- Near the theta point, the measured amplitude ratio mixes the gel and osmotic moduli [78] — a resolvable slow mode is itself the warning that $Kc/\Delta R$ is contaminated.

A per-mode intensity is one ill-posed inversion's amplitude times one intensity, the same footing as NNLS/CONTIN themselves (§6.5.2). The Amplitude Guinier fit is different: its points are independent angles unless two share a source measurement. Its size is the root of a coefficient ratio — a nonlinear transform — so its SE is Monte-Carlo-validated (§6) in the amplitudes' own noise regime: it is conservative, meeting or slightly exceeding the sampling spread (measured $\times 1.15$ at 3 % amplitude scatter, $\times 1.38$ at 20 %), never under-reporting — the same safe direction as HC3 on the short A_2 ladder (§6.1).

In the program: every result on a mode's share is labelled apparent and mode-resolved, with the three caveats stated on it rather than policed; per-mode intensities carry no ±; the Amplitude Guinier $R_{\mathrm{g,app}}$ carries its HC3 SE across independent angles — statistical only, and excluding the inversion's own systematics, regularization and integration bounds — and withholds it where two points share a source measurement; a per-mode M_{w}/A_2 says its ± excludes those systematics; the partition's provenance — mode set, method, α, bounds, source runs, date — travels into the affected export columns' Comments, so no per-mode number can be read without its origin.

4.3 Apparent Analyses: One Concentration or One Angle

At one concentration, a **Debye** plot [79] of $Kc/\Delta R_\theta$ against q^2 is linear; the intercept gives an apparent molecular weight and the slope an apparent radius of gyration. These are apparent

because the intercept still contains the concentration term,

$$\left.\frac{Kc}{\Delta R_\theta}\right|_{q\to 0} = \frac{1}{M_\mathrm{w}} + 2A_2 c, \tag{77}$$

i.e. $1/M_\mathrm{w} + 2A_2 c$, not $1/M_\mathrm{w}$. This single-concentration route still resolves angle, so it gives an apparent R_g; reducing further to a single angle leaves only an apparent M_w and no R_g, both distinct from the thermodynamic values of §4.4.

4.3.1 One angle over a concentration series

The single-angle ladder the Derived Intensity route materializes (§4.1), or a single-angle instrument's, is the same line fitted against c instead of q^2 — Debye's own construction [79]:

$$\frac{Kc}{\Delta R_\theta} = \frac{1}{M_{\mathrm{w,app}}(\theta)} + 2A_{2,\mathrm{app}}\, c, \tag{78}$$

with the apparent molecular weight from the intercept and the apparent second virial coefficient from the slope; the calibration-free product $2A_2 M_\mathrm{w}$ = slope/intercept (§4.5) follows from the same line and survives an uncalibrated measurement. Both are apparent: at a finite angle the intercept is $1/[M_\mathrm{w} P(q)]$, so $M_{\mathrm{w,app}} = M_\mathrm{w} P(q) \le M_\mathrm{w}$, and the slope carries the same angle dependence; a multi-angle Zimm or Berry set (§4.4) is the route to the thermodynamic values. $M_{\mathrm{w,app}}$ is the reciprocal of the fitted intercept, a nonlinear transform, and $A_{2,\mathrm{app}}$ the slope halved, a linear one.

${R_\mathrm{g}}^2$ is a ratio of two coefficients of the same regression, so a design too ill-conditioned for M_w is too ill-conditioned for R_g as well.

In the program: a Debye result is labeled apparent, and the single-angle ladder reports the calibration-free product beside its apparent values; the chapter's non-positive-point rule applies to both lines; the ill-conditioning flag of §6.6 reaches R_g as well, and the flagged value is shown rather than promoted silently into $\rho = R_\mathrm{g}/R_\mathrm{h}$. The ± is the HC3 regression SE of §6.1 propagated by §6.2, statistical only: on $M_{\mathrm{w,app}}$ the delta method, Monte-Carlo-validated and conservative on a short ladder (§6.1), and on $A_{2,\mathrm{app}}$ the slope's ± halved, a linear pass-through.

4.4 Zimm and Berry Double Extrapolation

The full analysis combines several concentrations and angles and extrapolates to both zero angle and zero concentration. The first-order Zimm equation [6], [80],

$$\frac{Kc}{\Delta R_\theta} = \frac{1}{M_\mathrm{w}}\left(1 + \frac{q^2 R_\mathrm{g}^2}{3}\right) + 2A_2 c, \tag{79}$$

is linear in $(q^2,\ c)$, so a single global regression $Kc/\Delta R_\theta = a + b\ q^2 + d\ c$ gives all three at once:

$$M_\mathrm{w} = \frac{1}{a}, \qquad R_\mathrm{g} = \sqrt{\frac{3b}{a}}, \qquad A_2 = \frac{d}{2}. \tag{80}$$

The **Berry** construction [81], [82] uses the square-root ordinate $\sqrt{Kc/\Delta R_\theta} = a' + b' q^2 + d' c$, giving

$$M_{\mathrm{w}} = \frac{1}{a'^2}, \qquad R_{\mathrm{g}} = \sqrt{\frac{6b'}{a'}}, \qquad A_2 = d'\,a'. \tag{81}$$

Berry linearizes the high-qR_{g} regime (roughly M_{w} above ~1 MDa, or where qR_{g} approaches 1) better than Zimm. Either construction extrapolates to both $q \to 0$ and $c \to 0$, so its values are **thermodynamic**, not apparent. The classic Zimm grid spreads each concentration's points along the abscissa by $q^2 + kc$ with an aesthetic spacing constant k that changes only how spread out the curves look, never the fitted values (**User-Manual** §9.1.2).

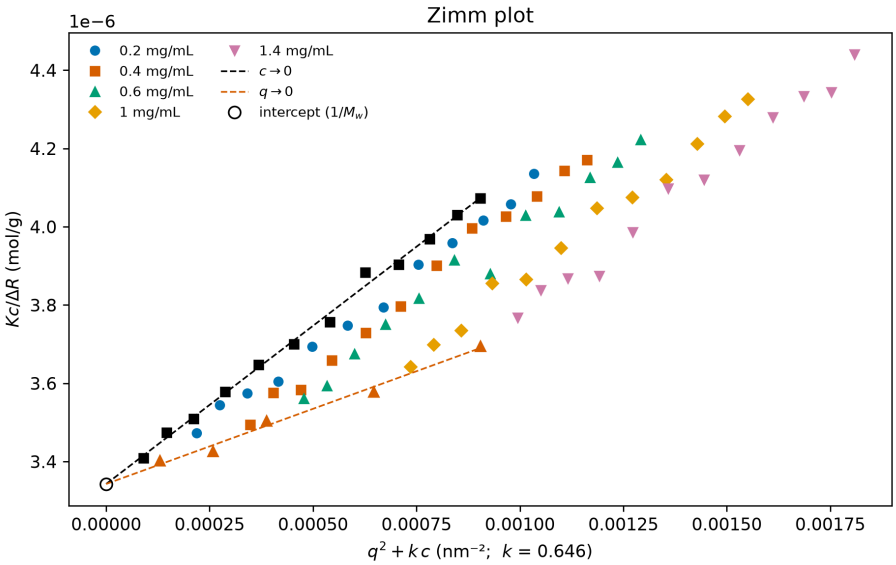


Figure 12: The Zimm plot on the **Analysis** sub-tab after a Zimm run on the synthetic PEG 300k set: each concentration's points spread along $q^2 + kc$, with the spacing constant k printed on the axis label; the $c \to 0$ and $q \to 0$ extrapolation lines are drawn to the axis and meet at the common intercept, $1/M_{\mathrm{w}}$.

In the program: the three coefficients come from one global regression over every unmasked (q^2, c) point; the result names the construction used and is marked thermodynamic; the Zimm and Berry lines are drawn to both axes so the two extrapolations meet at the common intercept, and the two-route gap of §6.3 is reported beside the result. The ± on M_{w}, R_{g} and A_2 is the HC3 covariance of the multilinear design (Eq. 111) propagated by the delta method (§6.2), statistical only, excluding the calibration and $\mathrm{d}n/\mathrm{d}c$ systematics (§6, the statistical-only rule).

4.5 Calibration-Free Second Virial Coefficient

A_2 can be obtained without any absolute calibration, from ratios of excess intensities across concentration at a fixed angle. Because only ratios enter, the optical constant K and the Rayleigh-ratio calibration cancel:

$$Y(c) = \frac{I(c_{\mathrm{ref}})/c_{\mathrm{ref}}}{I(c)/c}, \qquad \frac{\text{slope}}{\text{intercept}} = 2A_2M_{\mathrm{w}}. \tag{82}$$

where $Y(c)$ is fitted as a straight line against c, and the slope and intercept are that line's two coefficients. This yields the product $2A_2M_w$ directly; if M_w is known on an absolute scale, A_2 follows. It is valuable when absolute calibration is uncertain, as in low-dn/dc systems.

That qualification is the whole point of the method, and it is easy to lose. The product survives an uncalibrated measurement (§1.7.1). Dividing by M_w puts the scale back in: an M_w from an uncalibrated measurement is not a molar mass but a number on an arbitrary scale, and A_2 computed from it is not an A_2, which is why the program withholds such an A_2 rather than print a value that inherits the arbitrary scale.

A supplied M_w brings its own uncertainty, which enters A_2 linearly: a 5 % error on it is a 5 % error on A_2. And a ratio of two fitted coefficients read off a short concentration ladder is where HC3's small-N conservatism is most pronounced (§6.1); more concentration points, or repeat measurements (§6.4), give a tighter, unbiased ±.

In the program: $2A_2M_w$ is reported on every fit; A_2 itself only when the M_w behind it is calibrated or hand-entered, and withheld otherwise. A non-positive point cannot form $Y(c)$ and is dropped by the chapter's rule; the exported concentration and Y columns hold the remaining points, so the drop is stated in their Comments cells and the product's. The ± on $2A_2M_w$ is the HC3 covariance of the line (Eq. 109) propagated through the ratio by the delta method (Eq. 114), conservative on a short ladder (§6.1); the A_2 ± is that ± divided by $2M_w$, a linear pass-through with M_w treated as exact, so it excludes a supplied M_w's own uncertainty.

4.6 Guinier Plot

A single-concentration route to R_g using the log intensity — the counterpart to the Debye plot of §4.3. The Guinier approximation $\Delta R(q) = \Delta R(0)\,\exp(-q^2R_g^2/3)$ linearizes as

$$\ln[\Delta R(q)] = \ln[\Delta R(0)] - \frac{R_g^2}{3}\,q^2, \tag{83}$$

so a straight-line fit of $\ln(\Delta R)$ vs q^2 gives $R_g = \sqrt{-3\,\text{slope}}$ and an apparent M_w from the intercept ($\Delta R(0) = Kc\ M_w$). Like Debye's, the Guinier R_g survives an uncalibrated measurement (§1.7.1); only the intercept-derived M_w needs calibration. The approximation is first-order — exact to fourth order in q for near-spherical particles [83] — so it is used strictly in the Guinier regime $qR_g \lesssim 1$ [80]. Three cases need care:

- **Past the limit.** A fit past a *measured* limit is a failed precondition of the method, for the slope R_g and the intercept M_w alike.
- **A flat slope.** When the angular dependence is too small to measure, the fitted slope is not negative, no R_g can be read, and qR_g is therefore unknown. That is the small-qR_g limit Eq. 83 assumes, so the intercept M_w is sound and only the size is missing.
- **A rising slope.** No size is available there either, but the intercept is extrapolated against the trend, so the M_w read from it is likely low. Interparticle interference — a structure factor suppressed as $q \to 0$, as in concentrated or charged systems — and an over-corrected solvent blank both produce that shape. For larger species, Berry or Zimm linearize the high-qR_g regime better [6], [80].

In the program: the practical cutoff is $qR_g \lesssim 1.3$, slightly looser than the strict regime and a Settings default you can override (**User-Manual** §12.4), and the largest qR_g used is shown; a fit

past it flags both R_g and M_w, still reports them and names the limit applied; a slope that is not negative withholds R_g, keeps the intercept's flag and says which case occurred; a rising slope is a note rather than a flag, naming interference and an over-corrected blank as its causes. The ± is the HC3 SE on the slope propagated to R_g by §6.2's table, statistical only.

4.7 Depolarization Correction (Anisotropic Scatterers)

An optically anisotropic scatterer — one whose polarizability differs along different molecular axes (rigid rods, many nanoparticles, small anisotropic molecules like toluene; flexible coils are very nearly isotropic) — scrambles the polarization of part of the light it scatters. With the vertically polarized incident light used by essentially every modern instrument, this appears as a horizontally polarized (*depolarized*) component. Measuring it both enables the **Cabannes correction** that keeps M_w from being overestimated, and yields the **optical anisotropy** as a result in its own right. This is the static side of depolarized light scattering; the dynamic side (rotational diffusion D_r from the depolarized correlogram) is described in §3.6.

4.7.1 The Cabannes correction (vertical incidence)

The convention for I_{VV}, I_{VH} and the depolarization ratios ρ_v and ρ_u is §1.6.1's (Eq. 9). The anisotropic part inflates the measured VV Rayleigh ratio, so an uncorrected M_w comes out too high. The isotropic excess Rayleigh ratio — the one the Zimm/Debye/Berry analysis (§4.3–4.4) should use — is recovered by [14]

$$\Delta R_{iso} = \Delta R_{VV}\left(1 - \tfrac{4}{3}\,\rho_v\right). \tag{84}$$

For an isotropic scatterer ($\rho_v = 0$) this is the identity, so applying it is always safe. The removed anisotropic fraction is $(4/3)\rho_v$. A single VV/VH intensity pair is one measurement, not an ensemble (§6).

In the program: the correction is a display-only check in the SLS tab's Calibration panel — ρ_v entered directly or as I_{VV} and I_{VH} — which shows ρ_v, the optical anisotropy δ^2 of §4.7.2, the Cabannes factor and the corrected M_w, each without a ±, none of them written to the sample or exported; to use the corrected M_w downstream, enter it as the Manual M_w (**User-Manual** §9.1.3). A ρ_v outside its physical range follows §1.6.1's rule.

4.7.2 The optical anisotropy

In one standard parametrization [6] the apparent molecular weight and the depolarized Rayleigh ratio carry the mean-square optical anisotropy δ^2,

$$M_{app} = M_w\left(1 + \tfrac{4}{5}\delta^2\right), \qquad \delta^2 = \frac{5\rho_v}{3 - 4\rho_v}, \tag{85}$$

so δ^2 is 0 for an isotropic scatterer and grows without bound as $\rho_v \to 3/4$. Eliminating δ^2 between M_{app} and ρ_v reproduces Eq. 84 exactly — the two routes agree. (Anisotropy normalizations differ between conventions; convert through the depolarization ratio, which is convention-independent, rather than by equating differently-normalized anisotropies.)

4.7.3 The natural-light form

For unpolarized incident light the classical Cabannes factor [84] recovers the isotropic part from the total at 90°,

$$\Delta R_{\mathrm{iso}} = \Delta R_{\mathrm{total}} \cdot \frac{6 - 7\rho_{\mathrm{u}}}{6 + 6\rho_{\mathrm{u}}}. \tag{86}$$

It is given here for reading natural-light data; modern instruments use Eq. 84.

In the program: the natural-light form is not offered; the check of §4.7.1 reads vertical incidence only.

4.8 Full-Angular Form-Factor Fit: Sphere and Coil

The Zimm/Berry/Guinier expansions of §4.3–4.6 use only the small-qR_{g} limit of the particle form factor, $P(q) \approx 1 - q^2 {R_{\mathrm{g}}}^2/3$. The **form-factor fit** instead fits the *full* angular form factor to the ordinate [6], Eq. 2.1.8

$$\frac{Kc}{\Delta R(q,c)} = \frac{1}{M_{\mathrm{w}}\, P(q; R_{\mathrm{g}})} + 2A_2 c, \tag{87}$$

so it stays valid to larger qR_{g}, where the linear-in-q^2 plots are biased by the form-factor curvature. Two models are offered [85]:

The **solid sphere** returns the geometric radius R, with ${R_{\mathrm{g}}}^2 = (3/5)R^2$, and also reports R_{g} [83], [85]:

$$P(u) = \left[\frac{3(\sin u - u\cos u)}{u^3}\right]^2, \qquad u = qR. \tag{88}$$

The **Gaussian (Debye) coil** returns R_{g} [79], [85]:

$$P(v) = \frac{2}{v^2}(e^{-v} - 1 + v), \qquad v = (qR_{\mathrm{g}})^2. \tag{89}$$

Both reduce to the Guinier limit $P \to 1 - q^2 {R_{\mathrm{g}}}^2/3$ as $qR_{\mathrm{g}} \to 0$, so this fit extends, rather than replaces, the Guinier/Zimm slope. Because $P(q)$ is nonlinear in R_{g}, the parameters ($1/M_{\mathrm{w}}$, R_{g}, and — across concentrations — A_2) need a nonlinear least-squares fit. A single-concentration fit folds the $2A_2c$ term into the intercept, as in §4.3, and is apparent; a fit across concentrations resolves A_2 and is thermodynamic. As with the other SLS routes, R_{g} (and R) survive an uncalibrated run (§1.7.1), while M_{w}/A_2 need calibration. The fit's points are distinct angles, genuinely independent, and the reciprocal that turns the fitted $1/M_{\mathrm{w}}$ into M_{w} is Monte-Carlo-validated (§6): on a well-determined multi-angle × multi-concentration design the reported SE tracks the sampling spread to a few percent (mildly conservative).

In the program: the parameters are found by multi-start least squares seeded from the linear estimate; a single-concentration fit is labeled apparent and a fit across concentrations thermodynamic; the two bounds below — the sphere's first minimum (§4.8.1) and the size floor (§4.8.2) — flag the

result. With distinct angles the fit reports the HC3 SE of the model Jacobian at the optimum (§6.1), propagated by the delta method (Eq. 114) from the fitted $1/M_w$ to M_w and from R_g to $R = R_g\sqrt{5/3}$, statistical only (§6's statistical-only rule).

4.8.1 The sphere's first minimum

The sphere $P(u)$ has its first minimum where its amplitude vanishes, $u = qR = 4.4934$ (the first root of $\tan u = u$, equivalently the first zero of the spherical Bessel j_1), i.e. $qR_g = 4.4934\sqrt{3/5} \approx 3.48$; this sphere intensity — first minimum near $qR \approx 4.5$, multi-lobe structure beyond — is plotted in [83, Fig. 6]. Past it $P(q)$ is multi-lobed and a single-sphere fit is ill-conditioned — the amplitude (M_w) and the zero location trade off, so M_w can be badly wrong even at $R^2 = 1$ (R^2 does not flag it), while the size itself stays sound. Dropping the highest angles, or a full Mie treatment for high-contrast spheres (§4.10), keeps a fit out of that regime. The multi-start least-squares (several qR_g-spread starts, lowest-cost kept) finds the global minimum within this range. The Debye coil $P(q)$ is monotonic, so it carries no such limit.

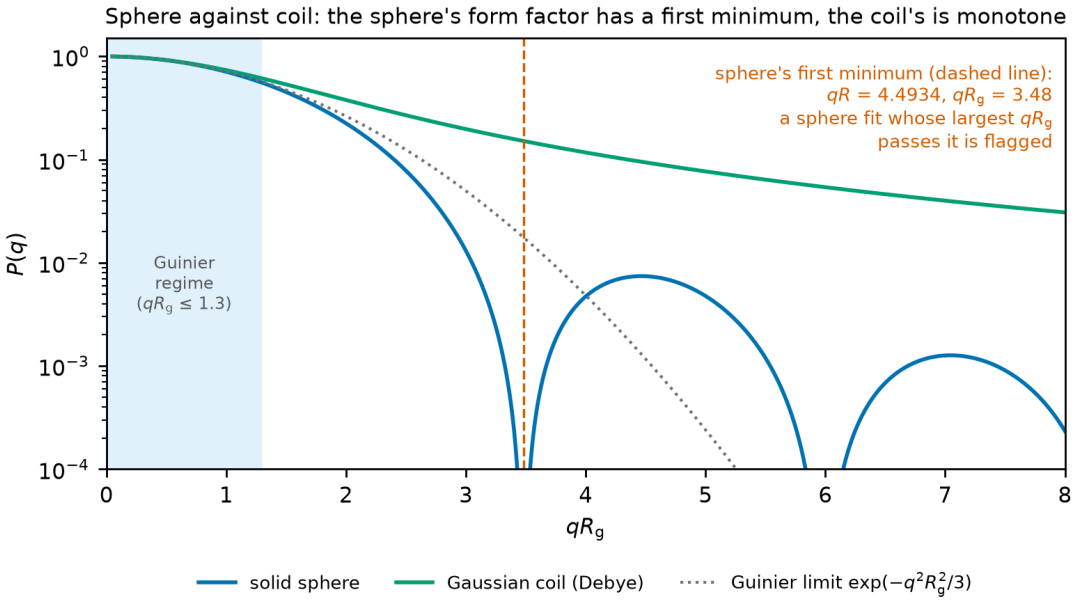


Figure 13: The solid-sphere form factor (Eq. 88) against the Gaussian coil's (Eq. 89) on a log scale, with the Guinier limit dotted: the sphere's $P(q)$ falls to its first zero at $qR_g = 3.48$ (dashed), beyond which it is multi-lobed and a single-sphere fit trades amplitude against the zero's position, while the coil's is monotone and carries no such limit; the shaded band is the Guinier regime $qR_g \leq 1.3$ that §4.6 fits.

In the program: a sphere fit whose largest qR_g passes the first minimum, 3.48, is flagged and its M_w/A_2 marked unreliable; R_g is still reported.

4.8.2 The size floor

The size floor's claim is only that the body is smaller than any particle the measurement could have seen — its value as an R_g is a 0.4 nm sphere diameter, narrower than a water molecule. It is deliberately *not* "$P(q)$ is flat here, so M_w is meaningless": $P > 0.995$ across the whole angular range for any R_g below ≈ 3 nm, and there M_w is the ordinary, correct intercept.

Where $P \to 1$ *does* matter is in explaining why neither bound above catches it: as the size goes to zero the shape term drops out of the ordinate, the largest qR_g is 0 and the Mie $P(q)$ is flat at 1, so a collapsed fit satisfies both. The floor closes that detection gap, for every model, the monotonic coil included; for the core–shell fit (§4.11) it must bind the outer radius, since that model's contrast-weighted R_g is undefined whenever the contrasts cancel. The value means the same physical size as the DLS path's size floor (§3.1.5), a *radius* converted to an R_g by $\sqrt{3/5}$.

Unlike the two bounds above, below the floor the size *is* the defect: past a minimum it is the *amplitude* that trades off while the size stays sound.

In the program: a fit whose size falls below the floor — an R_g of 0.155 nm (§3.1.5's floor, converted) for the sphere, coil, Mie and ellipsoid fits, the outer radius for the core–shell fit — is flagged (the DLS path's rule is §3.1.5's), and the size is withheld as an R_g candidate for $\rho = R_g/R_h$ and the scaling plots, while M_w, A_2 and the fitted numbers are still shown, flagged. A flat $P(q)$ above the floor flags nothing.

4.9 Ellipsoid Form Factor and Shape Fitting

The sphere and coil of §4.8 are isotropic in size. An ellipsoid of revolution adds a shape parameter — the axial ratio p (polar ÷ equatorial semi-axis, the same p as §3.6.6; Figure 11 shows a, b and the symmetry axis) — fitted from the angular $P(q)$. For semi-axes (R, R, pR) the form factor is the orientation average of the sphere amplitude F, the bracket of Eq. 88 (so that $P = F^2$), computed numerically (Guinier; [85], Eq. 61):

$$P(q;R,p) = \int_0^{\pi/2} F^2[q\,r(R,p,\vartheta)]\,\sin\vartheta\,\mathrm{d}\vartheta, \qquad r(R,p,\vartheta) = R\sqrt{\sin^2\vartheta + p^2\cos^2\vartheta}. \tag{90}$$

The fit is parameterized by the radius of gyration (comparable to the other models), with the equatorial radius formed internally,

$$R_g^2 = \frac{R^2(2+p^2)}{5}, \qquad R = R_g\sqrt{\frac{5}{2+p^2}}. \tag{91}$$

At $p = 1$ the integrand is constant and $\int_0^{\pi/2}\sin\vartheta\,\mathrm{d}\vartheta = 1$, so P collapses exactly onto the sphere (Eq. 88) with $R = R_g\sqrt{5/3}$; $p > 1$ is prolate, $p < 1$ oblate. The inverse is not single-valued (§4.9.1), so a user must choose among its solutions (§4.9.2), and at small qR_g the axial ratio is not identifiable at all (§4.9.3).

4.9.1 Degeneracy

A shape-from-scattering inverse is not single-valued: at low-to-moderate qR_g a prolate and an oblate ellipsoid (and a sphere) reproduce the *same* $P(q)$, so every axial ratio whose fit is comparable to the best is a solution. This mirrors the depolarized ellipsoid inverse (§3.6). As qR_g grows the prolate/oblate curves diverge and the fit resolves to one solution.

In the program: the fit returns a list of solutions — every axial ratio whose extra RMS misfit is below 2 % of the data RMS, a fixed default — flagged degenerate when more than one fits, never silently choosing one; a fit that does not converge returns no solution.

4.9.2 Selecting a degenerate solution

The scattering data cannot pick among a degenerate fit's solutions; only a user who has resolved the shape by an independent method can. The solutions can still agree on R_g when the shape is not resolved (the contrast-weighted size is determined while the radii are not — the common core–shell case), and then their mean is a sound single value. A relative agreement test alone is needlessly strict at small sizes, which is why an absolute tolerance stands beside it; and a mean over degenerate solutions is not a regression estimate.

In the program: the solutions are offered in the Cross-Sample source pickers as explicit, flagged choices, each labelled by its distinguishing shape parameters (**User-Manual** §10), and a picked solution is flagged wherever it propagates. Where the solutions agree on R_g — a peak spread max R_g − min R_g within 10 % of the mean, or within 1 nm absolute — their mean is offered as well, as a consensus value without a ±.

4.9.3 Axial-ratio identifiability

Because every shape shares the Guinier limit $P \to 1 - q^2 {R_g}^2/3$ as $qR_g \to 0$, the axial ratio is unidentifiable at small qR_g. The fit detects this from the sensitivity of the residual to p (the p column of the Jacobian). A well-determined size should not be held hostage to an unresolvable shape, which is why R_g and M_w are judged on the core $(1/M_w,\ R_g)$ conditioning alone.

The degeneracy is structural — a list of solutions, not a spread about one — so it has no place inside a ±; and the solutions' SEs are Monte-Carlo-validated in a resolved regime (§6).

In the program: each converged solution carries its own HC3 SEs (§6.1) on R_g, the axial ratio p, M_w (from the fitted $1/M_w$, as in §4.8) and A_2, statistical only, the degeneracy reported as the list of flagged solutions and never folded into them; an unresolved axial ratio withholds its ± and is flagged, while R_g and M_w, gated only on the core conditioning, are reported.

4.10 Full Mie Form Factor

The analytic sphere (Eq. 88) is the Rayleigh–Debye–Gans limit — valid only at low contrast ($|m - 1| \ll 1$) and small phase shift ($2x|m - 1| \ll 1$, §3.7). The full Mie theory ([73], their §4.3–4.4) is the *general* homogeneous-sphere solution — any size parameter $x = 2\pi nR/\lambda$ (with λ the vacuum wavelength, as in §1.2) and any relative index $m = n_{\text{solute}}/n$ — so it covers the classic aqueous-nanoparticle case (e.g. polystyrene latex in water, $m \approx 1.19$) that RDG and Rayleigh refuse. Here m enters the scattering arithmetic, not merely a validity gate, so the fit cannot run without the solute refractive index (§3.7). §4.10.1 states how the result composes with the sphere and RDG laws.

The Mie scattering coefficients are, in Riccati–Bessel form (permeability-matched; [73], Eqs. 4.56/4.57),

$$a_n = \frac{m\,\psi_n(mx)\,\psi_n'(x) - \psi_n(x)\,\psi_n'(mx)}{m\,\psi_n(mx)\,\xi_n'(x) - \xi_n(x)\,\psi_n'(mx)}, \qquad b_n = \frac{\psi_n(mx)\,\psi_n'(x) - m\,\psi_n(x)\,\psi_n'(mx)}{\psi_n(mx)\,\xi_n'(x) - m\,\xi_n(x)\,\psi_n'(mx)}, \tag{92}$$

with $\psi_n(z) = z\,j_n(z)$ and $\xi_n(z) = z\,h_n^{(1)}(z) = \psi_n(z) - i\,\chi_n(z)$ $(\chi_n(z) = -z\,y_n(z))$. The vertically-polarized amplitude function is ([73], Eq. 4.74)

$$S_1(\theta) = \sum_{n=1}^{n_{\max}} \frac{2n+1}{n(n+1)}(a_n\,\pi_n + b_n\,\tau_n), \tag{93}$$

with the angle functions $\pi_n = P_n^1(\cos\theta)/\sin\theta$ and $\tau_n = \mathrm{d}P_n^1/\mathrm{d}\theta$ by their upward recurrences, and the series truncated at $n_{\max} = \lceil x + 4x^{1/3} + 2 \rceil$ ([86]). Because a homogeneous sphere does not depolarize, the per-particle scattered intensity is $\propto |S_1|^2$ under either the VV or the VU (no-analyzer) geometry set in the Calibration panel — the two coincide for a non-depolarizing scatterer — and the form factor is the normalized angular intensity

$$P(q) = \frac{|S_1(\theta)|^2}{|S_1(0)|^2}, \qquad \cos\theta = 1 - \tfrac{1}{2}(q/k)^2, \quad k = \frac{2\pi n}{\lambda}, \quad x = kR, \tag{94}$$

with $R = R_{\mathrm{g}}\sqrt{5/3}$ (a solid sphere) formed internally so the fit stays parameterized by R_{g}. This is fit to $Kc/\Delta R = 1/(M_{\mathrm{w}}\,P(q)) + 2A_2 c$ exactly as in §4.8.

The Mie fit's SEs are Monte-Carlo-validated in both the thermodynamic and the apparent single-concentration regimes (§6). As for the analytic models, R_{g} and R survive an uncalibrated measurement (§1.7.1) while M_{w}/A_2 need calibration.

The same Mie per-particle intensity $|S_1|^2$ also drives the DLS intensity → volume/number reweighting in the Mie regime (§3.7), replacing the RDG refusal for high-contrast particles.

In the program: the Mie fit carries HC3 SEs (§6.1) on M_{w} (from the fitted $1/M_{\mathrm{w}}$, as in §4.8), R_{g}, R and A_2, statistical only; the relative index m is a fixed user input, treated as exact, and contributes no ± term.

4.10.1 Continuity: the ladder

As $m \to 1$, Eq. 94 collapses onto the sphere form factor (Eq. 88) with $u = qR$, and the per-particle intensity onto the RDG $R^6\,P(qR)$ law of §3.7; as $x \to 0$ it collapses onto the Rayleigh R^6 law. So Mie composes as the top of the Rayleigh → RDG → Mie ladder of §3.7, and within the RDG contrast limit the cheaper RDG model agrees with it. Mie extends the reliable range to high contrast (latex in water) that RDG and Rayleigh refuse, and its multi-start reaches the global fit basin even for large particles whose $P(q)$ oscillates fast. But when the fitted $P(q)$ dips to a deep minimum within the angular range, that angle's ordinate $1/(M_{\mathrm{w}}\,P)$ is a noisy small difference of large intensities, so M_{w}/A_2 are ill-conditioned there (a wrong fit basin passes the condition-number and R^2 checks alone), while the size R_{g} stays constrained by the whole curve — the general-case analog of the sphere's first minimum (§4.8.1).

In the program: the hand-entered solute refractive index is required and never guessed; a fit whose $P(q)$ dips below a fixed default floor of 0.05 at a fitted angle flags M_{w}/A_2 unreliable while reporting R_{g}; where the contrast is within the RDG limit the result notes that the cheaper model agrees.

4.11 Core–Shell (Coated-Sphere) Form Factor

A **coated (core–shell) sphere** is a core of relative index $m_1 = n_{\mathrm{core}}/n$ and radius $R_1 = R_{\mathrm{core}}$ inside a concentric shell of index $m_2 = n_{\mathrm{shell}}/n$ out to $R_2 = R_{\mathrm{outer}}$ (size parameters $x = kR_1$, $y = kR_2$; $k = 2\pi n/\lambda$). The fit needs both refractive indices. §4.11.1 gives the limit in which the coated

sphere becomes the homogeneous one, §4.11.2 the size the fit reports, §4.11.3 its degeneracy and ±. The scattered field has the same form as the homogeneous sphere (§4.10): only the scattering coefficients change, from the Aden–Kerker generalization of the Mie coefficients ([87]; [73], their §8.1, Eq. 8.2),

$$a_n = \frac{\psi_n(y)\,[\psi_n'(m_2y) - A_n\chi_n'(m_2y)] - m_2\,\psi_n'(y)\,[\psi_n(m_2y) - A_n\chi_n(m_2y)]}{\xi_n(y)\,[\psi_n'(m_2y) - A_n\chi_n'(m_2y)] - m_2\,\xi_n'(y)\,[\psi_n(m_2y) - A_n\chi_n(m_2y)]}, \tag{95}$$

$$b_n = \frac{m_2\,\psi_n(y)\,[\psi_n'(m_2y) - B_n\chi_n'(m_2y)] - \psi_n'(y)\,[\psi_n(m_2y) - B_n\chi_n(m_2y)]}{m_2\,\xi_n(y)\,[\psi_n'(m_2y) - B_n\chi_n'(m_2y)] - \xi_n'(y)\,[\psi_n(m_2y) - B_n\chi_n(m_2y)]}, \tag{96}$$

where the ratio functions A_n, B_n carry the core contribution and depend on the core size x only,

$$\begin{aligned} A_n &= \frac{m_2\,\psi_n(m_2x)\,\psi_n'(m_1x) - m_1\,\psi_n'(m_2x)\,\psi_n(m_1x)}{m_2\,\chi_n(m_2x)\,\psi_n'(m_1x) - m_1\,\chi_n'(m_2x)\,\psi_n(m_1x)}, \\ B_n &= \frac{m_2\,\psi_n(m_1x)\,\psi_n'(m_2x) - m_1\,\psi_n(m_2x)\,\psi_n'(m_1x)}{m_2\,\chi_n'(m_2x)\,\psi_n(m_1x) - m_1\,\psi_n'(m_1x)\,\chi_n(m_2x)}. \end{aligned} \tag{97}$$

These are evaluated by Bohren & Huffman's BHCOAT recurrence ([73], Appendix B), with the logarithmic derivatives $\mathcal{D}_n = \psi_n'/\psi_n$ (their D_n) by the downward recurrence of their Eq. (4.89) ([73], Appendix A, p. 478) and the series truncated at $n_{\max} = \lceil y + 4y^{1/3} + 2 \rceil$ on the outer y. The amplitude function $S_1(\theta)$ (Eq. 93) and the form factor are then exactly as in §4.10:

$$P(q) = \frac{|S_1(\theta)|^2}{|S_1(0)|^2}, \qquad \cos\theta = 1 - \tfrac{1}{2}(q/k)^2, \quad x = kR_1, \quad y = kR_2. \tag{98}$$

4.11.1 Continuity

When $m_2 \to m_1$ (or $R_1 \to 0$), $A_n = B_n = 0$ and Eq. 98 collapses onto the homogeneous Mie form factor (Eq. 94) of the outer sphere — so core–shell composes on top of the Rayleigh → RDG → Mie ladder.

4.11.2 The reported size: the contrast-weighted radius of gyration

A two-density object's small-q (Guinier) radius of gyration is the contrast-weighted second moment [88] (core contrast $\propto m_1 - 1$, shell contrast $\propto m_2 - 1$; the common n factor cancels):

$$R_{\mathrm{g}}^2 = \frac{3}{5}\,\frac{(m_1 - 1)\,R_1^5 + (m_2 - 1)\,(R_2^5 - R_1^5)}{(m_1 - 1)\,R_1^3 + (m_2 - 1)\,(R_2^3 - R_1^3)}. \tag{99}$$

This is the R_{g} the coated $P(q)$ exhibits in its Guinier limit — directly comparable to the Zimm/sphere R_{g}. It is undefined for a contrast-matched object, where the denominator (the net excess scattering length) is near zero or $R_{\mathrm{g}}^2 \le 0$; there a single scattering size is not physically defined.

4.11.3 Degeneracy and uncertainty

Two radii from one angular curve are frequently under-determined: at low qR_g every geometry shares the Guinier limit, and several (R_1, R_2) pairs reproduce the same $P(q)$, so several solutions fit comparably (like the ellipsoid, §4.9), and the degeneracy lifts only at high qR_g where the coated oscillations separate core from outer.

The fit's radii are the core radius and the shell thickness t, so $R_2 = R_1 + t$ is a linear combination of them, and the solutions' SEs are Monte-Carlo-validated in both the thermodynamic and apparent regimes (§6). The contrast-weighted R_g (Eq. 99) is a nonlinear combination of a structurally degenerate radius pair, so its uncertainty is the solution list, not a symmetric interval. The radii and R_g survive an uncalibrated measurement (§1.7.1) while M_w/A_2 need calibration.

In the program: both refractive indices are hand-entered on the Data tab — the shell's as a second solute refractive index — and neither is ever guessed, and the contrasts m_1, m_2 are treated as exact, with no ± term; every comparably-fitting (R_1, R_2) pair is returned and flagged degenerate, each with HC3 SEs (§6.1) on M_w (from the fitted $1/M_w$, as in §4.8), the two radii and A_2, statistical only; the contrast-weighted R_g reads *undefined (contrast profile)* in the results table for a contrast-matched object and carries no ±; a $P(q)$ spanning a deep form-factor minimum flags M_w/A_2 by §4.10.1's rule, and the size floor binds the outer radius (§4.8.2).

4.12 What This Chapter Hands On

Chapter 5 takes R_g — thermodynamic from §4.4 or §4.8–§4.11 across concentrations, apparent from §4.3, §4.6 or a single-concentration form-factor fit — with its apparent / thermodynamic label attached, and M_w and A_2 for the scaling laws with their calibration status. Chapter 6 states the estimator behind every ± this chapter reported (§6.1), the propagation each section named (§6.2), and the Zimm two-route check (§6.3); §1.7.1 remains the one statement of what an uncalibrated measurement keeps.

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5. Derived and Cross-Sample Quantities

The quantities of this chapter are formed from results the earlier chapters produced — R_h from chapter 3, R_g, M_w and A_2 from chapter 4 — and they are the first place the program combines two measurements of one sample, or several samples of one series. §5.1 defines the shape ratio $\rho = R_g/R_h$ and what it can and cannot say about architecture; §5.2 the power laws a homologous series follows; §5.3 the branching contraction factor g and the branching index read from it; §5.4 its hydrodynamic sibling h, with §5.4.1 on a factor that reads above unity. Each carries the apparent / thermodynamic label of its inputs, and each carries a ± only where §6.2 or §6.4 gives it one; §5.5 closes on what chapter 6 owes them.

5.1 The Shape Ratio $\rho = R_g/R_h$

The shape parameter links the radius of gyration (R_g, from SLS) and the hydrodynamic radius (R_h, from DLS) of the same sample:

$$\rho = \frac{R_g}{R_h} \tag{100}$$

Both should be infinite-dilution values of one sample for ρ to be meaningful. ρ is a *qualitative* architecture fingerprint — the θ-solvent reference values below are from [89], Table 4 (primary derivation [90]), with the classical coil/sphere limits [6], [91]. Branched and star objects sit *below* the linear-coil value, but their ranges overlap the sphere-to-coil span, so ρ narrows the possibilities rather than naming one architecture.

ρ (θ-solvent unless noted)	Architecture
0.3–0.6	Microgel / internally cross-linked sphere
≈ 0.78	Hard sphere (compact, near-spherical)
≈ 0.98	Dendrimer / "soft sphere"
≈ 1.33–1.08	Regular star — falls as arm number rises ($\approx$ 1.33 four-arm, ≈ 1.08 many-arm)
≈ 1.23–1.25	Hyperbranched / cyclic
≈ 1.50 (θ) – 1.78 (good solvent)	Linear random coil
≈ 1.73	Broad (polydisperse) coil, or randomly branched
≈ 2.05	Broad (polydisperse) coil in a good solvent — the upper coil range
≥ 2.2	Extended / rigid-rod (ρ grows as ln N), or aggregated

Two caveats travel with these numbers [89]: polydispersity raises ρ (a θ-coil shifts $1.50 \rightarrow 1.73$ for a broad distribution), and the Kirkwood–Riseman theory behind the tabulated values underestimates the hydrodynamic interaction, so measured ρ tends to run ~10–14 % *below* them. They remain guides, not hard boundaries. The practical consequence is that a real sample can read one row too low: a narrow θ-solvent coil, tabulated at 1.50, commonly measures 1.29–1.35 and so falls in the star/branched range. The two caveats work against each other, so the offset bites hardest on narrow samples, where no polydispersity shift is there to cancel it. It applies only to the rows the Kirkwood–Riseman treatment produces — the coil, branched/star and dendrimer values. The hard sphere is exact geometry ($R_g = \sqrt{3/5}\,R$, $R_h = R$, no hydrodynamic-interaction approximation) and the microgel range is empirical, so neither carries the offset. The hard-sphere band does, however, sit under an overlap: the same offset puts a dendrimer (tabulated 0.98) near 0.84–0.88, inside the hard-sphere band. For a physically meaningful shape parameter, prefer the thermodynamic (infinite-dilution) source on both axes.

In the program: ρ is formed only from an R_g and an R_h of the same sample (solute, solvent and temperature), and is flagged apparent if *either* input is apparent — a single-condition R_g or R_h is never silently combined with a thermodynamic one. The reading names the architecture row it falls in, states the Kirkwood–Riseman offset on the rows that carry it, and names the dendrimer overlap on a reading in the hard-sphere band. When ρ carries a $\pm$ is §6.2's rule.

5.2 Scaling Laws

Across a homologous series (one polymer/solvent, varying molecular weight), size and the second virial coefficient follow power laws in M_w:

$$R_g = k_g\, M_w^{\nu} \tag{101}$$

$$A_2 = k_{\mathrm{a}}\, M_{\mathrm{w}}^{-a} \tag{102}$$

and the hydrodynamic radius follows the same form,

$$R_{\mathrm{h}} = k_{\mathrm{h}}\, M_{\mathrm{w}}^{\nu_{\mathrm{h}}} \tag{103}$$

with its own prefactor and exponent: k_{g}, k_{a} and k_{h} are the prefactors and ν, a and ν_{h} the exponents. The branching factor h of §5.4 reads its linear reference from this law fitted to the linear-marked samples alone, as g reads its from Eq. 101.

In log–log space each law is a straight line whose slope, the exponent, is the physically meaningful quantity and is independent of the (calibration-dependent) prefactor. Typical flexible-coil values: $\nu \approx 0.588$ in a good solvent, 1/2 at the θ-point, 1/3 for a collapsed globule or hard sphere; $a \approx$ 0.2–0.3 [92], [93].

In the program: each law is fitted as a straight line in log–log space, unweighted (the treatment of §3.5.3), over the samples that carry the relevant pair, needing at least two; the exponent is reported with its regression SE (Eq. 110 on the logarithms, §6.2), the prefactor depends on the calibration and carries no reading. Below each plot the Cross-Sample tab's **Scaling** sub-tab prints a plain-language reading of the fitted exponent (e.g. "$\nu \approx 3/5 \rightarrow$ swollen coil, good solvent") — a guide, not a hard boundary, as the **Architecture** sub-tab's architecture hint is.

5.3 Branching Contraction Factor g

For flexible chains of the same chemistry and molar mass, a branched molecule is more compact than its linear analog. The **scattering branching contraction factor** compares their mean-square radii of gyration at matched M_{w} [94]:

$$g = \left.\frac{\langle R_{\mathrm{g}}^2\rangle_{\mathrm{branched}}}{\langle R_{\mathrm{g}}^2\rangle_{\mathrm{linear}}}\right|_{M_{\mathrm{w}}} = \left(\frac{R_{\mathrm{g,br}}}{R_{\mathrm{g,lin}}}\right)^2 \leq 1. \tag{104}$$

The linear reference $R_{\mathrm{g,lin}}$ is the linear-only R_{g}–M_{w} power law (Eq. 101) evaluated at the branched sample's M_{w}. For a **regular star of f_{a} arms** the contraction factor and its inverse are [89], [94]:

$$g = \frac{3f_{\mathrm{a}} - 2}{f_{\mathrm{a}}^2}, \qquad f_{\mathrm{a}} = \frac{3 + \sqrt{9 - 8g}}{2g} \quad (0 < g \leq 1). \tag{105}$$

so a measured g returns an **effective arm number** $f_{\mathrm{a,eff}}$ (the functionality f of [94]) — a *regular-star equivalent*, not a literal count (it assumes uniform arm length, and blurs for polydisperse arms and high f_{a}). g survives an uncalibrated measurement when the M_{w} scale is consistent across the samples (§1.7.1). This is guidance, like the ρ labels: g itself is model-free; only the arm number assumes a star.

In the program: the linear reference law of §5.2 is fitted to the samples you mark *linear*; the ± of g is §6.4's replicate rule, and $f_{\mathrm{a,eff}}$ carries no ± of its own.

5.3.1 Random branching: the branching index m_b

Real branched materials (radical/condensation polymers, hyperbranched, crosslinked gels) are rarely regular stars: their branch points fall at random, so a measured g maps not to an arm number but to a **branching density**. For a **randomly-branched fraction** — molecules of one molar mass with branch units distributed at random — Zimm and Stockmayer give closed forms for f_a-functional branch units, with m_b the average number of branch units per molecule (their m; the program's tables and exports label it n (branch)) [94], Eqs. 47a/47b, [89], Eqs. 65/66:

$$g_3 = \left[\left(1+\tfrac{m_b}{7}\right)^{1/2} + \tfrac{4m_b}{9\pi}\right]^{-1/2}, \qquad g_4 = \left[\left(1+\tfrac{m_b}{6}\right)^{1/2} + \tfrac{4m_b}{3\pi}\right]^{-1/2}. \tag{106}$$

Each $g_f(m_b)$ decreases monotonically from 1 ($m_b = 0$) toward 0, so it can be inverted numerically for $m_b \geq 0$ (a bracketing root-find [95]). The headline caveat is model-dependence: the same g returns a *different* m_b under a different functionality (and a different $f_{a,eff}$ as a regular star). These closed forms are the θ-solvent (ideal-chain) result and apply in good solvents too: because g is a *ratio* of branched-to-linear sizes at matched conditions, excluded-volume swelling largely cancels and g depends only weakly on solvent quality [96], [97]. m_b is a deterministic transform of the reported g, so its uncertainty is g's (§6.4).

In the program: the inversion runs once you select a **tri-** or **tetra-functional** model, and the reading always names the model it assumed; m_b, like $f_{a,eff}$, carries no separate $\pm$.

5.4 Hydrodynamic Contraction Factor h

The hydrodynamic sibling of g compares hydrodynamic radii instead of radii of gyration. Where g uses R_g (SLS), h uses R_h (DLS) [90]. It is defined through the translational diffusion coefficients (equivalently, the inverse-radius averages), so at matched M_w:

$$h = \left.\frac{D_{linear}}{D_{branched}}\right|_{M_w} = \frac{R_{h,br}}{R_{h,lin}} \leq 1. \tag{107}$$

The linear reference $R_{h,lin}$ is read from the linear-only R_h–M_w power law (Eq. 103, fit to the *linear*-marked samples) at the branched M_w. For a regular star of f_a arms the hydrodynamic factor follows its own relation — not g's $(3f_a-2)/{f_a}^2$ — the Kirkwood/Zimm pre-averaged, non-draining, θ result [90], Table II:

$$\begin{aligned} h &= \frac{\sqrt{f_a}}{(2-f_a)+\sqrt{2}\,(f_a-1)} = \frac{\sqrt{f_a}}{(\sqrt{2}-1)(f_a+\sqrt{2})}, \\ f_a &= \left(\frac{1+\sqrt{1-4\sqrt{2}\,u_h^2}}{2u_h}\right)^2, \; u_h = h(\sqrt{2}-1) \quad (0<h\leq 1). \end{aligned} \tag{108}$$

Because the hydrodynamic contraction is weaker than the geometric one, a star gives $h > g$ at every f_a (e.g. four-arm: $g = 0.625$, $h = 0.892$). Comparing g and h (or their ratio) is itself a topology / draining indicator. A factor within a few percent of unity is ordinary measurement scatter rather than architecture, which is why the program reads h against a **dead band** around

unity (the marked line) — a reporting tolerance, not a value from the literature — inside which h means little or no hydrodynamic contraction, *consistent with a linear (unbranched) chain.*

The band is a tolerance on a *radius*: h is already a radius ratio, while g is a squared radius ratio, so one number used on both would be half as wide on the radius for g, the noisier of the two measurements. The same argument applies to any comparison that mixes the two factors, g vs h itself included. h survives an uncalibrated measurement on the same terms as g (§1.7.1). g and h are separate quantities with separate star relations and are never conflated.

In the program: h is read against a ±5 % dead band around unity, taken *multiplicatively* — $1/1.05 = 0.952$ to 1.05 — and only $h < 0.952$ is turned into an equivalent arm number through Eq. 108; g is read against the same radius tolerance in its own units, $1.05^2 \approx 1.10$, and every comparison of g with h is made in radius units; the ± of h is §6.4's replicate rule.

5.4.1 Above unity

For a polydisperse or randomly-branched architecture the z-average ratio can rise *above* 1 even though each molecule is more compact: the broad size distribution inflates $\langle R^2 \rangle_z$ against a narrow linear reference, exactly as polydispersity raises ρ (§5.1). A random polycondensate of f_a-functional monomers (Flory's and Stockmayer's "A_f" type) reaches $g \approx 2$ and $h \approx 1.2$–1.4 over $f_a = 3$–6 [90], Table II. A factor above unity can also be an input error — a wrong linear reference law, or M_w values on scales that do not match. Both readings are therefore live whenever g or h exceeds unity, while the regular star never rises above $h = 1$.

In the program: a g or h above unity carries the branched-architecture reading beside the advice to check the linear reference and the M_w, with the regular star excluded from it.

5.5 What This Chapter Hands On

Every quantity here is a ratio or an exponent, so each inherits the scale-independence of §1.7.1 and the apparent / thermodynamic label of its inputs. Chapter 6 gives the ± of ρ (Eq. 115), of the exponents (Eq. 110 on the logarithms) and of g and h (the replicate SEM of §6.4), and says why m_b and $f_{a,eff}$ carry no ± of their own.

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6. Uncertainty

The program reports the **statistical** (regression) uncertainty of a fit, where the fitted points are genuinely independent measurements. Where no honest uncertainty exists, none is reported. Two scope rules follow from this:

1. **Statistical only.** Every ± below is the spread implied by the scatter of the fitted points, so it excludes systematic error — calibration (the constant k_c) and dn/dc in particular. Because $M_w \propto 1/(dn/dc)^2$ and scales with k_c, those systematics usually dominate the true error of an absolute M_w or A_2, and the ± is a lower bound on the total uncertainty. R_g, R_h, the size exponent ν, and the calibration-free $2A_2M_w$ are scale-independent (§1.7.1), so their statistical ± is the whole statistical story; when dn/dc is absent or the measurement is uncalibrated (§1.7.1), the ± on M_w and A_2 describes only the scatter about an arbitrary scale.
2. **No uncertainty from a single correlogram.** A correlogram's lag channels are strongly correlated [27], so an ordinary least-squares covariance from one correlogram assumes independence it does not have and under-reports the truth. ISO 22412 [31] therefore derives DLS

dynamic uncertainty from repeat measurements: the spread of the per-replicate fits of true repeats gives an ISO 22412 SD/SEM (§6.4).

In the program: every ± on a calibration-dependent quantity is labeled *"± statistical; excludes calibration & dn/dc"*, and no uncertainty is invented for a missing calibration or dn/dc. The single-correlogram dynamic fits — the cumulants, the single and double exponentials, KWW and the distribution methods (NNLS, CONTIN and lognormal) — report no ±; the dynamic ± comes from replicate averaging (§6.4), and the cumulant PDI and the fit RMS remain as width/goodness diagnostics. A displayed ± is rounded to one significant figure — two if its leading digit is 1 — and its value to the same decimal place, so 40.5 ± 0.83 nm reads 40.5 ± 0.8 nm; a value with no honest ± is shown at a fixed number of significant figures, and exports carry full precision (**User-Manual** §12.8).

This chapter is the consolidation the earlier chapters point at. The ± the program reports come from three estimators, each at home here: a coefficient's regression standard error under the HC3 estimator (§6.1) or the classical opt-in (§6.1.1); a propagation of a covariance through a derived quantity by the delta method (§6.2), whose table lists every SLS quantity; and a replicate standard error of the mean (§6.4). Beside them the chapter holds one diagnostic that is not an uncertainty at all, the Zimm two-route check (§6.3). §6.5 is the roster of where a ± is (§6.5.1) and is not (§6.5.2) reported, and §6.6 the ill-conditioning flag that withholds a whole covariance. Three uncertainty statements live with their sections because they are those sections' own: the per-mode Amplitude Guinier SE (§4.2.2), an HC3 SE; the ± of the DDLS shape models (§3.6.7), a Monte-Carlo propagation; and the solvent library's tolerances (§1.1.2), which are display-only and enter no ±.

Wherever this guide says a ± is **Monte-Carlo-validated**, the claim is this: many synthetic data sets are drawn from the same forward model with known parameters and noise, each is fitted, and the standard deviation of the fitted values across the draws — the true sampling spread — is compared with the ± the estimator reports on one draw. The comparison is made in a regime that stresses the approximation, so a nonlinear propagation (a reciprocal, a ratio, a root) is checked where its first-order form is weakest. The guarantee it gives is that the reported ± tracks the sampling spread within the tolerance each section states, and — for a short, high-leverage design — that it meets or exceeds it rather than under-reporting. It says nothing about systematics (the statistical-only rule above), and a ± that is a linear pass-through of an already-validated covariance inherits the validation without its own check.

6.1 The Estimator: HC3 Heteroscedasticity-Consistent Standard Errors

For a linear model $\mathbf{y} = \mathbf{X}\mathbf{b}$ with N points, p parameters and residuals e_i, the parameter covariance is the **HC3 "sandwich"** estimator — the simplified form of Long & Ervin [98] (MacKinnon & White [99] coined "HC3" as the jackknife variant with an extra $(N-1)/N$ factor, which the program does not use):

$$\mathrm{Cov}(\mathbf{b}) = (\mathbf{X}^\top\mathbf{X})^{-1}\left[\sum_i \frac{e_i^2}{(1-h_{ii})^2}\,\mathbf{x}_i\mathbf{x}_i^\top\right](\mathbf{X}^\top\mathbf{X})^{-1}, \qquad h_{ii} = \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i. \tag{109}$$

where h_{ii} is the leverage of point i. Unlike the classical covariance $s^2(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}$ of §6.1.1 (s^2 the residual variance), HC3 does not assume the points share one error variance, so it holds up when point precision varies across angles or concentrations (e.g. a per-angle Γ whose precision changes

with q), which makes unweighted least squares with this covariance the canonical treatment of a line whose points' variances are unknown — every pooled DLS line (§3.5.3) and every SLS fit (chapter 4). It reduces to the ordinary formula under equal-variance errors and is mildly conservative at small N (Monte-Carlo-validated; the passage above). With too few residual degrees of freedom one residual cannot honestly support a spread (and HC3 blows up as $N - p \to 1$); p counts every parameter the fit consumed, including one later profiled out (the ellipsoid form factor's unresolved axial ratio), under both estimators. The standard error of a coefficient is the square root of its diagonal, e.g. for a straight line $y = a + b\,x$:

$$\mathrm{SE}(b) = \sqrt{\mathrm{Cov}_{bb}}, \qquad \mathrm{SE}(a) = \sqrt{\mathrm{Cov}_{aa}}. \tag{110}$$

A standard error of exactly zero would claim infinite precision, which no measurement supports. A variance reaches exactly zero for a fit through exactly collinear data, whose residuals are zero, or at an abscissa too extreme for the arithmetic.

The conservatism of the HC3 SE at small N is largest for a ratio of two fitted coefficients on a short, high-leverage design — the calibration-free $2A_2M_\mathrm{w}$ = slope/intercept (§4.5) is the program's clearest case: at a 5-point ladder the endpoints carry leverage ≈ 0.5–0.7, HC3 up-weights their residuals by $(1-h_{ii})^{-2}$, and the reported ± runs ~20–40 % above the true sampling spread. This is *only* the residual-based variance estimate being cautious — the underlying delta-method propagation is exact. The ± is therefore honest-but-loose on the cautious side; it tightens toward exact as the concentration ladder lengthens, and an unbiased ± comes from replicates (§6.4). Where a single-shot SE rests on few, high-leverage points, the achievable honest target is *conservative and disclosed*, with more points or replicates the route to an unbiased one.

A through-origin slope $y = b\,x$ (used for $\Gamma = D\,q^2$) uses the same sandwich with the one-column design $\mathbf{X} = \mathbf{x}$, giving $\mathrm{SE}(b)$ from the scalar covariance.

The Zimm/Berry global fit is multilinear — ordinate $= a + b\,q^2 + d\,c$ with design $\mathbf{X} = [1,\ q^2,\ c]$ — and uses the same $\mathrm{Cov}(\mathbf{b})$ of Eq. 109 for the full 3×3 covariance:

$$\text{ordinate} = a + b\,q^2 + d\,c, \qquad \mathbf{X} = [\,\mathbf{1}\ \ \mathbf{q^2}\ \ \mathbf{c}\,]. \tag{111}$$

Every equation above is written with $(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}$ for legibility; the covariances are computed from the singular value decomposition of the column-scaled design instead, which loses no precision where the normal equations would [46] (the singular values of $\mathbf{X}^\mathsf{T}\mathbf{X}$ are the *squares* of those of $\mathbf{X}$). With $\mathbf{X} = \hat{\mathbf{X}}\mathbf{N}_\mathrm{c}$, $\mathbf{N}_\mathrm{c}$ the diagonal of column norms, and $\hat{\mathbf{X}} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$:

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \mathbf{N}_\mathrm{c}^{-1}\mathbf{V}\mathbf{S}^{-2}\mathbf{V}^\top\mathbf{N}_\mathrm{c}^{-1}, \qquad h_{ii} = \|\mathbf{U}_i\|^2. \tag{112}$$

The leverages come from the same decomposition and satisfy $0 \le h_{ii} \le 1$ and $\Sigma h_{ii} = p$ by construction; the estimator is Eq. 109 either way.

In the program: HC3 is the default estimator behind every regression ±; no SE is reported when $N - p < 2$, and a variance of exactly zero is withheld, not reported as ± 0 — an empty cell in the export — the one zero-variance rule, applied to each diagonal entry as it is taken, to a zero spread across repeats (§6.4) and to a covariance that cancels to zero or below (§6.5.2).

6.1.1 Classical OLS standard errors (opt-in, for comparability)

The **classical homoscedastic OLS** covariance [39] is the one reported by textbooks, classical light-scattering software and spreadsheets:

$$\mathrm{Cov}_{\mathrm{OLS}}(\mathbf{b}) = s^2\,(\mathbf{X}^\top\mathbf{X})^{-1}, \qquad s^2 = \frac{\sum_i e_i^2}{N-p}. \tag{113}$$

This assumes every point shares one error variance. It is intended for reproducing a result from classical software or a published table *like-for-like*, rather than as a general improvement. On a short, high-leverage design it under-reports: on the 5-point calibration-free A_2 ladder above, OLS returns ≈ 0.90 of the true sampling spread, where HC3 sits at ≈ 1.3 on the cautious side; the OLS deficit shrinks toward 1.0 as points are added. Under-reporting is the direction the program avoids by default, which is why OLS is an opt-in. Point estimates (M_w, R_g, D, R_h) are identical under either estimator; only the $\pm$ moves, and everything downstream (the delta method of §6.2, the ratio and power-law SEs) propagates whichever covariance was selected. The cumulant fit's unweighted residual opt-in (§3.2.11) follows the same discipline, with one difference: a weighting moves the point estimates, where the estimator moves only the $\pm$.

In the program: OLS is a labeled opt-in, chosen in Settings (**User-Manual** §12.6); the selected estimator is recorded on every result that carries a regression $\pm$ and written into the export, an OLS $\pm$ column's *Comments* cell reading `SE: classical OLS` (HC3 is the silent default); the switch changes only the $\pm$.

6.2 Propagating a Covariance to a Derived Quantity (Delta Method)

A scalar function $f(\mathbf{b})$ of the fitted coefficients has, to first order, the standard error given by the **law of propagation of uncertainty** [100]:

$$\mathrm{SE}(f) = \sqrt{\mathbf{J}^\top\,\mathrm{Cov}(\mathbf{b})\,\mathbf{J}}, \qquad \mathbf{J} = \nabla_{\mathbf{b}} f. \tag{114}$$

This delivers every SLS SE from Eqs. 109, 111 with the appropriate Jacobian $\mathbf{J}$:

Quantity	Relation	Jacobian $\mathbf{J}$ (order of params)
M_w (Zimm)	$M_\mathrm{w} = 1/a$	$[-1/a^2,\ 0,\ 0]$
A_2 (Zimm)	$A_2 = d/2$	$[0,\ 0,\ 1/2]$
R_g (Zimm)	$R_\mathrm{g} = \sqrt{3b/a}$	$[-R_\mathrm{g}/(2a),\ R_\mathrm{g}/(2b),\ 0]$
M_w (Berry)	$M_\mathrm{w} = 1/a'^2$	$[-2/a'^3,\ 0,\ 0]$
A_2 (Berry)	$A_2 = d'a'$	$[d',\ 0,\ a']$
$M_{\mathrm{w,app}}$ (Debye)	$= 1/\mathrm{intercept}$	$[-1/\mathrm{intercept}^2,\ 0]$
$R_{\mathrm{g,app}}$ (Debye)	$= \sqrt{3\,\mathrm{slope}/\mathrm{intercept}}$	$[-R_\mathrm{g}/(2{\cdot}\mathrm{intercept}),$ $R_\mathrm{g}/(2{\cdot}\mathrm{slope})]$
R_g (Guinier)	$= \sqrt{-3\,\mathrm{slope}}$, slope of $\ln \Delta R$ vs q^2	$[-3/(2R_\mathrm{g})]$ (slope only)
$2A_2M_\mathrm{w}$ (cal-free)	$= \mathrm{slope}/\mathrm{intercept}$	$[-\mathrm{slope}/\mathrm{intercept}^2,$ $1/\mathrm{intercept}]$

The Berry rows' a' and d' are the coefficients of the square-root ordinate of §4.4.

For a **ratio** of two *independent* quantities $r = y_1/y_2$ (used for $\rho = R_g/R_h$, with R_g from SLS and R_h from DLS — measured independently; never for a ratio of two coefficients of one fit, such as slope/intercept, whose covariance the table above carries), Eq. 114 reduces to the familiar fractional sum in quadrature:

$$\mathrm{SE}(r) = \left|\frac{y_1}{y_2}\right| \sqrt{\left(\frac{\mathrm{SE}(y_1)}{y_1}\right)^2 + \left(\frac{\mathrm{SE}(y_2)}{y_2}\right)^2}. \tag{115}$$

For a **power map** $f = k\,x^{\mathrm{p}}$ (used for $R_h \propto D^{-1}$, so $p = -1$), Eq. 114 gives a fractional relation:

$$\frac{\mathrm{SE}(f)}{|f|} = |p|\,\frac{\mathrm{SE}(x)}{|x|}, \qquad \text{so } \frac{\mathrm{SE}(R_h)}{R_h} = \frac{\mathrm{SE}(D)}{D}. \tag{116}$$

Both the ratio (Eq. 115) and the reciprocal power map (Eq. 116) are first-order results, exact only while the *denominator's* fractional error is small. Monte-Carlo checks against the sampling spread confirm they track it to a few percent at realistic precision (a denominator coefficient of variation $\lesssim 0.05$, where a multi-angle Γ vs q^2 fit sits) and turn mildly anti-conservative — under-reporting by roughly 5–10 % — once the denominator CV reaches ≈ 0.15, with the heavy tail of $1/(\text{denominator})$ taking over beyond ≈ 0.3. The direction is set by which quantity is noisy: $\rho = R_g/R_h$ at a viscous-solvent R_h CV of 0.10–0.15 under-reports a few percent, whereas $\rho_v = I_{VH}/I_{VV}$ is immune (its noisy channel is the numerator). As everywhere in §6, the honest remedy for a weak denominator is more points or replicates — a smaller denominator CV — not a wider algebraic formula. The reciprocal R_h SE (Eq. 116) is carried unchanged wherever R_h follows from D: the Γ vs q^2 and D vs c extrapolations, and the depolarized translational R_h from D_t.

The cross-sample scaling exponents (ν in $R_g \propto {M_w}^{\nu}$, ν_h in $R_h \propto M_w^{\nu_h}$, and $-a$ in $A_2 \propto {M_w}^{-a}$, §5.2) are slopes of a log–log regression, so their SE is Eq. 110 applied to ($\log M_w$, $\log y$).

In the program: every derived SE is Eq. 114 with the Jacobian of the table, through whichever covariance §6.1 selected. ρ is given a $\pm$ only when both R_g and R_h carry an SE — both from a regression, a Zimm/Berry R_g with a Γ vs q^2 or D vs c R_h; with a single-cumulant R_h, which has none, ρ prints without a $\pm$.

6.3 A Consistency Check, Not an Uncertainty: the Zimm Two-Route M_w

The Zimm grid is extrapolated to the intercept along two independent routes — the $c \to 0$ line and the $q \to 0$ line — which must meet at the same $1/M_w$. Their relative gap is the amount by which the two routes' intercept-derived weights differ:

$$\mathrm{gap} = \frac{\left|M_w^{(c\to 0)} - M_w^{(q\to 0)}\right|}{\frac{1}{2}\left(M_w^{(c\to 0)} + M_w^{(q\to 0)}\right)}. \tag{117}$$

A gap of zero means the routes agree exactly. This is a diagnostic of extrapolation quality (curvature, an over-masked grid), not a statistical uncertainty. The two weights carry the calibration scale and their gap, a ratio, does not (§1.7.1). The Zimm/Berry plot's two extrapolation lines, drawn to the axis and meeting at the common intercept, show the same check end to end (Figure 12).

In the program: both weights and their gap are reported, and a gap above ~10 % is flagged for inspection, printed as the two routes "differ by X %"; on an uncalibrated measurement the two weights are withheld and the gap is still reported.

6.4 Repeat Measurements: a DLS Dynamic Uncertainty from Replicates

A single correlogram cannot supply an honest dynamic ± (the single-correlogram rule above), so the program gets one the way ISO 22412 [31] prescribes — from repeat measurements: a set of true replicates, the same sample re-measured back-to-back, each fitted independently and the fitted values summarized across the set. A repeat set is defined by what it holds constant: a mean across a concentration ladder describes no real sample, an angular scan has its own analysis (§3.5), and an averaged measurement adds no scatter while raising N. The actions that average a selection, and how they refuse one that is not a repeat set, are the **User-Manual** §7.1.1.

Each replicate is fitted independently with the method you pick, and for a parameter x (e.g. R_h) the N fitted values are summarized as a mean and a standard error of the mean [100, Sec. 4.2]:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2}, \qquad \mathrm{SEM} = \frac{s}{\sqrt{N}}. \tag{118}$$

No spread is defined for $N < 2$. Replicates that agree exactly, $s = 0$, agree for real, but that is a claim about the arithmetic rather than about the sample, and in practice it means the set is not made of independent repeats — a zero-noise synthetic set, or the same measurement loaded twice. Unlike a within-correlogram covariance, the x_i from separate measurements are independent draws, so Eq. 118 does not under-report; it also captures genuine drift between measurements (alignment, dust, number fluctuations) that one curve cannot see. This SE is Monte-Carlo-validated (the passage above): the reported SD/$\sqrt{N}$ matches the actual SD of the set-means to about 1 %, and the mean is unbiased.

The branching contraction factors g and h of §5.3–§5.4 take their ± from the same SEM over replicate branched measurements. The arm number $f_\mathrm{a,eff}$ and the branching index m_b are deterministic transforms of the reported g (or h), so their uncertainty is the one already shown on it. That ± reflects the branched replicate spread only: it treats the fitted linear reference law as exact and does not propagate that law's own fit uncertainty.

The distribution methods (NNLS / CONTIN / lognormal) resolve a *variable* number of size-distribution peaks, so their replicates can only be aligned by size rank, and two cautions apply:

- When measurements disagree on the peak count, a measurement that misses a peak shifts the positional alignment, so inspect the distribution overlay before trusting the per-peak averages.
- NNLS and CONTIN peaks carry §3.4's caution: the ± across measurements is a reproducibility spread, not a calibrated error.

In the program: a repeat set must agree on solute and solvent, temperature, scattering angle, wavelength, concentration, the Data tab's **Mw fraction** label (the molecular-weight fraction a series is split into) and analyzer geometry, and hold no measurement that was itself made by averaging. The reported value is $\bar{x} \pm \mathrm{SEM}$, none for $N < 2$; a zero spread withholds the ± by §6.1's zero-variance rule, the mean and N still reported. g and h carry a ± only from replicate branched measurements, the h one a linear pass-through of the R_h replicate SEM, never a single-shot analytic SE, and the reference law is treated as exact; $f_\mathrm{a,eff}$ and m_b carry no separate ±. The averaged R_h

± SEM is written into the sample, where ρ can use it, only when at least two replicates fitted completely and the method resolves one R_h mode; it is not written — and the summary says which — when the fits are too few, the mean is not finite, the sample already carries a hand-entered value, or any averaged fit returned a size outside the range DLS can establish (§3.1.5), because a size beyond what the technique resolves must not reach ρ carrying a ±. The mean and its spread are still reported. A hand-entered R_h is never overwritten; a picked one is replaced, and the summary says so (**User-Manual** §7.1.1). The distribution methods' replicates are averaged by size rank: each replicate's peaks are sorted by R_h, and position 1 (smallest R_h) is averaged across all measurements, position 2 across all, and so on, each reported as mean ± SD/$\sqrt{N}$ (Eq. 118) with its mean intensity weight and how many of the N measurements resolved it; a disagreement on the peak count raises a warning, and the per-peak averages are report-only, never written to the sample's R_h. The action that averages the correlograms themselves carries no ± of its own; it, and the route from it to a multi-population R_h that can feed ρ, are the **User-Manual** §7.1.1.

6.5 Where Uncertainties Are and Are Not Reported

Two rosters, drawn from the sections' marked lines: §6.5.1 the quantities that carry a regression, propagated, Monte-Carlo or replicate ±, and §6.5.2 those that carry none, and why.

6.5.1 Reported: independent points, a valid regression SE

- **SLS lines:** Zimm/Berry M_w, R_g, A_2; Debye/Guinier apparent M_w, R_g; the single-angle ladder's $M_{w,app}$ and $A_{2,app}$ (§4.3.1); the calibration-free $2A_2M_w$ (and A_2 if an absolute-scale M_w is supplied — §4.5).
- **SLS form-factor fits** (§4.8–§4.11, HC3 at the optimum): M_w and R_g of the sphere, coil, Mie and ellipsoid, with the sphere's and Mie's R and the ellipsoid's axial ratio; M_w and the core and outer radii of the core–shell; and A_2 wherever the fit spans concentrations. The Mie and core–shell radii are Monte-Carlo-validated.
- **Per-mode SLS:** the Amplitude Guinier $R_{g,app}$ across independent angles, and the M_w and A_2 of an SLS analysis on a mode's share, each excluding the inversion's own systematics (§4.2.2).
- **DLS lines:** from Γ vs q^2, D and R_h; from D vs c, D_0, $R_{h,0}$ and k_D; the mode map's per-family D, R_h, a and b (§3.5.4).
- **DDLS:** D_t and D_r across angles, and the shape-model dimensions they give, by Monte Carlo (§3.6.2, §3.6.7).
- **Cross-sample:** the scaling exponents; $\rho = R_g/R_h$ when both inputs have an SE; g and h from replicate branched measurements (§5.3, §5.4).
- **Replicates:** the replicate-averaged DLS dynamic parameters (Eq. 118 — the DLS dynamic ± obtainable at a single angle and concentration, where no regression is available).
- **Beside the ±:** the solvent library's display-only tolerances on n and η (§1.1.2), shown with the proposed values and folded into no ±.

6.5.2 Not reported: no honest statistical uncertainty

- every single-correlogram dynamic fit (the single-correlogram rule above), *unless* repeats are averaged via §6.4;
- the NNLS/CONTIN distributions (ill-posed inversions — no defensible covariance; the "fine structure is not always reproducible" caveat stands) and the reweighted distributions built on them (§3.7);
- per-mode intensities (§4.2.2);

- a single-angle M_w (one datum, no fit);
- a single-angle DDLS pair (§3.6.2), and the ρ_v, δ^2, Cabannes factor and corrected M_w of the anisotropy check (§4.7.1) — one measurement is not an ensemble;
- the core–shell contrast-weighted R_g (§4.11.2) and a degenerate fit's consensus mean (§4.9.2);
- m_b and $f_{a,eff}$, deterministic transforms of g (§5.3);
- every parameter of a fit whose design is too near-collinear to carry a covariance — the ill-conditioning flag of §6.6 — and of a fit whose variance cancels to zero or below or cannot be represented: a covariance is reported whole or not at all.

6.6 Numerical Health: the Ill-Conditioning Flag

An extrapolation design can be *numerically* degenerate while remaining physically sensible — near-duplicate concentrations, or angles clustered so tightly that $[1,\ q^2,\ c]$ is close to collinear. The program measures this with the **condition number of the column-normalized design**, $\mathrm{cond}(\hat{\mathbf{X}})$, each column scaled to unit norm so that it reflects genuine collinearity and not the arbitrary units of q^2 and c:

$$\mathrm{cond}(\hat{\mathbf{X}}) = \frac{s_{\max}(\hat{\mathbf{X}})}{s_{\min}(\hat{\mathbf{X}})}, \tag{119}$$

with $s_{\max}$ and $s_{\min}$ its largest and smallest singular values. Above the limit fewer than about four significant digits of the covariance survive, and an exactly rank-deficient design ($\mathrm{cond} = \infty$, parameters not separately identifiable) is the limiting case. A well-posed design sits at $\mathrm{cond}(\hat{\mathbf{X}}) \approx O(1\text{–}100)$, so the limit is far above anything healthy; it is a numerical criterion, not a physics one, and does not gate on the magnitude of M_w. Reaching this state takes a genuinely collapsed ladder; it is a backstop rather than a routine warning.

In the program: the flag fires at $\mathrm{cond}(\hat{\mathbf{X}}) > 10^{12}$, the value still shown: it marks M_w and the absolute A_2 unreliable in the SLS tab and the export Comments cell, with the advice to *add concentrations or angles, or widen their spread*, marks R_g unreliable too (it comes from the same fit's coefficients, §4.3.1), and withholds every ± of that fit (§6.5). A fit whose M_w overflows to non-finite is flagged unreliable by the same rule, rather than reported as a finite-looking number.

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7. Synthetic Data Generation

The built-in generator exists so every analysis in this guide can be checked against a known answer. Its forward model is the same physics the analyses use — Stokes–Einstein, q, the Siegert relation — so what it is *not* modeling matters as much as what it is. Two aspects of a real correlator's output are modeled: the lag axis of §2.2.2 (§7.1) and the per-channel noise of §2.3 (§7.2); §7.3 adds a control for correlated channel noise; §7.4 closes the chapter. The controls are the **User-Manual** §11.3.

7.1 The Lag Axis

The generator lays its lags on the multi-tau staircase of Eq. 27 or on a uniform log or linear grid.

This matters for validation rather than for physics. Every instrument the program supports emits such an axis, so a distribution method exercised only on a uniform grid has been tested on data systematically easier than any measurement.

Some instruments report each block-boundary lag twice, once from each adjacent block. That is an export that has not applied the discard the multi-tau scheme prescribes [101], Eq. (11), p. 4014.

In the program: the **Lag axis** block chooses the staircase, with the channels per octave and the finest sample time you set, or a uniform grid and, on the staircase, whether block-boundary lags are duplicated.

7.2 The Per-Channel Noise

The generator draws the exact form — the Biganzoli–Ferri variance [28], Eq. (9), p. 29378, Eq. 40 of §3.2.8, the form the Nonlinear fit weights by — for the reason §2.3 gives: the closed forms hold only for $\Theta_k \ll 1$. The **Synthetic Generator**'s two per-channel models, **Per-channel** and **Per-channel + shot noise** (Biganzoli–Ferri), are its classical limit and its full form with the shot terms (which value each preset consumes is the **User-Manual** §11.3).

In the program: Per-channel draws the exact form's classical limit and Per-channel + shot noise the full form at the **Mean count rate** entered beside it; the absolute scale is the Noise level, which sets the first channel's SD, a count rate changing the shape and never the scale; a multi-angle set is drawn at a fixed run duration of 60 s, which the file it writes states as its **Duration**, and a single correlogram carries no run duration and takes the $T_{\mathrm{run}} \to \infty$ form.

Because the generator is a validation tool, its limits matter as much as its model:

- **The run duration** enters as §2.3 and §3.2.9 state — common to every channel in the closed forms, kept by the exact form.
- Only the *shape* is modeled: an absolute prediction would also need the acquisition's normalization convention, which a generator cannot supply.
- The model is grounded on the multi-tau axis, where Δt_k is a block's real accumulation time. The generator also allows it on a uniform *log* or *linear* axis, where Δt_k is only a stand-in read off the axis — the nearest sample time that divides each lag a whole number of times, the same one the cumulant fit uses (§3.2.9) — so $M_k = T_{\mathrm{run}}/\Delta t_k$ and "the coarsest blocks" have no physical referent there. That combination is legal, but the resulting noise is a shape, not a model of any correlator.
- Real correlogram noise is not independent from channel to channel. It is positively correlated between neighboring channels (a lag-1 correlation of a few tenths), where an independent draw's is zero. Correlated noise is what defeats a regularized fit's smoothing-parameter selection, and Peters found that synthetic photon noise reproduced none of the pathology measured on real data [29, Ch. 3]. The **Channel noise correlation** control below exists so that failure mode can be exercised at all; read what it is, and is not, before drawing a correctness conclusion from it.
- Under the classical limit alone, the per-channel SD *span* is far too narrow over the channels that carry fit weight: the retained range is the short-lag region where, as Peters notes, shot noise still matters [29, p. 152], and Eq. 29 restores the two terms the limit drops (§2.3).

7.3 Correlated Noise

Setting Channel noise correlation to a_1 applies a first-order autoregression over the channel index,

$$e_0 = z_0, \qquad e_k = a_1\, e_{k-1} + \sqrt{1 - a_1^2}\, z_k, \qquad z_k \sim \mathcal{N}(0,1), \tag{120}$$

and adds $\sigma_k\, e_k$ to channel k, so every channel keeps its own SD from Eq. 28 or Eq. 29 and the lag-1 correlation is exactly a_1 (a few tenths is the size real replicate groups show). This is a flagged convenience simplification, not the physics: it reproduces one statistic — the lag-1 correlation — and nothing else about the covariance — the correlation length, its dependence on the decay rate and on the block structure are all absent. The canonical form is the full covariance matrix [27], Eq. (48), p. 296, generalized across multi-tau sample times by [29, Sec. 3.3.1], Eqs. (3.9) and (3.11), p. 160 and applied through a Cholesky factor; the program draws only the first-order form, because a generator's noise is a far weaker claim than a fit weight, and Biganzoli and Ferri state that the closed forms cannot be used across multi-tau stages [28]. Read a generated correlation as *a* correlation of the right size, never as the instrument's structure.

In the program: the control is off by default, so the draws are independent unless you set a lag-1 correlation.

7.4 What This Chapter Hands On

The generator hands nothing on: it consumes §2.2.2's axis, §2.3's variance and the evaluation point of §3.2.8–§3.2.9, and every analysis above can be run on what it draws with the answer known.

Appendix: Equations Index

#	Name	Equation	Section
(1)	Cauchy dispersion	solvent Cauchy $n(\lambda,T) = A_\mathrm{n} + B_\mathrm{n}/\lambda^2 + C_\mathrm{n}/\lambda^4 + (\mathrm{d}n/\mathrm{d}T)(T-T_\mathrm{ref})$	§1.1.1
(2)	Vogel–Fulcher–Tammann	solvent VFT viscosity $\eta(T) = A_\eta \exp(B_\eta/(T-T_0))$	§1.1.1
(3)	solvent n tolerance	per-condition solvent $\sigma_\mathrm{n}(\lambda,T)$ quadrature	§1.1.2
(4)	solvent η tolerance	per-condition solvent $\sigma_{\eta,\mathrm{rel}}(T)$ quadrature	§1.1.2
(5)	scattering vector	$q = (4\pi n/\lambda)\sin(\theta/2)$	§1.2
(6)	Stokes–Einstein	$R_\mathrm{h} = k_\mathrm{B}T/(6\pi\eta D)$	§1.3
(7)	Coumou correction	$f_\mathrm{n} = (n_\mathrm{s}/n_\mathrm{std})^2$	§1.4
(8)	optical constant	$K = 4\pi^2 n^2 (\mathrm{d}n/\mathrm{d}c)^2/(N_\mathrm{A}\lambda^4)$	§1.5
(9)	depolarization ratios	depolarization ratios ρ_v, ρ_u	§1.6.1
(10, 11)	toluene Rayleigh ratio	toluene $R_\mathrm{VV}(T)$, R_VU, R_VH	§1.6.2
(12)	excess Rayleigh ratio	excess Rayleigh ratio	§1.7
(13)	calibration constant	calibration constant k_c	§1.7

#	Name	Equation	Section
(14)	computed solvent blank	solvent blank computed from a typed 90° intensity	§1.7.2
(15)	Hampel identifier (MAD band)	robust (median/MAD) outlier band	§2.1.1
(16)	windowed Hampel filter	windowed Hampel decomposition (rolling median/MAD)	§2.1.2
(17)	Fano factor	Fano factor $F = \sigma^2/\mu$	§2.1.3
(18)	blocking analysis	blocking analysis: SE of the mean vs block size	§2.1.4
(19)	Allan variance	Allan variance ${\sigma_y}^2(T_a)$ (fractional, contiguous gates)	§2.1.5
(20)	Welch's method	Welch averaged modified periodogram, as a one-sided density	§2.1.6
(21)	Welch E.D.F.	Welch equivalent degrees of freedom	§2.1.6
(22)	chi-square band	chi-square confidence band on the Welch PSD	§2.1.6
(23)	shot-noise floor	shot-noise floor $2\bar{r}$ (cps^2/Hz)	§2.1.7
(24)	Scargle false alarm	periodogram peak false-alarm probability	§2.1.8
(25)	baseline removal	$g_2{-}1 = (G - G_\infty)/G_\infty$ (baseline removal)	§2.2.1
(26)	Siegert relation	$g_2{-}1 = \beta\lvert g_1\rvert^2$ (Siegert)	§2.2.1
(27)	multi-tau axis	multi-tau lag axis $\Delta t_j = 2^j \Delta t_0$, $\tau_{j,k} = k\,\Delta t_j$	§2.2.2
(28)	Schätzel variance (classical)	per-channel var($g_2{-}1$), Schätzel classical limit	§2.3
(29)	Schätzel variance (full)	per-channel var($g_2{-}1$), Schätzel full six-term form with the count rate	§2.3
(30)	Koppel normalization	baseline as a determined normalization, $g_2{-}1 = C(\tau)/G_\infty - 1$ (Koppel)	§3.1.2
(31)	ISO 22412 A.16	the Linear fit with the baseline G_∞ (ISO's A) pinned (ISO 22412 Formula A.16)	§3.1.2
(32)	pinned offset	offset ε plus $\beta\lvert g_1\rvert^2$ for the exponential / KWW fits	§3.1.3
(33)	decay-completeness check	the decay-completeness check $\Gamma{\cdot}\Delta\tau < ½ \ln 50$ (ISO 22412)	§3.1.4
(34)	size range	DLS size-range bound $R_h > {R_h}^{max}$ (sedimentation, ISO 22412)	§3.1.5
(35)	Koppel cumulant expansion	cumulant log-expansion	§3.2.1
(36)	polydispersity index	PDI $= \mu_2/\bar{\Gamma}^2$	§3.2.1
(37)	ISO 22412 A.14	cumulant size as the scattered-intensity-weighted harmonic mean (ISO 22412 Formula A.14)	§3.2.2
(38)	Frisken cumulant	the Nonlinear fit (Frisken, floating baseline)	§3.2.5
(39)	fourth cumulant	fourth-order terms: $\varkappa_4 = \mu_4 - 3{\mu_2}^2$ (linear) vs μ_4 (nonlinear)	§3.2.6
(40)	Biganzoli–Ferri variance	per-channel var($g_2{-}1$), exact in Θ_k (Biganzoli & Ferri)	§3.2.8

#	Name	Equation	Section
(41)	cumulant residual weight	the Nonlinear fit's residual weight: the exact-in-Θ_k variance evaluated on the first-pass single exponential (Koppel)	§3.2.8
(42)	single exponential	single exponential	§3.3.1
(43)	double exponential	double exponential	§3.3.2
(44)	KWW	KWW stretched exponential	§3.3.3
(45)	KWW mean time	mean relaxation time $\langle\tau\rangle = (\tau_c/s)\cdot\Gamma_E(1/s)$ (Lindsey & Patterson)	§3.3.3
(46)	Laplace inversion	distribution (A x = y)	§3.4.1
(47)	distribution residual weight	distribution residual weight $w_k \propto {g_1}^2/\sigma^2$, the delta-method inverse variance of the recovered g_1	§3.4.2
(48)	scale anchor	scale anchor a_w matching $\mathrm{tr}(\mathbf{A}^\mathsf{T}\mathbf{M}\mathbf{A})$ to the ${g_1}^2$ form's	§3.4.3
(49)	CONTIN (Tikhonov)	CONTIN (Tikhonov)	§3.4.5
(50)	effective degrees of freedom	Tikhonov effective DOF $\mathrm{NDF}(\alpha) = \mathrm{tr}(\mathrm{hat}) = \Sigma\ {s_j}^2/({s_j}^2+\alpha^2)$	§3.4.5
(51)	GCV	CONTIN GCV score $N_y\,V/(N_y-\mathrm{NDF})^2$ (α selection, classical; $\gamma_r = 1$ of robust GCV)	§3.4.6
(52)	robust GCV (Lukas)	CONTIN robust GCV score $[\gamma_r + (1-\gamma_r)t_2]\cdot\mathrm{GCV}(\alpha)$ (Lukas's RGCV; $\gamma_r = 1$ is classical GCV)	§3.4.7
(53)	L-curve (Hansen)	CONTIN L-curve curvature $\kappa(\alpha)$ (max-curvature corner on the log residual vs log $\|\mathbf{x}\|$ curve)	§3.4.8
(54)	Provencher F-test	CONTIN F-test statistic $F(\alpha)$ at NDF_0 (probability to reject)	§3.4.9
(55)	lognormal	lognormal DLS weighting law $w_i \propto \exp[-(\ln R_{h,i}-\mu)^2/2\sigma^2]$	§3.4.10
(56)	Γ vs q^2	$\bar{\Gamma} = Dq^2$	§3.5.1
(57)	D vs c	$D_{app}(c) = D_0(1 + k_D c)$	§3.5.2
(58)	Sedlák–Amis slow mode	${\tau_s}^{-1} = aq^2 + bq^4$	§3.5.4
(59)	DDLS rates	DDLS decay rates $\Gamma_{VV} = q^2 D_t$, $\Gamma_{VH} = q^2 D_t + 6D_r$	§3.6.1
(60)	rotational diffusion	rotational diffusion $D_r = (\Gamma_{VH} - \Gamma_{VV})/6$; τ_{rot}	§3.6.1
(61)	Stokes–Einstein–Debye	sphere $D_r = k_B T/(8\pi\eta R^3)$; R from D_r	§3.6.4
(62)	Tirado–García de la Torre	rod D_t, D_r (Tirado cylinder)	§3.6.5
(63)	rod end effects	rod end-effect corrections ν, $\delta_\perp$	§3.6.5
(64)	Perrin shape factor	ellipsoid Perrin shape factor $aS(\phi)$	§3.6.6
(65)	Perrin translational friction	ellipsoid D_t (Tanford/Perrin friction)	§3.6.6

#	Name	Equation	Section
(66)	Koenig rotational friction	ellipsoid $D_{\mathrm{r}} = k_{\mathrm{B}}\,T / C_2$ (Koenig)	§3.6.6
(67)	intensity → number	intensity→number reweighting $N \propto w_{\mathrm{I}}/I_1$	§3.7
(68)	surface / volume weights	surface/volume moments $N{\cdot}R^2$, $N{\cdot}R^3$	§3.7
(69)	Rayleigh law	Rayleigh per-particle $I_1 \propto R^6$	§3.7
(70)	Rayleigh–Debye–Gans	RDG per-particle $I_1 \propto V^2 P(qR)$	§3.7
(71)	derived intensity (trace)	derived intensity $\bar{I}(\theta) = (1/N)\Sigma I_k$	§4.1
(72)	derived intensity (header)	header-derived intensity $\bar{I} = N_{\mathrm{counts}}/(N_{\mathrm{samples}}\Delta t)$	§4.1
(73)	Sedlák partition	Sedlák partition of the normalized excess (two modes)	§4.2
(74)	Sedlák partition (n modes)	n-mode closure ⇒ $\Delta R_i = \hat{A}_i \Delta R$	§4.2
(75)	Amplitude Guinier	Amplitude Guinier: $A_{\mathrm{s}}(0)/A_{\mathrm{s}}(q) \cong 1 + (1/3){R_{\mathrm{g,app}}}^2 q^2$	§4.2.1
(76)	resolution floor	per-mode R_{g} resolution floor $\sqrt{3\delta_{\mathrm{A}}/(q_{\max}^2 - q_{\min}^2)}$	§4.2.1
(77)	Debye plot	apparent Debye intercept	§4.3
(78)	single-angle ladder	single-angle ladder: apparent M_{w}, A_2 from $Kc/\Delta R_\theta$ vs c	§4.3.1
(79)	Zimm	Zimm equation	§4.4
(80)	Zimm results	Zimm M_{w}, R_{g}, A_2	§4.4
(81)	Berry	Berry M_{w}, R_{g}, A_2	§4.4
(82)	calibration-free $A_2 M_{\mathrm{w}}$	calibration-free $2A_2 M_{\mathrm{w}}$	§4.5
(83)	Guinier	Guinier $\ln(\Delta R)$ vs q^2	§4.6
(84)	Cabannes (vertical)	Cabannes isotropic correction (vertical) $\Delta R_{\mathrm{iso}} = \Delta R_{\mathrm{VV}}(1 - 4\rho_{\mathrm{v}}/3)$	§4.7.1
(85)	optical anisotropy	optical anisotropy δ^2 and M_{app} inflation	§4.7.2
(86)	Cabannes (natural light)	natural-light Cabannes factor $(6 - 7\rho_{\mathrm{u}})/(6 + 6\rho_{\mathrm{u}})$	§4.7.3
(87)	form-factor ordinate	form-factor ordinate $Kc/\Delta R = 1/(M_{\mathrm{w}} P(q)) + 2A_2 c$	§4.8
(88)	sphere form factor	solid-sphere form factor $P(u)$, $u = qR$	§4.8
(89)	Debye coil form factor	Gaussian-coil (Debye) form factor $P(v)$, $v = (qR_{\mathrm{g}})^2$	§4.8
(90)	ellipsoid form factor	ellipsoid-of-revolution form factor $P(q;\, R,\, p)$ (orientation average)	§4.9
(91)	ellipsoid R_{g}	ellipsoid ${R_{\mathrm{g}}}^2 = R^2(2 + p^2)/5$	§4.9
(92)	Mie coefficients	Mie scattering coefficients a_n, b_n (Riccati–Bessel)	§4.10
(93)	Mie amplitude	Mie amplitude function $S_1(\theta)$	§4.10
(94)	Mie form factor	Mie form factor $P(q) = \lvert S_1(\theta)\rvert^2 / \lvert S_1(0)\rvert^2$	§4.10

#	Name	Equation	Section
(95)	Aden–Kerker a_n	coated-sphere (Aden–Kerker) coefficient a_n	§4.11
(96)	Aden–Kerker b_n	coated-sphere (Aden–Kerker) coefficient b_n	§4.11
(97)	Aden–Kerker A_n, B_n	coated-sphere ratio functions A_n, B_n (core)	§4.11
(98)	core–shell form factor	core–shell form factor $P(q)$ (coated $\lvert S_1\rvert^2/\lvert S_1(0)\rvert^2$)	§4.11
(99)	core–shell R_g	core–shell contrast-weighted ${R_g}^2$	§4.11.2
(100)	shape ratio ρ	$\rho = R_g/R_h$	§5.1
(101)	R_g–M_w scaling	R_g–M_w scaling law	§5.2
(102)	A_2–M_w scaling	A_2–M_w scaling law	§5.2
(103)	R_h–M_w scaling	R_h–M_w scaling law	§5.2
(104)	Zimm–Stockmayer g	branching contraction factor $g = \langle {R_g}^2\rangle_{br}/\langle {R_g}^2\rangle_{lin}$	§5.3
(105)	regular star (g)	regular f_a-arm star $g = (3f_a{-}2)/{f_a}^2$ and its inverse	§5.3
(106)	random branching (Zimm–Stockmayer)	random-branching fraction $g_3/g_4(m_b) \to$ branching index m_b	§5.3.1
(107)	hydrodynamic factor h	hydrodynamic contraction factor $h = R_{h,br}/R_{h,lin}$	§5.4
(108)	regular star (h)	regular f_a-arm star $h = \sqrt{f_a}\,/\,[(2-f_a)+\sqrt{2}\,(f_a-1)]$ and its inverse	§5.4
(109)	HC3	HC3 sandwich covariance $\mathrm{Cov}(\mathbf{b})$	§6.1
(110)	coefficient SE	OLS / through-origin slope & intercept SE	§6.1
(111)	Zimm design	Zimm/Berry multilinear design	§6.1
(112)	equilibrated SVD	$(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}$ and leverages from the equilibrated SVD	§6.1
(113)	classical OLS	classical OLS covariance $s^2(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}$	§6.1.1
(114)	delta method (GUM)	delta-method propagation $\mathrm{SE}(f)=\sqrt{\mathbf{J}^\top\Sigma\,\mathbf{J}}$	§6.2
(115)	ratio propagation	ratio propagation ($\rho = R_g/R_h$)	§6.2
(116)	power-law propagation	power-law propagation ($R_h \propto D^{-1}$)	§6.2
(117)	two-route gap	Zimm two-route M_w gap	§6.3
(118)	replicate SEM (ISO 22412)	replicate mean $\pm$ SD/$\sqrt{N}$ (ISO 22412 repeats)	§6.4
(119)	condition number	design condition number $\mathrm{cond}(\hat{\mathbf{X}})$ and the ill-conditioning limit	§6.6
(120)	AR(1) channel noise	correlated channel noise, first-order autoregression (house)	§7.3

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